Learning to Replenish and Audit under Limited Inventory Traceability

Recep Bekci
Department of Management Science and Engineering, University of Waterloo, recep.bekci@uwaterloo.ca
Mehmet Gumus
Desautels Faculty of Management, McGill University, mehmet.gumus@mcgill.ca

Abstract. Digital technologies have lowered the cost of entering retail, but they have not eliminated the operational burden of maintaining accurate item-level inventory for most small retailers. These retailers maintain a book-inventory record by updating recorded orders and sales. Unrecorded shrinkage, such as spoilage, damage, misplacement, or theft, makes this record drift above the sellable stock until a stockout or a costly physical count restores information. The inventory planner can either order against this book inventory or discount it by an expected shrinkage rate that must be learned. Motivated by a small online grocer, we develop KM–Audit (*Kaplan–Meier Audit*), an online policy that combines retention learning from exact and right-censored observations with adaptive replenishment and physical counting. It learns the order-up-to level, chooses between the two record-keeping conventions, and determines when a physical count is worth its cost. We prove that KM-Audit's cumulative cost gap relative to the best base-stock policy that knows the true shrinkage distribution grows sublinearly over time, so its average cost converges to that of this benchmark. Numerical experiments with both synthetic and real retailer demand data show the value of adaptive audits and how the trade-off among cost, service, and waste varies across operating environments.

Key words: inventory record inaccuracy; inventory auditing; hidden shrinkage; censored learning; online inventory control

1. Introduction

Cloud e-commerce management platforms (CEMPs), such as Shopify and nopCommerce, have dramatically lowered the cost of entering online retail. For a modest subscription fee,[1] an entrepreneur can assemble a fully functional storefront from off-the-shelf modules for product-category management, shopping-cart formation, online payment, shipping-cost calculation, and shipment tracking. These tools have largely commoditized the customer-facing side of e-commerce. The back-end side has not kept pace. In larger firms, item-level inventory control is typically governed by enterprise resource planning (ERP) and warehouse management systems (WMS) whose licensing, integration, and maintenance costs remain prohibitive for most small online retailers (OECD, 2021).[2] Many small retailers therefore rely on the basic stock counter built into their CEMP, supplemented by spreadsheets or simple databases that are updated manually rather than automatically. Growth on the consumer side can thus outpace the back-end infrastructure needed for

[1] Third-party comparisons report that standard subscription plans for popular e-commerce management platforms range from roughly $25 to $400 per month (Aviso, 2025).

[2] Much of the cost of these systems lies in deployment rather than software. Industry estimates place an enterprise WMS deployment at roughly $200,000 to more than $1 million, with implementation timelines of 6 to 18 months (Grasshopper Labs, 2026; Monetizely, 2025).

replenishment, as retailers can add products and orders far more quickly than they can acquire the systems required to track them.

CEMPs do provide a basic form of inventory tracking. For each SKU, the platform keeps a running stock count that is decremented automatically with every recorded sale and incremented when the retailer enters a receipt. The platform therefore constructs an inventory level from transaction records rather than observing it directly. We refer to this constructed quantity as the retailer's *book inventory*. The book record is only as accurate as the transactions behind it. Namely, even when every sale and receipt is recorded faithfully, the quantity available for sale is eroded by events that never pass through the platform. For instance, a case of produce that spoils in the warehouse generates no transaction, so the book record continues to count it as available for sale. Book inventory therefore need not coincide with the quantity actually available for sale. The distinction is operationally consequential. A book record describes what should be available, whereas replenishment depends on what can actually be found and sold. Theft, damage, misplacement, unrecorded disposal, and spoilage can reduce availability without a corresponding database update. We use *inventory shrinkage* as an umbrella term for these unrecorded losses. Since the surviving inventory carries into later periods, an unrecorded loss affects not only the current sale but also subsequent inventory estimates, orders, and stockout risk. Much of classical dynamic inventory theory nevertheless takes the physical inventory state as observable at each decision epoch. For a retailer facing such record drift, however, this benchmark is not directly implementable because standard replenishment decisions require knowledge of the physical inventory level, which is precisely what the retailer cannot observe.

One might argue that periodic counting resolves this problem. Under a periodic-review policy, the retailer could count its stock just before each replenishment order, resetting the record to the true level at every review epoch. This argument breaks down as the assortment grows. Counting every SKU before every replenishment decision may be feasible for a small assortment and prohibitive for one containing thousands of items—a scale that is increasingly common even among small online operators. The alternative is to rely on routine operations to restore visibility. A stockout reveals that sellable inventory has reached zero, but only after the retailer risks unserved demand. The managerial question is therefore how to substitute between exact information purchased through a targeted SKU-level physical count and the imperfect signals generated by routine operations. Between counts, the retailer must replenish from an imperfect state estimate and learn from those signals.

This paper is motivated by our collaboration with a small online grocer in Eastern Canada, founded by an individual entrepreneur, that connects local farmers and specialty food-and-beverage producers directly with end consumers. Building on a CEMP storefront and a network of local suppliers, the founder grew the assortment from a few hundred SKUs to more than 10,000 in under three years. The COVID-19 pandemic added a tailwind, as consumers sought fresh, locally sourced products that large supermarket chains, facing supply-chain disruptions and capacity restrictions, struggled to provide. As order volumes grew, the

firm rented a local warehouse to consolidate inbound shipments and fulfill consumer orders. The warehouse turned a problem that had been manageable at small scale into a persistent one. Perishable food-and-beverage inventory is subject to stochastic spoilage that varies across suppliers and product types, and the retailer's limited data infrastructure made item-level visibility difficult to maintain. Unrecorded spoilage is one source of shrinkage, alongside breakage, theft, damage, and misplacement. These mechanisms differ physically but create the same information problem: the retailer's operational inventory estimate may exceed the quantity that can be located and sold. For this retailer, restoring visibility requires either a labor-intensive manual count or a stockout that risks lost sales. These are the two information channels our model formalizes.

Motivated by this setting, we take the perspective of a resource-constrained small online retailer and develop an inventory framework that respects the traceability limitations it faces in practice. Specifically, we study a periodic-review lost-sales inventory system in which sellable inventory is hidden and evolves through replenishment, shrinkage, and sales. We model shrinkage using a random retention factor, defined as the fraction of inventory that remains sellable after unrecorded loss in a period. The factor is an aggregate representation of inventory loss. The state is known at stockout boundaries and after audit reconciliation, but remains hidden during routine within-cycle replenishment. This information structure creates a linked learning problem: estimating the shrinkage process, choosing a replenishment target and record-updating rule, and deciding when a physical count of the focal SKU is worth its cost.

Observing retailer behavior and prior literature on this topic, we retain the familiar order-up-to logic of replenishing toward a target and consider two ways to update the inventory record. The expected-inventory (EI) rule updates its internal record using the learned mean retention factor, whereas the stockout-reconciled book-inventory (BI) rule updates its book balance from recorded orders and sales and resets it to zero when a reliable stockout reveals that no sellable inventory remains. Both rules rely on routine CEMP records, and we focus on them because a CEMP-based planner can implement either without installing additional inventory-tracking infrastructure. To learn the base-stock target and choose between these rules while deciding when exact inventory information is worth purchasing, we develop **KM–Audit** (*Kaplan–Meier Audit*), an online policy that pools exact audit observations with the exact and right-censored retention signals generated during ordinary operation. Audits therefore help determine not only the magnitude of shrinkage, but also whether shrinkage should alter replenishment. Our benchmark is the better of the clairvoyant non-auditing EI and BI target policies, each optimized over its base-stock level under the true retention law. All performance guarantees compare KM–Audit with this benchmark.

The joint choice is economically nontrivial because record accuracy is an input to replenishment rather than an objective by itself. EI corrects the systematic overstatement created by shrinkage and can protect service, but it can also induce larger orders and greater inventory exposure. BI may therefore be preferable when the value of a leaner operating posture exceeds the cost of the additional shortages it creates. Likewise,

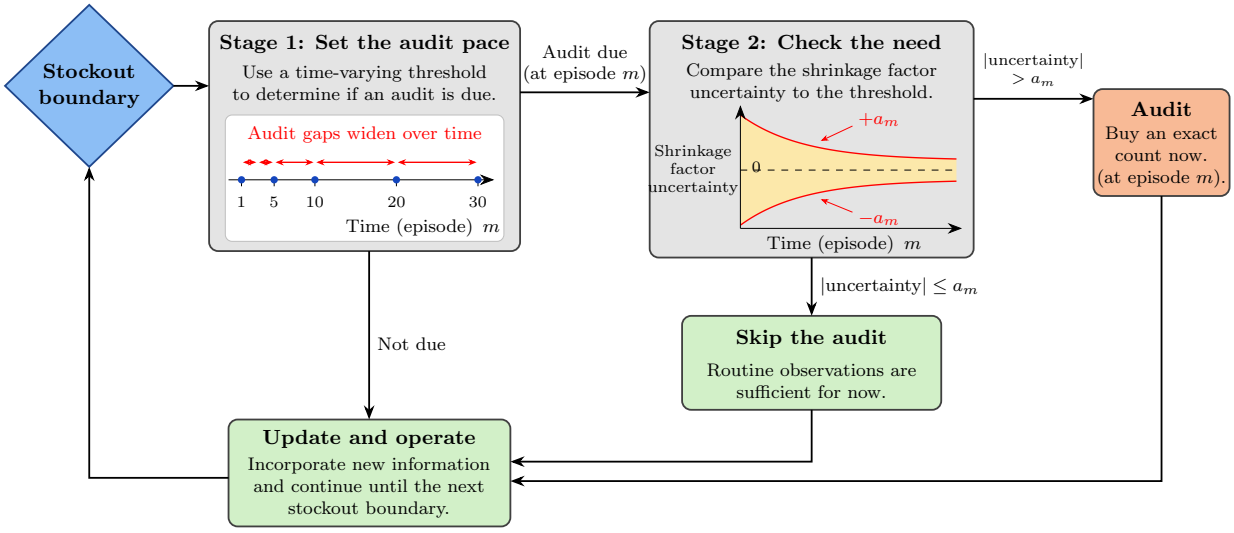


Figure 1 **The two-stage audit logic of KM–Audit. Stage 1 determines when another audit may be worthwhile. Stage 2 conducts the audit only if routine operating observations have not already reduced retention uncertainty sufficiently.**

an exact count is valuable only when the information is likely to change the preferred target or tracking rule enough to justify the count and its operating consequences. Audit intensity should therefore depend not only on how much shrinkage occurs, but also on how sensitive the replenishment decision is to the remaining uncertainty and how much information routine operation has already supplied. Figure 1 summarizes this two-stage information-acquisition logic.

Main Contributions. Our main results and contributions can be summarized as follows.

1. **A tractable learning formulation under hidden shrinkage.** Although true period costs are hidden, a completed stockout cycle reveals total sales and cumulative orders net of sales. We show that their per-period rates form an observable pseudo-cost whose ranking coincides with that of average operating cost, up to the target-independent term $b\mathbb{E}[D_t]$. This regenerative representation makes it possible to compare candidate replenishment policies using the retailer's operational records.
2. **A unified algorithm for replenishment and information acquisition.** KM-Audit jointly learns the base-stock target and whether to use EI or BI. It pools exact audit observations with the exact and censored retention signals generated at ordinary cycle starts, and constructs complementary confidence intervals that adapt to the informativeness of routine operation. The resulting confidence set supports both policy evaluation and EI implementation, while the algorithm decides adaptively whether residual retention uncertainty warrants another paid count.
3. **Regret guarantees and cost-adjusted audit decisions.** We establish a $\widetilde{O}(T^{2/3})$ regret upper bound against the better policy across the continuous EI and BI target classes (Theorem 1). The bound separates policy-arm learning from retention learning and makes the information-acquisition trade-off

explicit: counts buy exact information, whereas unresolved retention uncertainty distorts policy evaluation and EI implementation. Its comparative statics show that more costly audit episodes make the policy more tolerant of uncertainty, while greater operating-cost sensitivity to retention error makes exact information more valuable. We also prove an $\Omega(\sqrt{T})$ lower bound showing that retention uncertainty alone creates a nontrivial regret floor (Theorem 2).

4. **Evaluation using real-world retailer data.** Using two years of data, the learning policies are calibrated on 2023 records and tested on the subsequent 2024 fulfilled-order sequence. An externally specified shrinkage scenario generates policy-specific inventory states, counts, stockouts, and lost sales. The analysis preserves observed order timing and operational heterogeneity.

2. Literature Review

This paper contributes to two closely related streams: inventory control under imperfect operational visibility and data-driven inventory control with censored observations.

Inventory record inaccuracy and operational visibility. Inventory record inaccuracy has long been recognized as a central obstacle in inventory management. In an early contribution, Iglehart and Morey (1972) study inventory systems in which the recorded stock level may differ from the physical stock because of delays in information flow, pilferage, counting errors, and other sources of discrepancy. Raman et al. (2001) argue that inventory record inaccuracy and misplaced inventory are important execution failures in retail operations, while DeHoratius and Raman (2008) provide large-scale empirical evidence on the prevalence and systematic variation of inventory record inaccuracy across stores and product categories. Kang and Gershwin (2005) show, through analytical and simulation models, that even small rates of undetected stock loss can accumulate over time and substantially degrade replenishment performance. Related work quantifies the value of inventory visibility technologies, such as RFID, by comparing policies that operate with inaccurate records to policies with more accurate physical inventory information (Lee and Özer, 2007). Our proportional-retention model is also related to deterioration and random-yield systems (Bakker et al., 2012; Henig and Gerchak, 1990; Yano and Lee, 1995), but differs in that the unobserved factor applies to total pre-shrinkage inventory rather than unit age or procurement output.

A closely related stream studies replenishment and auditing decisions under inaccurate inventory records. Kök and Shang (2007) analyze joint inspection and replenishment policies when inventory records are inaccurate and inspections are costly. DeHoratius et al. (2008) develop a Bayesian inventory management tool that maintains a belief over the physical inventory level and uses sales observations to update that belief. Mersereau (2013) studies information-sensitive replenishment, showing how record inaccuracy changes the value of replenishment decisions because sales observations carry information about the hidden state. Mersereau (2015) further shows that demand estimation from censored sales data can be systematically biased when the firm ignores inventory record uncertainty. More recent work studies inventory auditing using point-of-sale data (Bassamboo et al., 2020), data-driven shelf inspection for phantom stockouts

(Chen, 2021), simultaneous learning of inventory level and shrinkage rate (Li et al., 2022), continuous-time Bayesian models of restocking and auditing under record inaccuracy (Chehrazi, 2025a), and the distinct roles of transaction errors and unobservable loss in inaccurate inventory systems (Chehrazi, 2025b). Other recent work develops large-scale anomaly-detection methods for inventory inaccuracies (Farias et al., 2024).

The closest learning-based study in this stream is Li et al. (2022), who jointly learn hidden inventory and an unknown shrinkage rate within a Bayesian partially observed control model. Our distinction is not audit endogeneity alone, since inspection and audit decisions are also optimized in prior work (Kök and Shang, 2007; Bassamboo et al., 2020; Chehrazi, 2025a). Instead, we combine nonparametric learning of the retention law from exact and predictably censored observations with regenerative comparison of EI and BI target policies and finite-horizon regret guarantees. This shifts the focus from solving a specified belief-state control model to learning both the operating policy and the value of additional inventory information online.

The closest structural predecessors are Bensoussan et al. (2013) and Bensoussan et al. (2016), who study periodic-review lost-sales control under cash-register information: sales are observed, whereas demand, shrinkage, and physical inventory are only partially observed. Bensoussan et al. (2013) use unnormalized conditional probabilities to linearize the hidden-state evolution and characterize the associated Bellman equation. Bensoussan et al. (2016) allow recorded and unrecorded demand to be dependent, update the inventory belief from observed sales, and develop an iterative approximate-control method for the resulting infinite-dimensional problem. Other structural work under partial observation includes Bensoussan et al. (2008), who study a backlogging model with rain-check observations; Bensoussan et al. (2010), who consider an information structure in which the manager observes only whether inventory is positive or nonpositive; and average-cost analyses with a partially observed demand regime (Avci et al., 2020) or inventory observed only upon reaching zero (Bai et al., 2023).

These papers provide the closest belief-state control foundations for our model. Rather than solving the corresponding infinite-dimensional belief-state control problem, we exploit regenerative stockout cycles to construct an observable pseudo-cost for comparing implementable EI and BI target policies. This reduction makes it possible to learn the operating policy online and establish finite-horizon regret guarantees without continuously reconstructing the hidden inventory state.

Inventory learning with censored demand. Our paper also relates to the literature on inventory learning with censored demand. In lost-sales inventory systems, sales data are censored because the firm observes demand only up to the available inventory level. The Bayesian learning literature studies how firms should balance immediate inventory costs with the informational value of stocking more inventory. Lariviere and Porteus (1999) analyze Bayesian inventory management with unobserved lost sales, and Chen (2010) develops bounds and heuristics for Bayesian inventory control with unobserved lost sales. The censored newsvendor literature further shows that censoring fundamentally alters the value of information and the need for experimentation (Ding et al., 2002; Besbes and Muharremoglu, 2013).

A large stream of nonparametric and online-learning papers studies adaptive inventory control when the demand distribution is unknown. Huh and Rusmevichientong (2009) develop nonparametric adaptive policies for inventory planning with censored demand. Huh et al. (2009a) study adaptive learning of the optimal base-stock policy in lost-sales systems with censored demand and establish a cubic-root convergence rate. Huh et al. (2011) use the Kaplan–Meier product-limit estimator to construct data-driven policies for censored-demand inventory problems. More recent work closes regret gaps for lost-sales systems with lead times (Zhang et al., 2020), studies multi-location inventory learning with censored demand (Bekci et al., 2023), incorporates feature information into censored-demand inventory control (Ding et al., 2024), and studies joint pricing and inventory learning when demand is price-sensitive and censored by inventory availability (Chen et al., 2020, 2021, 2024). Base-stock policies thus provide a standard policy benchmark in lost-sales inventory learning. In fully observed lost-sales models, order-up-to policies also enjoy asymptotic guarantees and strong numerical performance when lost-sales penalties are large relative to holding costs (Huh et al., 2009b).

The key distinction is that in this literature the inventory state is typically observable or correctly trackable, while demand is censored by that state. In our model, censoring is compounded by a second information friction: shrinkage makes the inventory state itself unobservable. This changes both the statistical and control problems. Estimating mean retention μ_ρ is needed to evaluate both record-updating rules through our cycle-level performance statistic and to implement EI, but BI remains fixed at its book-inventory update. Thus, unlike standard censored-demand learning, parameter uncertainty affects policy evaluation and changes the trajectory of the EI policy being evaluated. Prior inventory-learning and censored-bandit applications use the Kaplan–Meier estimator (Kaplan and Meier, 1958) primarily to recover demand distributions or censored rewards (Huh et al., 2011; Lyu et al., 2024). In contrast, KM–Audit pools exact audit observations with policy-dependent, predictably censored operating observations and derives an explicit finite-sample confidence guarantee for the latent mean-retention factor that governs both EI implementation and the EI–BI policy comparison.

Accordingly, our comparator is the best non-auditing fixed-target policy across the EI and BI classes. It knows the retention law and the optimal target from the outset and therefore requires no physical counts for learning. Regret measures the finite information cost that KM–Audit incurs to identify and approach this clairvoyant ordinary-operation policy.

3. Model

This section formalizes the setting described in Section 1. We consider a single SKU managed by an *inventory planner* at a small online retailer. The retailer operates its storefront through a CEMP that records replenishment orders and customer sales but does not directly observe unrecorded shrinkage or the sellable inventory on hand. The planner therefore does not continuously observe the physical inventory available

for sale. Two events restore that information: a stockout, which reveals that no sellable unit remains, and a costly physical count, which we call an *audit*. Section 3.1 describes the true inventory dynamics and what the planner observes. Section 3.2 introduces the planner's two ordering policies, and Section 3.3 shows how their performance can be evaluated from CEMP records alone.

3.1. True dynamics and observations

Time is indexed by $t = 1, 2, \ldots$. There is no lead time. We initialize the system at a stockout state:

$$X_1 = 0.$$

Let $X_t \geq 0$ denote the true sellable and locatable on-hand inventory at the beginning of period t, before ordering. Without observing the true on-hand inventory, the inventory planner places an order $q_t \geq 0$, so the inventory before shrinkage becomes $X_t + q_t$. A random retention factor $\rho_t \in [0, 1]$ is then realized. We interpret $1 - \rho_t$ as the fraction of this inventory rendered unavailable by an unrecorded loss, such as theft, damage, misplacement, disposal, or spoilage. The usable inventory available for sale in period t is therefore

$$A_t := \rho_t(X_t + q_t).$$

Demand in period t is denoted by $D_t \geq 0$. Sales are censored by usable inventory:

$$Y_t := \min\{A_t, D_t\}.$$

Lost demand and next-period true inventory are

$$L_t := D_t - Y_t = (D_t - A_t)^+, \qquad X_{t+1} := (A_t - D_t)^+.$$

Unsold surviving inventory carries into the next period together with the next order. The retention factor is an aggregate representation of shrinkage. It does not track product age, expiration, or the mechanism that produced the loss, and a misplaced unit is treated as lost to the sellable state. This proportional, age-independent specification is most accurate for losses that scale with the amount of stock exposed, such as handling damage, misplacement, and spoilage of short-life items that are replenished frequently.

Under ordinary operation, the observable data in period t are

$$(Y_t, Z_t), \qquad Z_t := \mathbf{1}\{D_t > A_t\},$$

where Z_t is the stockout indicator. These are exactly the data available from the CEMP: each fulfilled customer order is recorded as a sale, and a stockout becomes known when an order cannot be fulfilled because no sellable unit can be found. Away from stockout boundaries and the state immediately reconciled by an audit count, the true inventory X_t, usable inventory A_t, lost demand L_t, and realized retention factor ρ_t are unobserved.

Within the modeled SKU environment and horizon, we assume that $\{D_t\}_{t\geq 1}$ and $\{\rho_t\}_{t\geq 1}$ are i.i.d. over time and mutually independent. Their distributions are unknown to the inventory planner. Thus, observable drivers of nonstationarity, such as supplier, batch, or seasonal conditions, are held fixed in the present analysis. The mean retention factor is denoted by

$$\mu_\rho := \mathbb{E}[\rho_t].$$

The one-period operating cost is

$$C_t := hX_{t+1} + bL_t = h\big(\rho_t(X_t + q_t) - D_t\big)^+ + b\big(D_t - \rho_t(X_t + q_t)\big)^+,$$

where $h > 0$ is the holding-cost parameter and $b > 0$ is the lost-sales penalty.

The planner can also purchase exact information through a manual count of the SKU (hereafter referred to as an *audit*), which is accurate but costly in labor and time. An audit may be initiated only at a stockout boundary, where $X_t = 0$ is already known, and before the current demand and shrinkage realizations are observed. We restrict counts to stockout boundaries because a count that starts from a known zero state isolates a single retention draw and therefore yields an exact, uncensored observation. On the other hand, a count at an arbitrary time would instead reveal a product of several draws confounded with the hidden carried-over inventory. Fix a known audit-start order

$$s_{\mathrm{aud}} \in (0, \bar{S}].$$

If period t is designated as an audit period, then $X_t = 0$, the inventory planner places the order

$$q_t = s_{\mathrm{aud}},$$

and the period otherwise follows the same operating dynamics as above. At the end of the period, in addition to observing (Y_t, Z_t), the inventory planner conducts an exact physical count of the focal item's ending on-hand inventory X_{t+1}. Because the audit begins from zero true inventory,

$$\rho_t s_{\mathrm{aud}} = Y_t + X_{t+1}.$$

Consequently, the audit reveals the current retention realization exactly:

$$\rho_t = \frac{Y_t + X_{t+1}}{s_{\mathrm{aud}}}. \tag{1}$$

This identity holds whether or not the audit period stocks out.

We treat the audit-start order as an exogenous operational design parameter and use only the uniform bound $s_{\mathrm{aud}} \leq \bar{S}$; its service–carryover trade-off is discussed in Section A of the Electronic Companion.

Let

$$J_t^{\mathrm{aud}} := \mathbf{1}\{\text{a physical inventory audit is conducted in period } t\}.$$

Each physical count incurs a known direct cost

$$\kappa_{\mathrm{aud}} > 0,$$

representing counting labor, operational disruption, and reconciliation effort. The total cost incurred by the learning algorithm in period t is therefore

$$C_t^{\mathrm{alg}} := C_t + \kappa_{\mathrm{aud}} J_t^{\mathrm{aud}}. \tag{2}$$

The ordinary holding and lost-sales consequences of the audit-start order and any subsequent recovery periods are already included in C_t, so κ_{aud} represents only the direct cost of conducting the physical count.

The physical count itself is completed in one period, but the corresponding audit episode need not end after that period. If the audit period stocks out, then $X_{t+1} = 0$ and the next period is a new stockout boundary. If it does not stock out, the exactly counted inventory $X_{t+1} > 0$ is carried into a post-audit recovery tail. The audit episode then continues under a bounded recovery policy until the next stockout. No new statistical episode begins during this recovery tail.

3.2. Book-inventory and expected-inventory base-stock policies

Base-stock policies are widely used in retail practice, and our partner's planner follows the same logic: replenish each SKU up to a target level. Implementing such a policy requires an inventory figure to order against. Because physical stock is hidden, the planner orders against an internal record $\hat{X}_t$ rather than X_t. The natural record is the CEMP's own stock counter. As discussed in Section 1, the CEMP updates the book-inventory record from observed transactions by adding each replenishment order and subtracting each sale. We call this simulated quantity the *book inventory*. Because shrinkage is never recorded, the book inventory drifts above the true sellable inventory between stockouts. When a stockout occurs, however, the planner knows that no sellable unit remains and can reset the record to zero. The record starts at zero, as does the true inventory:

$$\hat{X}_1 = 0.$$

To model the setting faced by our industry partner, we give the planner two options for maintaining $\hat{X}_t$. Under option (i), the planner treats the CEMP's figure as correct and orders against it. We call this the *book-inventory* (BI) policy. Under option (ii), the planner discounts the recorded stock by the expected retention factor before ordering. We call this the *expected-inventory* (EI) policy. Both options share one update, indexed by a coefficient $\nu \in (0, 1]$. After ordering q_t and observing (Y_t, Z_t), the record is updated according to

$$\hat{X}_{t+1} = (1 - Z_t)\big(\nu(\hat{X}_t + q_t) - Y_t\big)^+. \tag{3}$$

The reset after a stockout is exact: if $Z_t = 1$, then all usable inventory has been sold, so both the true inventory and the internal record are zero at the next period. Away from stockouts and audit reconciliations, the recursion substitutes the coefficient ν for each unobserved retention draw. With $\nu = 1$, the recursion reproduces the CEMP's book inventory, reconciled to zero at each stockout. With $\nu < 1$, the planner removes an expected fraction $1 - \nu$ of the recorded stock each period to account for unrecorded shrinkage.

For a base-stock level $S \in [0, \bar{S}]$, the corresponding policy orders

$$q_t = (S - \hat{X}_t)^+. \tag{4}$$

We denote this policy by π_S^ν.

Let

$$\mathcal{C}_{\mathrm{tr}} := \{\mathrm{EI}, \mathrm{BI}\}, \qquad \nu_{\mathrm{EI}} := \mu_\rho, \qquad \nu_{\mathrm{BI}} := 1.$$

The two modes represent distinct inventory-record updating conventions. Within an ordinary decision cycle, $\hat{X}_t + q_t = S$; hence, following a nonstockout under target S, their next-period orders are

$$q_{t+1}^{\mathrm{EI}} = \min\{S, Y_t + (1 - \mu_\rho)S\}, \qquad q_{t+1}^{\mathrm{BI}} = \min\{S, Y_t\}.$$

Thus, following a nonstockout, BI replenishes observed sales, whereas EI also adds $(1 - \mu_\rho)S$, an adjustment for average shrinkage at target S. In either case, the order is capped at S. In either mode, the reliable stockout signal reconciles the tracker to zero. The inventory planner jointly selects the base-stock target and which rule to use according to their operating costs.

Implementing EI requires the mean retention factor μ_ρ, which the planner does not know. Two channels supply information about it. First, an audit reveals the current retention draw exactly through (1), at the direct cost κ_{aud}. Second, routine operation supplies partial information at no additional cost. Right after a stockout, the record and the true inventory are both zero, so the planner orders exactly S, and the sales in that period yield the lower bound $\rho_t \geq Y_t/S$. These are the planner's two information sources: a paid count and the sales observed after a stockout. The KM-Audit algorithm in Section 4 pools both channels to learn μ_ρ, decides when another audit is worth its cost, and identifies the base-stock target and policy with the lowest operating cost.

After a nonstockout audit period, the physical count permits the exact reconciliation $\hat{X}_{t+1} := X_{t+1}$. The inventory planner then follows a prespecified bounded recovery policy $(S^{\mathrm{rec}}, \nu^{\mathrm{rec}}) \in [0, \bar{S}] \times (0, 1]$ until the next stockout. This recovery tail remains part of the audit episode and does not generate a policy-arm performance observation.

For any $S \in [0, \bar{S}]$ and $\nu \in (0, 1]$, define the long-run average operating cost

$$g(S; \nu) := \limsup_{T \to \infty} \frac{1}{T} \mathbb{E}^{\pi_S^\nu} \left[\sum_{t=1}^{T} C_t \right]. \tag{5}$$

For $c \in \mathcal{C}_{\mathrm{tr}}$, write

$$g_c(S) := g(S; \nu_c).$$

The benchmark selects the better optimized policy across the EI and BI classes:

$$(c^\star_\cup, S^\star_\cup) \in \operatorname*{arg\,min}_{\substack{c \in \mathcal{C}_{\mathrm{tr}} \\ S \in [0, \bar{S}]}} g_c(S), \qquad g^\star_\cup := g_{c^\star_\cup}(S^\star_\cup). \tag{6}$$

For an online algorithm, regret over a horizon of T periods is measured against this non-auditing benchmark:

$$R_T := \mathbb{E}\left[\sum_{t=1}^{T} C_t^{\mathrm{alg}}\right] - T g^\star_\cup. \tag{7}$$

The benchmark knows which rule has lower cost, the mean retention factor required by EI, and the best target in the selected mode. During execution it uses only the ordinary observations and does not audit. It therefore incurs C_t but no count fee, whereas the inventory planner's direct count fees and the operating consequences of audit episodes enter C_t^{alg} and hence regret.

3.3. Stockout cycles and observable pseudo-costs

The true period operating cost is not observable under ordinary operation. The CEMP records orders and sales, but not the units lost to shrinkage or the demand turned away during a stockout. Except for the exact ending count obtained in an audit period, the inventory planner observes sales and stockout indicators, but not the next true inventory X_{t+1}, lost demand L_t, usable inventory A_t, or the retention realization ρ_t. The key observation is that ordinary stockout cycles nevertheless contain enough observable aggregate information to evaluate policies up to a one-dimensional nuisance parameter. The pseudo-cost construction below concerns completed cycles generated by a fixed, non-auditing policy π^ν_S.

Stockouts create regeneration points. Define the stockout times recursively by

$$\tau_0 := 0, \qquad \tau_k := \inf\{t > \tau_{k-1} : Z_t = 1\}, \qquad k \geq 1,$$

and let

$$\mathcal{C}_k := \{\tau_{k-1} + 1, \ldots, \tau_k\}$$

be the kth completed stockout cycle. For a completed cycle, define its length, total replenishment, total sales, and observable inventory loss by

$$\Lambda_k := \tau_k - \tau_{k-1}, \qquad Q_k := \sum_{t \in \mathcal{C}_k} q_t, \qquad V_k := \sum_{t \in \mathcal{C}_k} Y_t, \qquad W_k := Q_k - V_k.$$

The quantity W_k is observable because it is the difference between total orders and total sales, both of which the CEMP records. Telescoping the inventory balance over a completed cycle, whose initial and terminal physical inventories are both zero, gives

$$W_k = \sum_{t \in \mathcal{C}_k} (1 - \rho_t)(X_t + q_t). \tag{8}$$

Thus W_k is exactly the aggregate quantity rendered unavailable by shrinkage over the cycle. Equivalently, W_k is the book inventory that the CEMP would still display at the stockout if the planner did not reset it. Moreover, the true cost can be rewritten as

$$C_t = bD_t + hA_t - (h+b)Y_t. \tag{9}$$

The sales term is observed, and the demand term contributes only the arm-independent constant $b\mathbb{E}[D_t]$ per period after normalization. The remaining task is to express expected usable inventory through observable cycle aggregates; the complete derivation appears in the proof of Proposition 1 in the Electronic Companion.

For a fixed base-stock level S and tracker parameter ν, write

$$(\Lambda_k^{S,\nu}, Q_k^{S,\nu}, V_k^{S,\nu}, W_k^{S,\nu})$$

for the kth stockout cycle generated by π_S^ν from a stockout state. These are population cycle random variables under the fixed policy (S,ν). Independence of ρ_t from pre-shrinkage inventory and (8) imply

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] = \frac{\mu_\rho}{1-\mu_\rho}\mathbb{E}[W_1^{S,\nu}]. \tag{10}$$

For $\mu \in (0,1)$, define

$$\alpha(\mu) := \frac{h\mu}{1-\mu}. \tag{11}$$

Thus, the corresponding cycle pseudo-cost random variable is

$$\widetilde{K}_k^{S,\nu}(\mu) := \alpha(\mu)W_k^{S,\nu} - (h+b)V_k^{S,\nu}. \tag{12}$$

When the pseudo-cost coefficient is the true one, define the population pseudo-cost rate

$$\gamma(S;\nu) := \frac{\mathbb{E}[\widetilde{K}_1^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda_1^{S,\nu}]}. \tag{13}$$

In particular, $\gamma(S;\nu_c)$ is the population pseudo-cost rate of rule c at base-stock level S.

The next proposition explains why this quantity is central.

Proposition 1 *Fix $S \in [0,\bar{S}]$ and a tracker parameter $\nu \in (0,1]$ for which the induced stockout cycles are positive recurrent. Let*

$$v(S;\nu) := \frac{\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}, \qquad w(S;\nu) := \frac{\mathbb{E}[W_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}.$$

Then

$$g(S;\nu) = b\mathbb{E}[D_t] + \alpha(\mu_\rho)w(S;\nu) - (h+b)v(S;\nu) = b\mathbb{E}[D_t] + \gamma(S;\nu). \tag{14}$$

In particular, minimizing $g_c(S)$ jointly over $c \in \mathcal{C}_{\mathrm{tr}}$ and $S \in [0,\bar{S}]$ is equivalent to minimizing $\gamma(S;\nu_c)$ over the same set.

Proposition 1 is the structural reduction on which the learning problem rests. The true period cost C_t is hidden because lost demand and remaining true inventory are hidden. A completed ordinary decision cycle, however, reveals the aggregate statistics V_k and W_k from CEMP records alone. The oracle pseudo-cost of that cycle is still not directly known because its coefficient $\alpha(\mu_\rho)$ depends on the unknown mean retention factor. Thus the learning problem separates into two coupled tasks: estimate the common unknown mean-retention parameter μ_ρ, and use realized decision-cycle records to learn which rule and base-stock level have the smallest population pseudo-cost rate. The first task is why μ_ρ matters even for BI: although BI can be executed without it, comparing BI with EI requires it.

We have the following assumptions.

Assumption 1 (Bounded demand) *There is a known constant $0 < \overline{D} < \infty$ such that*

$$0 \le D_t \le \overline{D}$$

almost surely.

Assumption 2 (Compact mean-retention interval) *The unknown mean retention factor satisfies*

$$\mu_\rho \in [\underline{\mu}, \bar{\mu}] \subset (0, 1).$$

Assumption 3 (Positive stockout probability) *Let $\mathcal{F}_t^-$ denote the full, possibly latent, system history after q_t is chosen but before (D_t, ρ_t) is realized. There exists a known constant $p_0 \in (0, 1]$ such that the bound below holds uniformly over every decision episode generated with $S \in [0, \bar{S}]$ and $\nu \in [\underline{\mu}, \bar{\mu}] \cup \{1\}$, including both comparator and learner policies, and over every audit episode, including its bounded recovery tail:*

$$\mathbb{P}(Z_t = 1 \mid \mathcal{F}_t^-) \ge p_0$$

almost surely in every period in which the current episode remains active.

Assumption 4 (Bounded demand density) *There exists a constant $\bar{f}_D < \infty$ such that, for every interval $I \subset \mathbb{R}_+$,*

$$\mathbb{P}(D_t \in I) \le \bar{f}_D |I|,$$

where $|I|$ denotes the interval length.

Assumptions 1–2 are boundedness conditions. Assumption 3 simply requires that, under any policy considered, a stockout has some positive probability in every period while an episode is active. This ensures that each episode eventually ends at a stockout and that episode lengths and residual times have geometrically decaying tails. The condition holds, for example, whenever sellable inventory is physically bounded and demand exceeds that bound with positive probability. Assumption 4 prevents demand from concentrating

near a stockout threshold. Consequently, a small change in the tracker parameter or base-stock level changes the stockout indicator only with proportionally small probability. The audit-start order and all recovery targets are likewise capped by $\bar{S}$, so audit episodes operate on the same bounded inventory scale as ordinary decision episodes.

4. Proposed Algorithm: Kaplan–Meier (KM) Audit

KM–Audit exploits stockouts as regenerative decision epochs. At each such epoch, it either tests a base-stock policy under the EI or BI tracking rule, or purchases an exact retention observation through a physical count. Both actions generate an episode-start observation of retention: routine operation may censor that observation, whereas an audit reveals it exactly. Pooling the two sources of information allows the algorithm to learn the mean retention factor, compare the EI and BI policy classes, and determine whether its cost-calibrated rule calls for another exact observation.

After one initialization audit, the procedure repeats three steps at each stockout boundary: update the retention and policy-arm confidence sets; audit if the retention-confidence radius exceeds its cost-calibrated threshold and otherwise run the most optimistic active arm; and, at the next stockout, append the new retention observation and any decision-episode performance record. We first define the common episode structure and then describe each learning and decision component.

4.1. Episode structure and candidate policies

Fix a finite grid of candidate base-stock levels,

$$\mathcal{S}_M := \{s_1, \dots, s_M\} \subset (0, \bar{S}].$$

Pairing each target with both tracking modes gives the $2M$ policy arms

$$\mathcal{P}_M := \mathcal{S}_M \times \mathcal{C}_{\mathrm{tr}}.$$

Note that the regret benchmark remains the better policy over the continuous EI and BI classes in (6).

Calendar periods are indexed by t, and episodes by $m = 1, 2, \dots$. Let t_m denote the first period of episode m. Set $t_1 := 1$, treating the initial zero-inventory state as the first regenerative boundary, and define

$$t_{m+1} := 1 + \inf\{t \geq t_m : Z_t = 1\}.$$

Thus episode m occupies

$$\mathcal{C}_m := \{t_m, \dots, t_{m+1} - 1\}, \qquad \Lambda_m := |\mathcal{C}_m| = t_{m+1} - t_m.$$

At a decision boundary, $m \geq 0$ denotes the number of completed episodes, so the next episode is $m + 1$, begins at t_{m+1}, and the pseudocode state satisfies $t = t_{m+1}$. For every episode $m \geq 1$, let $\mathcal{G}_{m-1}$ contain the

complete history through episode $m-1$, together with the action selected for episode m, but before that episode's demand and retention realizations.

Two statistical bookkeeping devices will also be useful: the truncation grid

$$\mathcal{A}_T := \left\{0, \frac{1}{T}, \frac{2}{T}, \ldots, 1\right\}$$

for the Kaplan–Meier estimator and the episode-wise confidence budget

$$\delta_m := \frac{1}{64T^4(m+1)^2}, \qquad m \geq 0.$$

Episode m begins in period t_m, immediately after a stockout, so physical inventory and the internal tracker are both zero, i.e., $X_{t_m} = 0, \hat{X}_{t_m} = 0$. The algorithm then selects an episode type

$$E_m \in \{\text{aud}\} \cup \big(\{1, \ldots, M\} \times \mathcal{C}_{\text{tr}}\big).$$

Since the tracker is zero, the episode-start order equals the selected target:

$$\xi_m := \begin{cases} s_{\text{aud}}, & E_m = \text{aud}, \\ s_i, & E_m = (i,c) \in \{1, \ldots, M\} \times \mathcal{C}_{\text{tr}}. \end{cases}$$

If $E_m = (i,c)$, the algorithm runs arm (i,c) until the next stockout. Its tracker coefficient is

$$\nu_{c,m-1} := \begin{cases} \hat{\mu}_{m-1}, & c = \text{EI}, \\ 1, & c = \text{BI}, \end{cases}. \tag{15}$$

A completed decision episode contributes a performance record for arm (i,c) and an episode-start retention observation.

If $E_m = \text{aud}$, the episode follows the audit and exact-reconciliation logic in Section 3.1. The count reveals $\rho_{t_m} = (Y_{t_m} + X_{t_m+1})/s_{\text{aud}}$ exactly. If the audited period does not stock out, the tracker is reconciled to the counted inventory and the bounded recovery policy $(S^{\text{rec}}, \nu^{\text{rec}})$ is followed until the next stockout. The recovery tail remains part of the audit episode and is excluded from all policy-arm performance records. Thus, decision and audit episodes share the same regenerative endpoint.

The two episode types admit a common censored-data representation. For analysis only, define the latent episode-start censoring threshold

$$C_m^\dagger := \begin{cases} +\infty, & E_m = \text{aud}, \\ D_{t_m}/\xi_m, & E_m = (i,c), \end{cases}. \tag{16}$$

The algorithm does not observe $C_m^\dagger$ but it records the pair

$$U_m := \begin{cases} \dfrac{Y_{t_m} + X_{t_m+1}}{s_{\text{aud}}}, & E_m = \text{aud}, \\ \dfrac{Y_{t_m}}{\xi_m}, & E_m = (i,c), \end{cases} \qquad \Delta_m := \begin{cases} 1, & E_m = \text{aud}, \\ Z_{t_m}, & E_m = (i,c). \end{cases} \tag{17}$$

Since $X_{t_m} = 0$, both cases satisfy

$$U_m = \min\{\rho_{t_m}, C_m^\dagger\}, \qquad \Delta_m = \mathbf{1}\{\rho_{t_m} < C_m^\dagger\}, \tag{18}$$

up to a zero-probability censoring tie. Thus an audit is uncensored, while an ordinary episode is exact when $\Delta_m = 1$ and otherwise gives only $U_m = C_m^\dagger \le \rho_{t_m}$. After m completed episodes, the pooled retention sample is

$$\mathscr{D}_\rho(m) := \{(U_\ell, \Delta_\ell) : 1 \le \ell \le m\}.$$

The episode counter advances only at the next stockout.

4.2. Retention learning from pooled censored observations

No single source of retention information is uniformly informative across sample paths. KM–Audit therefore constructs three valid confidence intervals for μ_ρ and intersects them, allowing the realized data to determine which source of information is most useful. The bracketing interval uses the pathwise bounds implied by every observation, the audit interval provides an exact-observation anchor, and the Kaplan–Meier interval extracts the additional information contained in censored observations. Their intersection preserves simultaneous coverage while inheriting the sharpest available restriction. In particular, the bracketing and audit intervals safeguard coverage when ordinary operation reveals little of the upper tail, whereas the Kaplan–Meier interval can be substantially tighter when the realized risk sets are informative.

Bracketing interval. For each observation, set $L_\ell := U_\ell$ and set $H_\ell := U_\ell$ when $\Delta_\ell = 1$ and $H_\ell := 1$ otherwise. Then $L_\ell \le \rho_{t_\ell} \le H_\ell$ almost surely. Writing $\bar{L}_m$ and $\bar{H}_m$ for their sample means, the bracketing interval is

$$\mathcal{C}_{\mathrm{br}}(m) := \big[\bar{L}_m - r_{\mathrm{br}}(m), \bar{H}_m + r_{\mathrm{br}}(m)\big] \cap [\underline{\mu}, \bar{\mu}], \qquad r_{\mathrm{br}}(m) := \sqrt{\frac{\log(6/\delta_m)}{2m}}. \tag{19}$$

This interval deliberately ignores the censoring distribution and uses only the pathwise bracket on each latent draw.

Audit interval. Let $\mathcal{P}_{\mathrm{aud}}(m)$ be the set of completed audit episodes, $N_{\mathrm{aud}}(m)$ its cardinality, and, when this set is nonempty, define

$$\widehat{\mu}_{\mathrm{aud}}(m) := \frac{1}{N_{\mathrm{aud}}(m)} \sum_{\ell \in \mathcal{P}_{\mathrm{aud}}(m)} U_\ell, \qquad \widetilde{\mu}_{\mathrm{aud}}(m) := \Pi_{[\underline{\mu}, \bar{\mu}]}\big(\widehat{\mu}_{\mathrm{aud}}(m)\big).$$

For $N_{\mathrm{aud}}(m) > 0$, set

$$\mathcal{C}_{\mathrm{aud}}(m) := [\widetilde{\mu}_{\mathrm{aud}}(m) - r_{\mathrm{aud}}(m), \widetilde{\mu}_{\mathrm{aud}}(m) + r_{\mathrm{aud}}(m)] \cap [\underline{\mu}, \bar{\mu}], \qquad r_{\mathrm{aud}}(m) := \sqrt{\frac{\log(6/\delta_m)}{2N_{\mathrm{aud}}(m)}}. \tag{20}$$

If $N_{\mathrm{aud}}(m) = 0$, set $\mathcal{C}_{\mathrm{aud}}(m) := [\underline{\mu}, \bar{\mu}]$. Because the audit decision is made before the current retention draw is observed, this interval remains valid under adaptive audit times (Lemma 4).

Kaplan–Meier interval. For a decision episode, the censoring threshold

$$C_\ell^\dagger = \frac{D_{t_\ell}}{\xi_\ell}$$

depends on the predictable order ξ_ℓ. Conditional on the episode-start history, the threshold is independent of the fresh retention draw; for an audit, it is infinite. Hence censoring may vary across episodes without invalidating the pooled product-limit estimator. The grouped construction below also permits point masses in the retention distribution. Lemma 7 in the Electronic Companion establishes the required concentration under this adaptive sampling scheme.

Let $0 \le r_{1,m} < \cdots < r_{J_m,m} \le 1$ be the distinct uncensored values. For each such value, let $d_{j,m}$ be its event count and let $n_{j,m}$ be the number of observations at risk; their explicit grouped definitions are given in Section A of the Electronic Companion. The Kaplan–Meier estimator of the survival function

$$\bar{F}_\rho(x) := \mathbb{P}(\rho_t > x)$$

is

$$\widehat{\bar{F}}_{\rho,m}(x) := \prod_{j: r_{j,m} \le x} \left(1 - \frac{d_{j,m}}{n_{j,m}}\right), \qquad x \in [0,1], \tag{21}$$

with the convention that an empty product equals one. For a truncation level $a \in [0,1]$, define the restricted-mean estimator

$$\widehat{\mu}_{\mathrm{KM}}(m;a) := \int_0^a \widehat{\bar{F}}_{\rho,m}(x)\, dx. \tag{22}$$

For $a \in \mathcal{A}_T$, the finite-sample radius $e_{\mathrm{KM}}(m;a)$ is an explicit function of the empirical risk set, the grouped Nelson–Aalen cumulative hazard, and δ_m; the full formula is given in (72) of the Electronic Companion. It equals a when the risk set is empty and becomes sharper with greater effective risk-set coverage. The truncated Kaplan–Meier confidence interval is

$$\mathcal{C}_{\mathrm{KM}}(m;a) := [\widehat{\mu}_{\mathrm{KM}}(m;a) - e_{\mathrm{KM}}(m;a), \widehat{\mu}_{\mathrm{KM}}(m;a) + e_{\mathrm{KM}}(m;a) + (1-a)] \cap [\underline{\mu}, \bar{\mu}]. \tag{23}$$

The term $1-a$ bounds the omitted tail. To make the selection well defined on every sample path, let

$$\mathcal{A}_T^{\mathrm{KM}}(m) := \{a \in \mathcal{A}_T : \mathcal{C}_{\mathrm{KM}}(m;a) \neq \varnothing\}.$$

This set is nonempty because $a = 0$ yields the prior interval $[\underline{\mu}, \bar{\mu}]$. The algorithm chooses the nonempty candidate with minimum width:

$$a_m^{\mathrm{KM}} \in \operatorname*{arg\,min}_{a \in \mathcal{A}_T^{\mathrm{KM}}(m)} \operatorname{diam}\big(\mathcal{C}_{\mathrm{KM}}(m;a)\big), \qquad \mathcal{C}_{\mathrm{KM}}(m) := \mathcal{C}_{\mathrm{KM}}(m; a_m^{\mathrm{KM}}). \tag{24}$$

Define the raw intersection

$$\mathcal{C}_\mu^\cap(m) := \mathcal{C}_{\mathrm{KM}}(m) \cap \mathcal{C}_{\mathrm{br}}(m) \cap \mathcal{C}_{\mathrm{aud}}(m).$$

To make the algorithm well defined on every sample path, set

$$\mathcal{C}_\mu(m) := \begin{cases} \mathcal{C}_\mu^{\cap}(m), & \mathcal{C}_\mu^{\cap}(m) \neq \emptyset, \\ \mathcal{C}_{\mathrm{aud}}(m), & \mathcal{C}_\mu^{\cap}(m) = \emptyset. \end{cases} \tag{25}$$

On the simultaneous confidence event, the intersection is nonempty and contains μ_ρ. The audit-only fallback makes the construction well defined on every sample path. One can summarize the interval by its midpoint and radius,

$$\hat{\mu}_m := \mathrm{mid}\big(\mathcal{C}_\mu(m)\big), \qquad r_\mu(m) := \frac{1}{2}\,\mathrm{diam}\big(\mathcal{C}_\mu(m)\big). \tag{26}$$

After m completed episodes, the next EI decision episode fixes its tracker parameter at $\hat{\mu}_m$ and a BI episode fixes it at one. Equivalently, episode ℓ uses the execution parameter indexed by $\ell - 1$ in (15).

4.3. Learning and eliminating policy arms

Confidence-based elimination is familiar in online inventory learning. However, the main difficulty in inaccurate inventory lies in constructing valid and comparable arm-level confidence bounds. Each arm is a base-stock–tracker pair, and arm-performance feedback arrives only at the end of a random-length regenerative episode. Actual cycle cost is hidden, so performance must be recovered from a plug-in pseudo-cost built from observable cycle statistics and the learned mean retention. The same unknown parameter is also essential for the implementation of EI. An EI episode executed under an earlier estimate therefore does not exactly implement the population arm being evaluated. Moreover, audit episodes sharpen the shared retention estimate but contribute no arm-performance record. Valid elimination must consequently control regenerative sampling error, current plug-in evaluation error, and historical implementation error simultaneously.

For $(i, c) \in \{1, \ldots, M\} \times \mathcal{C}_{\mathrm{tr}}$, define

$$\mathcal{I}_{i,c}(m) := \{\ell \leq m : E_\ell = (i, c) \text{ and episode } \ell \text{ is complete}\}, \qquad N_{i,c}(m) := |\mathcal{I}_{i,c}(m)|.$$

Audit episodes, including their recovery tails, are excluded from all arm logs. For each retained episode, the algorithm stores the observable statistics $(\Lambda_\ell, Q_\ell, V_\ell, W_\ell)$ defined in Section 3.3.

For a completed decision episode ℓ, define its realized pseudo-cost evaluated at coefficient μ by

$$\widetilde{K}_\ell^{\mathrm{obs}}(\mu) := \alpha(\mu) W_\ell - (h + b) V_\ell. \tag{27}$$

If $\ell \in \mathcal{I}_{i,c}(m)$, then this realized cycle was generated at target s_i with execution parameter

$$\nu_{c,\ell-1} := \begin{cases} \hat{\mu}_{\ell-1}, & c = \mathrm{EI}, \\ 1, & c = \mathrm{BI}. \end{cases}$$

Although the episode was generated using $\nu_{c,\ell-1}$, the arm's population target is $\gamma(s_i; \nu_c)$, where $\nu_{\mathrm{EI}} = \mu_\rho$ and $\nu_{\mathrm{BI}} = 1$. The implementation component of the confidence radius below accounts for this discrepancy.

The empirical pseudo-cost rate of policy arm (i,c) is

$$\hat{\gamma}_{i,c}(m) := \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \widetilde{K}_\ell^{\mathrm{obs}}(\hat{\mu}_m)}{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell}, \qquad N_{i,c}(m) > 0. \tag{28}$$

Since the population objective is a regenerative reward rate, the estimator weights episodes by their duration rather than averaging episode-level costs.

We attach a confidence radius

$$B_{i,c}(m) := B_{i,c}^{\mathrm{stat}}(m) + B_{i,c}^{\mu,\mathrm{eval}}(m) + B_{i,c}^{\mu,\mathrm{impl}}(m). \tag{29}$$

The three terms correspond to three different errors: sampling error in the cycle pseudo-costs, plug-in error from evaluating pseudo-costs at $\hat{\mu}_m$, and implementation error from having executed past cycles with earlier tracker estimates.

Sampling term. Define

$$L_{2M,T} := \log(128MT^5), \qquad \bar{\Lambda}_T := \left\lceil p_0^{-1} \log(8T^5) \right\rceil,$$
$$\bar{c}_{\mathrm{per}} := \alpha(\bar{\mu})\bar{S} + \big(\alpha(\bar{\mu}) + h + b\big)\overline{D}, \qquad c_{\mathrm{stat},T} := \frac{8\bar{c}_{\mathrm{per}}}{p_0}\left(1 + \sqrt{L_{2M,T}}\right).$$

The regenerative sampling radius is

$$B_{i,c}^{\mathrm{stat}}(m) := c_{\mathrm{stat},T}\sqrt{\frac{L_{2M,T}}{N_{i,c}(m)}}, \qquad N_{i,c}(m) > 0. \tag{30}$$

This term controls sampling fluctuations uniformly over arms and episode counts.

Current evaluation error. On $[\underline{\mu}, \bar{\mu}]$, the coefficient $\alpha(\mu) = h\mu/(1-\mu)$ is Lipschitz with constant

$$L_\alpha := \frac{h}{(1-\bar{\mu})^2}. \tag{31}$$

Since $W_\ell \le \bar{S}\Lambda_\ell$, the resulting plug-in error is bounded by

$$B_{i,c}^{\mu,\mathrm{eval}}(m) := L_\alpha \bar{S}\, r_\mu(m). \tag{32}$$

Past implementation error. Past EI episodes were executed using the retention estimate available at their start, rather than the unknown μ_ρ. Lemma 10 bounds the cost effect of this discrepancy by the primitive constant L_{impl}. Accordingly,

$$B_{i,c}^{\mu,\mathrm{impl}}(m) := \mathbf{1}\{c = \mathrm{EI}\} L_{\mathrm{impl}} \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell r_\mu(\ell - 1)}{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell}, \qquad N_{i,c}(m) > 0. \tag{33}$$

The indicator makes this term zero for BI, whose tracker coefficient is always one.

For an unsampled active arm, set

$$\mathrm{LCB}_{i,c}(m) := -\infty, \qquad \mathrm{UCB}_{i,c}(m) := +\infty,$$

so that every active arm is sampled at least once before it can be eliminated. For $N_{i,c}(m) > 0$, define

$$\mathrm{LCB}_{i,c}(m) := \hat{\gamma}_{i,c}(m) - B_{i,c}(m), \qquad \mathrm{UCB}_{i,c}(m) := \hat{\gamma}_{i,c}(m) + B_{i,c}(m). \tag{34}$$

At each stockout boundary, the algorithm eliminates any arm whose lower bound exceeds the smallest upper bound among active arms. It samples every active arm once and thereafter runs an active arm with the smallest lower bound.

4.4. Adaptive audit rule

An ordinary episode reveals retention exactly when it ends in a stockout immediately and otherwise yields only a right-censored observation, whereas a physical count guarantees an exact observation but incurs a cost. Two simple rules bracket the design space. Auditing at every stockout boundary makes the confidence interval for μ_ρ shrink at the familiar rate $m^{-1/2}$ after m episodes, but the number of audits, and hence their cost, grows linearly in m. Never auditing avoids that cost, but leaves EI and the pseudo-cost comparison to rely on censored observations alone, which may reveal little about the upper tail of the retention distribution. KM-Audit lies between these extremes. It begins with one initialization audit, which anchors the first tracker update with an exact retention observation. Its subsequent audit decisions are easiest to understand in two stages: the first sets the pace at which audits become due, and the second decides whether a due audit is actually executed.

Stage 1: a widening audit schedule. Think of the radius of the confidence interval for μ_ρ as a margin of error, as in an opinion poll. A target margin b has two costs. Reaching it from exact observations requires roughly L_T/b^2 audits, where L_T is the logarithmic confidence term defined below, so halving the margin quadruples the number of counts. Leaving a margin b unresolved costs a little in every decision episode, because both the pseudo-cost evaluation and the EI record rely on the estimate of μ_ρ. Over m episodes, this mismatch cost grows roughly like mb. The first cost falls and the second rises as b grows, and balancing them gives the cube-root target $b \propto (L_T/m)^{1/3}$. KM-Audit therefore tolerates a margin of error that shrinks as more episodes are completed, but only at the cube-root rate.

If counts were the only source of information about retention, this target would translate into an audit schedule whose gaps widen over time. Figure 2 shows a stylized version. Audit 1 is conducted immediately; the next two audits are spaced two episode boundaries apart, the next three are spaced three apart, the next four are spaced four apart, and so on. Over 30 episode boundaries, this schedule conducts 10 audits rather than 30, and the ratio keeps improving as the horizon grows. After m boundaries, the schedule has conducted on the order of $m^{2/3}$ audits, and its confidence interval has width on the order of $m^{-1/3}$. Auditing at every boundary would instead conduct m audits to reach width $m^{-1/2}$. The slower contraction is deliberate. It is just fast enough to keep the cumulative mismatch cost of the same order as the cumulative audit cost.

We now make the calibration of Stage 1 precise.

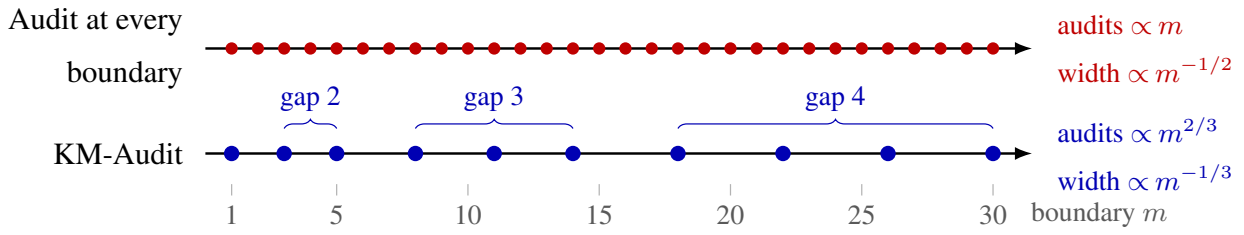


Figure 2 **Stylized audit schedule implied by the cube-root target when audits are the only source of retention information. Top: auditing at every stockout boundary. Bottom: KM-Audit, in which audit 1 is immediate and the next k audits are spaced k boundaries apart for $k = 2, 3, 4, \ldots$. In the example above, over 30 boundaries, KM-Audit conducts 10 audits rather than 30. In the actual policy, informative observations generated during ordinary operation can eliminate audits that would otherwise be required under this audit-only schedule (Stage 2).**

Recall from (2) that $C_t^{\mathrm{alg}} = C_t + \kappa_{\mathrm{aud}} J_t^{\mathrm{aud}}$ is the realized, though not necessarily observed, period cost. It includes the direct count fee in the audited period. An audit may also induce ordinary holding and lost-sales costs during that period and its recovery tail. Under the bounded-order and positive-stockout-probability assumptions, define

$$\bar{C}_{\mathrm{aud}} := \kappa_{\mathrm{aud}} + \frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0}. \tag{35}$$

If audit episode ℓ contains the period set $\mathcal{C}_\ell$ and begins with history $\mathcal{G}_{\ell-1}$, then

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_\ell} C_t^{\mathrm{alg}} \,\middle|\, \mathcal{G}_{\ell-1}\right] \le \bar{C}_{\mathrm{aud}}.$$

Thus $\bar{C}_{\mathrm{aud}}$ converts the number of initiated audits into an upper bound on their expected contribution to regret. It is therefore the coefficient attached to audit count in Theorem 1 and also supplies the audit-cost side of the threshold below. The quantity is neither an additional economic cost nor a pathwise bound on realized cost.

We use the following horizon-T coefficient to bound the regret contribution of residual uncertainty about μ_ρ:

$$C_{\mathrm{sh,T}} := \frac{2L_\alpha \bar{S} + L_{\mathrm{impl}}}{p_0} + \frac{2L_{\mathrm{impl}}\bar{\Lambda}_T \log(eT)}{p_0}. \tag{36}$$

The coefficient $C_{\mathrm{sh,T}}$ converts uncertainty about mean retention into a uniform performance-loss bound, including both pseudo-cost evaluation error and the effect of implementing EI with earlier retention estimates.

Let A_T^{init} denote the number of audit episodes initiated by the horizon and $\mathcal{U}_T^\mu$ the cumulative retention-confidence radius at decision-episode starts, both are defined formally in Section 5. The regret analysis isolates their trade-off as

$$\bar{C}_{\mathrm{aud}} A_T^{\mathrm{init}} + C_{\mathrm{sh,T}} \mathcal{U}_T^\mu.$$

The two terms capture the central information-acquisition trade-off. Audits incur an immediate episode cost, whereas deferring an audit leaves residual retention uncertainty that can impair both policy evaluation and EI implementation. Let

$$L_T := \log(384T^6),$$

which uniformly upper-bounds the logarithmic confidence terms $\log(6/\delta_m)$ over all episode boundaries up to horizon T. The cube-root balance described in Stage 1 therefore yields a threshold of the form

$$a_m = \min\left\{1, \vartheta(L_T/m)^{1/3}\right\}, \qquad \vartheta > 0.$$

Ordinary censored observations may reduce the realized radius without an audit. The calculation determines only the tolerance against which that radius is compared. It remains to choose the scale ϑ from the two economic cost bounds. Section 5 shows that, within this threshold family, choosing ϑ reduces to minimizing the strictly convex regret-bound coefficient

$$\frac{\bar{C}_{\text{aud}}}{2\vartheta^2} + \frac{3}{2}C_{\text{sh,T}}\vartheta.$$

The first term decreases with ϑ: a higher threshold triggers fewer audit episodes. The second increases with ϑ: a higher threshold permits more residual retention uncertainty during decision episodes. Its unique minimizer is

$$\vartheta_T := \left(\frac{2\bar{C}_{\text{aud}}}{3C_{\text{sh,T}}}\right)^{1/3}. \tag{37}$$

The audit threshold is therefore

$$a_m := \min\left\{1, \vartheta_T\left(\frac{L_T}{m}\right)^{1/3}\right\}, \qquad m \geq 1. \tag{38}$$

Thus, more costly audit episodes raise the threshold and make an audit harder to trigger, whereas greater sensitivity to retention error lowers the threshold and favors earlier information acquisition. The schedule is fixed from the horizon and known primitive bounds. The realized data affect the decision through the confidence radius $r_\mu(m)$ below.

Stage 2: execute an audit only when it is needed. The threshold a_m sets only the pace at which audits become due. Whether an audit is actually conducted depends on the realized confidence radius $r_\mu(m)$, which pools every source of retention information in Section 4.2, including the censored observations generated by routine operation. When those observations are informative, $r_\mu(m)$ can fall below the threshold without a count, and the audit that Stage 1 would have called for is skipped. Formally, at the beginning of episode $m+1$, after computing the combined mean-retention confidence interval, the algorithm applies the rule

$$\text{audit if and only if} \qquad r_\mu(m) > a_{m+1}. \tag{39}$$

Otherwise it starts a decision episode.

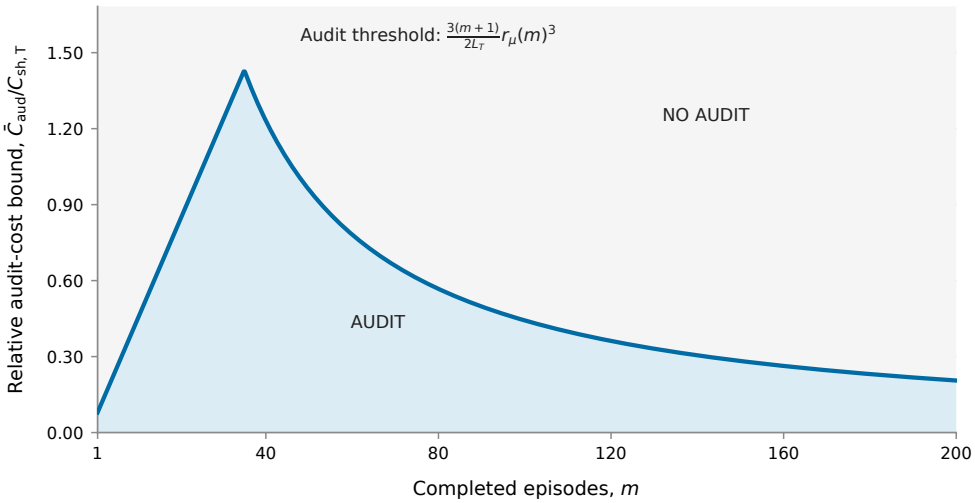


Figure 3 Audit decision region implied by (40). KM–Audit initiates an audit when the relative audit-cost bound $\bar{C}_{\mathrm{aud}}/C_{\mathrm{sh,T}}$ lies below the realized boundary and otherwise starts a decision episode.

Since $r_\mu(m) \le 1$, the same rule can be written as the direct comparison

$$\frac{3C_{\mathrm{sh,T}}(m+1)}{2L_T}\, r_\mu(m)^3 > \bar{C}_{\mathrm{aud}}. \tag{40}$$

Figure 3 displays this comparison after dividing both sides by $C_{\mathrm{sh,T}}$. The rule is more likely to trigger an audit when the bounded audit cost is small, residual uncertainty is large, or operating performance is more sensitive to retention uncertainty.

The rule is path dependent rather than calendar based. If routine operation does not narrow the confidence set, the declining threshold eventually triggers another audit. When pooled Kaplan–Meier information contracts the interval, the cubic dependence on $r_\mu(m)$ shrinks the audit region. Thus ordinary operation can postpone or eliminate a physical count, while the audit-only interval anchors inference when upper-tail information is weak. Consequently, two SKUs with the same mean retention and count fee need not receive the same audit schedule. A SKU whose ordinary episodes reveal more about retention can be counted less often. Audit frequency is therefore driven by the decision value of unresolved uncertainty, rather than by shrinkage or a fixed calendar alone. In short, Stage 1 widens the gaps between audits as the horizon grows, whereas Stage 2 uses the data to decide whether each due audit is worth executing.

4.5. Algorithmic summary

The pseudocode of KM-Audit is given in Algorithm 1. The AUDIT and RUN subroutines return a completion flag χ. When $\chi = 1$, t^+ is the first period after the episode's regenerative endpoint, whereas $\chi = 0$ means that the horizon is reached before completion. Both return (U, Δ) upon completion, and RUN also returns (Λ, Q, V, W). All model primitives and confidence-bound constants defined above are treated as

Algorithm 1: KM-Audit

Input: $\mathcal{S}_M,\ s_{\text{aud}},\ (S^{\text{rec}},\nu^{\text{rec}}),\ T,\ (a_m),\ (\delta_m),\ \mathcal{A}_T$

$\mathcal{A} \leftarrow \{1,\dots,M\}\times\mathcal{C}_{\text{tr}},\ \mathscr{D}_\rho \leftarrow \emptyset,\ \mathcal{I}_{i,c} \leftarrow \emptyset\ \forall (i,c)\in\mathcal{A},\ (t,m)\leftarrow(1,0)$

$E_1 \leftarrow \text{aud},\ (\chi,t^+,U,\Delta) \leftarrow \texttt{Audit}(t;s_{\text{aud}},S^{\text{rec}},\nu^{\text{rec}})$

if $\chi = 0$ **then**
  terminate

end

$\mathscr{D}_\rho \leftarrow \mathscr{D}_\rho \cup \{(U,\Delta)\},\ (t,m)\leftarrow(t^+,1)$

while $t \le T$ **do**

  Compute $\mathcal{C}_\mu(m)$, $\hat{\mu}_m$, and $r_\mu(m)$

  Compute $(\hat{\gamma}_{i,c},B_{i,c})$ for sampled $(i,c)\in\mathcal{A}$, and compute $(\text{LCB}_{i,c},\text{UCB}_{i,c})$ for all $(i,c)\in\mathcal{A}$ using (34)

$$\mathcal{A} \leftarrow \left\{(i,c)\in\mathcal{A} : \text{LCB}_{i,c}(m) \le \min_{(j,d)\in\mathcal{A}} \text{UCB}_{j,d}(m)\right\};$$

  if $r_\mu(m) > a_{m+1}$ **then**
    $E_{m+1}\leftarrow \text{aud},\ (\chi,t^+,U,\Delta)\leftarrow\texttt{Audit}(t;s_{\text{aud}},S^{\text{rec}},\nu^{\text{rec}})$

  else
    Choose unsampled $(i,c)\in\mathcal{A}$, if one exists; otherwise $(i,c)\in\arg\min_{(j,d)\in\mathcal{A}}\text{LCB}_{j,d}(m)$

    $\nu\leftarrow\hat{\mu}_m$ if $c=\text{EI}$, and $\nu\leftarrow 1$ if $c=\text{BI}$

    $E_{m+1}\leftarrow(i,c)$,

$$(\chi,t^+,U,\Delta,\Lambda,Q,V,W)\leftarrow\texttt{Run}(t,i,c,\nu);$$

  end

  if $\chi = 0$ **then**
    break

  end

  $\mathscr{D}_\rho \leftarrow \mathscr{D}_\rho\cup\{(U,\Delta)\}$

  if $E_{m+1}\neq\text{aud}$ **then**
    Store (Λ,Q,V,W) and $\mathcal{I}_{i,c}\leftarrow\mathcal{I}_{i,c}\cup\{m+1\}$

  end

  $(t,m)\leftarrow(t^+,m+1)$

end

global inputs to the pseudocode. The primitive bounds required by the algorithm can be calibrated conservatively using information available to a small retailer. $\overline{D}$ and $\bar{S}$ can be set from historical maximum demand and base-stock levels, a conservative lower bound from the empirical stockout frequency can be used for p_0, while $[\underline{\mu},\bar{\mu}]$ can be informed by category-level loss benchmarks such as the Food and Agriculture Organization (FAO) of the United Nations' *food Loss and Waste Database* used in Section 6.3.

5. Regret Analysis

This section quantifies how much the inventory planner loses, relative to the benchmark of Section 3.2, by learning shrinkage from audits and routine sales rather than knowing it in advance. The regret analysis separates three sources of error. First, within each inventory-tracking mode the algorithm searches over the finite grid $\mathcal{S}_M$, while the comparator optimizes the target continuously. Second, the $2M$ policy arms must

be learned from regenerative pseudo-cost observations. Third, uncertainty about μ_ρ affects the evaluation of both modes and the implementation of EI cycles.

5.1. Main upper bound

For the candidate grid $\mathcal{S}_M$ defined in Section 4.1, let

$$(i_M^\star, c_M^\star) \in \operatorname*{arg\,min}_{\substack{i\in\{1,\dots,M\}\\ c\in\mathcal{C}_{\mathrm{tr}}}} g_c(s_i), \qquad g_{\cup,M}^\star := g_{c_M^\star}(s_{i_M^\star}).$$

The corresponding approximation error relative to the continuous EI-and-BI benchmark is

$$\varepsilon_M := g_{\cup,M}^\star - g_\cup^\star. \tag{41}$$

To state the finite-horizon bound, let $\mathcal{D}_T^{\mathrm{init}}$ and A_T^{init} denote, respectively, the decision episodes and the number of audit episodes initiated by time T, including the episode active at the horizon. For analytical purposes, the terminal episode is conceptually continued under the action selected at its start until the next stockout. The algorithm itself stops at T and does not observe or use any post-horizon data. If (i_ℓ, c_ℓ) is the arm selected in decision episode ℓ, define

$$\mathcal{W}_T^{\mathrm{stat}} := \sum_{\substack{\ell\in\mathcal{D}_T^{\mathrm{init}}:\\ N_{i_\ell,c_\ell}(\ell-1)\ge 1}} B_{i_\ell,c_\ell}^{\mathrm{stat}}(\ell-1), \qquad \mathcal{U}_T^{\mu} := \sum_{\ell\in\mathcal{D}_T^{\mathrm{init}}} r_\mu(\ell-1),$$

and let

$$C_{\mathrm{bd}} := \frac{1}{p_0}\left(\frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0}\right)$$

bound the terminal-cycle continuation.

Theorem 1 *Under Assumptions 1– 4, the regret of the KM-Audit algorithm satisfies*

$$\begin{aligned} R_T \le C_{\mathrm{bd}} + \bar{C}_{\mathrm{aud}}\mathbb{E}[A_T^{\mathrm{init}}] + \frac{2\bar{c}_{\mathrm{per}}}{p_0}(2M) + \frac{2}{p_0}\mathbb{E}[\mathcal{W}_T^{\mathrm{stat}}] \\ + C_{\mathrm{sh,T}}\mathbb{E}[\mathcal{U}_T^{\mu}] + T\varepsilon_M + O(T^{-3}). \end{aligned} \tag{42}$$

The bound separates target approximation and policy-arm learning from retention learning. The terms $2M + \mathbb{E}[\mathcal{W}_T^{\mathrm{stat}}] + T\varepsilon_M$ account for initializing and comparing the policy arms and approximating the continuous target. The remaining two terms respectively bound the cost of initiated audit episodes and the effect of retention uncertainty on pseudo-cost evaluation and EI implementation. Since the audit count and cumulative retention radius are retained along the realized sample path, the bound adapts to information generated during ordinary operation.

Explicit rate on a uniform target grid. For the uniform grid $\mathcal{S}_M = \{j\bar{S}/M : j = 1, \dots, M\}$, Lemma 12 gives $\varepsilon_M \le L_S \bar{S}/M$ for a primitive-dependent constant L_S. Under the cost-calibrated threshold schedule (38), the audit count, cumulative retention radius, and selected-arm statistical widths satisfy, pathwise,

$$A_T^{\mathrm{init}} \le 1 + \frac{L_T^{1/3} T^{2/3}}{2\vartheta_T^2}, \qquad \mathcal{U}_T^{\mu} \le \frac{3}{2}\vartheta_T L_T^{1/3} T^{2/3}, \qquad \mathcal{W}_T^{\mathrm{stat}} \le 2c_{\mathrm{stat},T}\sqrt{2MTL_{2M,T}}.$$

The first two bounds are the worst-case counterparts of the widening schedule in Figure 2. When routine observations add no information about retention, the threshold permits on the order of $T^{2/3}$ audits and leaves a confidence radius of order $T^{-1/3}$ in each decision episode, so both the audit cost and the cumulative retention uncertainty grow like $T^{2/3}$. The choice of ϑ_T in (37) therefore bounds the combined audit and retention-uncertainty contribution by

$$\bar{C}_{\mathrm{aud}} + \left(\frac{3}{2}\right)^{5/3} \bar{C}_{\mathrm{aud}}^{1/3} C_{\mathrm{sh,T}}^{2/3} L_T^{1/3} T^{2/3}.$$

Substitution in Theorem 1 gives

$$R_T = \widetilde{O}\left(M + \sqrt{MT} + \frac{T}{M} + T^{2/3}\right).$$

Choosing $M_T = \lceil T^{1/3} \rceil$ yields $R_T = \widetilde{O}(T^{2/3})$. The complete constant-explicit specialization appears in the Electronic Companion.

Instance adaptation. The path-dependent decomposition can be substantially sharper than its worst-case specialization. When ordinary episode-start observations produce informative Kaplan–Meier risk sets, both the audit count and the cumulative retention radius may be much smaller than their uniform bounds. The theorem retains these realized quantities and therefore captures the information generated without physical counts. This is the role of Stage 2 of the audit rule. Whenever routine sales after stockouts shrink the confidence radius below the threshold, the corresponding audit is skipped, and the realized audit count falls below the worst-case schedule.

Economic interpretation. The decomposition distinguishes three managerial levers. The complete-episode bound $\bar{C}_{\mathrm{aud}}$ prices exact visibility, $C_{\mathrm{sh,T}}$ measures how costly it can be to act on the wrong retention estimate, and the realized confidence radius records how much uncertainty ordinary operation has already removed. High shrinkage alone therefore does not imply frequent counting. Audits become more attractive when replenishment performance is sensitive to retention error and routine observations are weak. They become less attractive when counts are expensive or when the preferred target and tracking rule are robust to the remaining uncertainty. For a retailer such as our partner, a SKU with substantial spoilage therefore need not be counted often if routine sales after stockouts already pin down its retention rate, whereas a SKU whose replenishment is sensitive to that rate may deserve more frequent counts even when spoilage is moderate.

5.2. Lower Bound

Existing inventory-learning lower bounds largely concern unknown demand (Besbes and Muharremoglu, 2013; Zhang et al., 2020). In what follows, demand is known and all uncertainty lies in the retention law, whose effect propagates through availability, future orders, and stockout timing.

Let $\Pi_{\cup}$ be the paper's nonanticipative tracker-based class with the stockout-episode structure and order bound $\bar{S} = 3$. Enlarge it to an oracle class Π_{or} that observes the current true inventory state and all past primitives, may choose any predictable randomized order in $[0, 3]$, and retains the original audit and recovery actions. Current demand and retention are revealed only after the period's order has been selected. Hence

$$\Pi_{\cup} \subseteq \Pi_{\mathrm{or}}. \tag{43}$$

The Electronic Companion constructs a fixed class $\mathfrak{M}_{\mathrm{sh}}$ satisfying all model assumptions. Its instances have the same known demand distribution and differ only in their retention laws.

Theorem 2 *For every horizon $T \geq 1$,*

$$\inf_{\pi \in \Pi_{\mathrm{or}}} \sup_{\mathsf{M} \in \mathfrak{M}_{\mathrm{sh}}} R_T^{\pi}(\mathsf{M}) \geq \frac{1}{64\sqrt{40}} \sqrt{T}. \tag{44}$$

Consequently, because $\Pi_{\cup} \subseteq \Pi_{\mathrm{or}}$,

$$\inf_{\pi \in \Pi_{\cup}} \sup_{\mathsf{M} \in \mathfrak{M}_{\mathrm{sh}}} R_T^{\pi}(\mathsf{M}) \geq \frac{1}{64\sqrt{40}} \sqrt{T}. \tag{45}$$

In both displays, regret is measured against the continuous non-auditing EI/BI benchmark $g_{\cup}^{\star}(\mathsf{M})$ defined in (6).

The proof is given in the Electronic Companion. The two nearby retention laws tilt the cost of physical pre-shrinkage inventory in opposite directions: one penalizes operating above a fixed threshold and the other penalizes operating below it. Each period supplies only $O(\Delta^2)$ information for distinguishing laws separated by Δ, whereas operating on the wrong side of the threshold incurs $\Theta(\Delta)$ expected loss. Taking Δ of order $T^{-1/2}$ yields the $\Omega(\sqrt{T})$ lower bound.

Thus, retention-law uncertainty alone imposes a nontrivial regret floor even under the oracle relaxation. Together with Theorem 1, the result places worst-case regret between $\Omega(\sqrt{T})$ and $\widetilde{O}(T^{2/3})$.

6. Numerical Analysis

This section evaluates KM–Audit using controlled synthetic instances and a chronological replay based on data from our retail collaborator. The synthetic experiments study adaptive auditing, inventory-record choice, and sensitivity to operating conditions; the retailer study examines their cost, service, and waste implications.

To compare performance across synthetic instances, we report the percentage relative regret of a policy π:

$$\mathrm{RR}_T(\pi) := 100\,\frac{\overline{C}_T(\pi) - g^\star_\cup}{g^\star_\cup}, \qquad \overline{C}_T(\pi) := \frac{1}{T}\mathbb{E}^\pi\left[\sum_{t=1}^{T}\left(C_t + \kappa_{\mathrm{aud}} J_t^{\mathrm{aud}}\right)\right],$$

where

$$g^\star_\cup := \min_{\substack{c\in\{\mathrm{EI},\mathrm{BI}\}\\ S\in[0,\bar{S}]}} g_c(S)$$

is the cost of the better separately optimized EI or BI base-stock policy. Numerically, we approximate this benchmark by optimizing both policy classes over the same base-stock grid and selecting the lower-cost solution. Thus, relative regret includes the operating consequences and direct fees of KM–Audit's physical counts, while the clairvoyant benchmark knows the problem primitives and does not audit.

Explore–then–commit (ETC). We compare KM–Audit with ETC, a nonadaptive alternative that concentrates learning at the beginning of the horizon and then commits to one base-stock–tracking pair. Its exploration duration is $H_\beta = \min\{T, \lceil T^\beta \rceil\}$, while its cost-adjusted audit budget is chosen separately before operations begin. ETC is free to optimize both EI and BI over the common base-stock grid, but it cannot revise its information-acquisition or replenishment decisions after commitment. The complete construction is given in Section D.1 of the Electronic Companion.

6.1. Experiment 1: Adaptive Auditing and Inventory-Record Choice

Our first experiment examines the value of adaptive audit timing, the choice between EI and BI, and the contribution of the three retention bounds. We compare KM–Audit with three ETC policies as the horizon grows, vary mean retention while restricting KM–Audit separately to the EI and BI families, and use a fixed-history leave-one-out diagnostic to measure how each bound changes retention uncertainty and the next audit decision.

All policies use the same base-stock grid and are evaluated using common demand and retention realizations. Total cost includes holding costs, lost-sales penalties, audit fees, and the operating costs incurred during exploration and audit episodes.

Impact of the planning horizon. We first consider mean retention levels $\mu_\rho \in \{0.85, 0.95, 0.98\}$ and horizons ranging from $T = 500$ to $T = 20{,}000$. Since replenishment frequency can differ substantially across retailers and SKUs, we measure the horizon in replenishment-review periods rather than calendar days. The range spans initial learning through mature operation and allows us to examine how the cost of learning dissipates over time, as predicted by the regret analysis. We evaluate performance over an actual calendar horizon separately in the 319-day retailer-data replay.

The comparison includes KM–Audit and ETC with $\beta \in \{0.5, 0.6, 0.7\}$, the exploration values tuned for the best performance. ETC receives the same information and may optimize both the base-stock level and the inventory-record convention, so the comparison isolates adaptive revision from one-shot commitment.

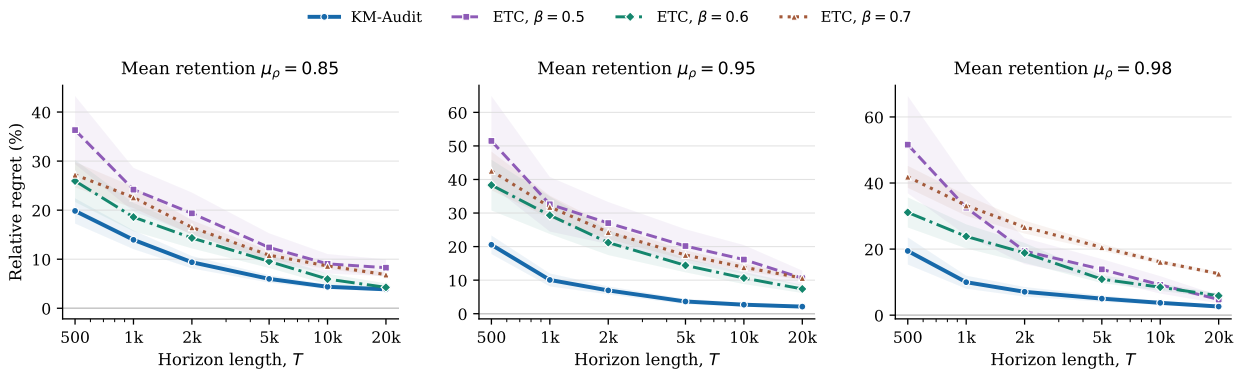


Figure 4 **Relative regret as a function of the planning horizon at three mean-retention levels. KM–Audit selects audits adaptively, whereas each ETC policy fixes its audit budget before operation and commits after an initial high-stock learning phase. Shaded regions represent pointwise 95% confidence intervals.**

Figure 4 shows that KM–Audit achieves the lowest mean relative regret in every displayed horizon–retention combination. At $T = 500$, its relative regret is approximately 20% in all three retention regimes. By $T = 20{,}000$, it has declined to 3.86%, 2.11%, and 2.62% at $\mu_\rho = 0.85$, 0.95, and 0.98, respectively. The ETC policies also improve with the horizon, but their performance reveals the difficulty of choosing a learning duration ex ante. Among the three ETC specifications, $\beta = 0.6$ generally has the lowest mean regret. Shorter exploration leaves the final decision more exposed to estimation error, whereas longer exploration purchases additional precision at the cost of more high-stock periods and delayed commitment. Nevertheless, even the best ETC specification remains behind KM–Audit throughout the figure. At $T = 20{,}000$, the best ETC regrets are 4.26%, 7.38%, and 4.75% in the three retention regimes.

The size of the advantage is itself informative. At $\mu_\rho = 0.85$, shrinkage is sufficiently pronounced that an initial block of observations identifies its operational importance relatively quickly. A well-chosen ETC policy therefore approaches KM–Audit at long horizons. At $\mu_\rho = 0.98$, shrinkage is limited and the book-inventory rule is often a competitive choice, again making the decision comparatively easier. The largest and most persistent advantage appears at the intermediate retention level $\mu_\rho = 0.95$. In this regime, shrinkage is economically consequential but not sufficiently pronounced to make the appropriate inventory-record convention obvious from a small initial sample. KM–Audit benefits from being able to revise both its information-acquisition and operating decisions as evidence accumulates.

Impact of mean retention and the choice between EI and BI. We next fix $T = 10{,}000$ and vary mean retention over the range shown in Figure 5. To isolate the role of the inventory record, we consider two restricted versions of KM–Audit. *KM–Audit, EI only* optimizes the base-stock level while propagating the inventory record using the learned mean retention. *KM–Audit, BI only* independently optimizes the base-stock level while fixing the tracker coefficient at one. The two policies retain the same learning environment

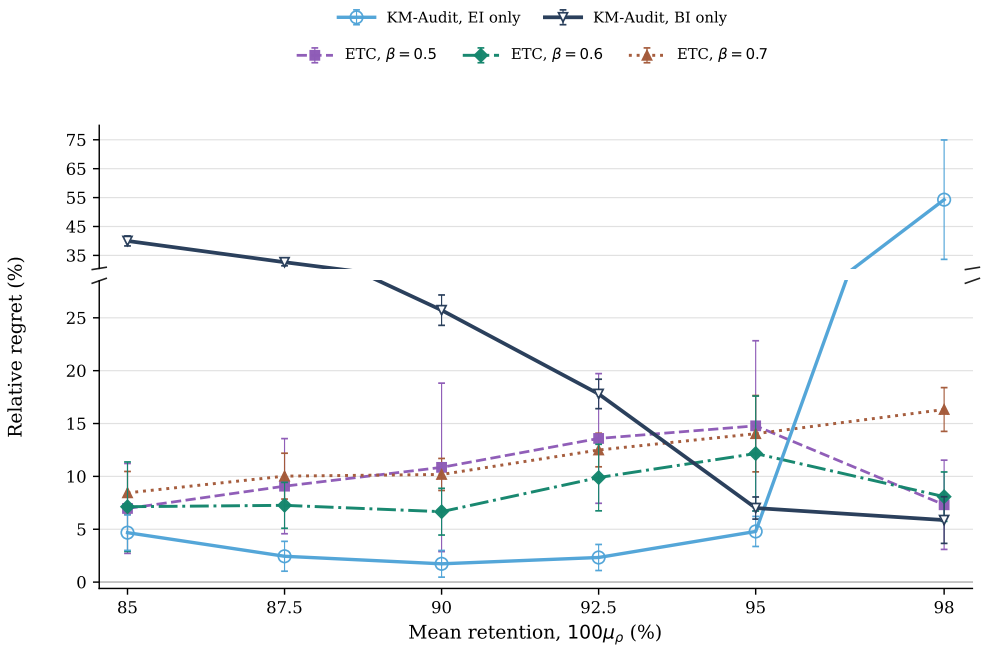


Figure 5 **Relative regret as a function of mean retention at $T = 10{,}000$. The EI-only and BI-only policies optimize their base-stock levels separately over the common grid. The ETC policies are free to select between the optimized EI and BI alternatives before commitment. Error bars represent pointwise 95% confidence intervals.**

and audit logic but only the admissible inventory-record convention differs. Optimizing the base-stock level separately is important because it ensures that the comparison reflects the choice between EI and BI rather than an arbitrary stock-level disadvantage imposed on either family.

The results reveal a sharp transition in the preferred inventory-record convention. For mean retention between 0.85 and 0.95, EI has the lowest mean regret among the displayed policies. At $\mu_\rho = 0.85$, its relative regret is 4.68%, compared with 40.00% for BI. When shrinkage is substantial, the unadjusted book record systematically overstates the inventory available for sale. The resulting underreplenishment produces a large service penalty, and explicitly discounting the inventory record using the learned retention rate is therefore valuable.

The advantage of EI narrows as retention rises. At $\mu_\rho = 0.95$, the mean regrets of EI and BI are 4.79% and 7.01%, respectively. At $\mu_\rho = 0.98$, however, the ranking reverses: BI achieves a relative regret of 5.87%, while EI rises to 54.29%. When almost all inventory survives, there is little systematic record error for EI to correct. Estimation error in the retention adjustment can then be propagated through repeated inventory updates, causing the policy to discount usable inventory unnecessarily and replenish too aggressively. BI avoids this additional estimation channel and becomes the more robust operating rule.

The ETC results reinforce this interpretation. Because ETC optimizes the base-stock level separately for EI and BI and commits to the estimated lower-cost pair, it is not mechanically tied to either convention.

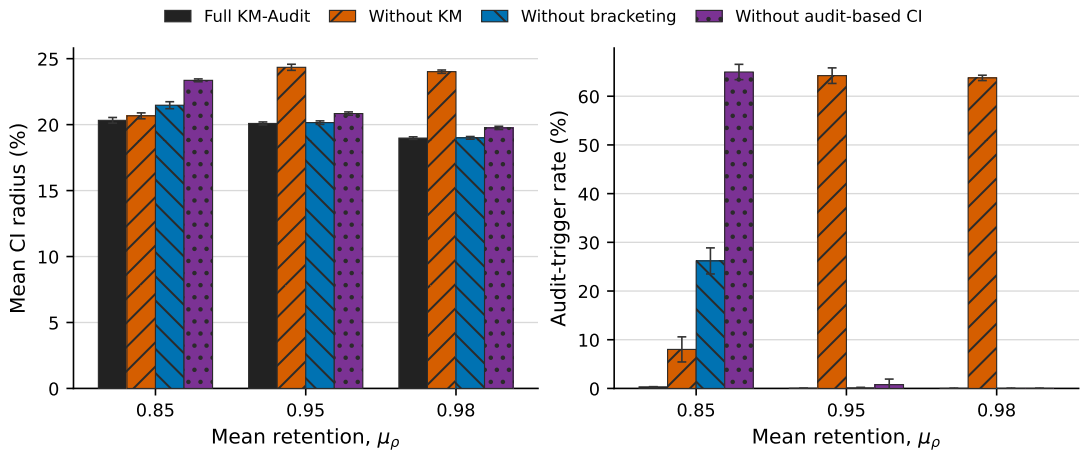


Figure 6 **Fixed-history leave-one-out diagnostic for the three components of the retention-confidence set. The left panel reports the mean confidence radius and the right panel the counterfactual audit-trigger rate under the common audit rule. All variants are evaluated on the same histories. Error bars are pointwise 95% confidence intervals.**

The intermediate exploration design, $\beta = 0.6$, generally performs best over the interior retention regimes, while the shorter $\beta = 0.5$ design is most competitive at the endpoints. At $\mu_\rho = 0.98$, for example, ETC with $\beta = 0.5$ attains a relative regret of 7.31%, close to the BI-only policy, whereas the more exploratory $\beta = 0.7$ policy incurs 16.33%. When shrinkage is limited, purchasing additional retention information and remaining at the high exploration stock for longer provide little corresponding operating benefit.

Taken together, the two comparisons show that adaptive revisability is most valuable when the appropriate response to shrinkage is not obvious, and that neither EI nor BI dominates uniformly.

Role of the three retention bounds. KM–Audit intersects Kaplan–Meier, observation-wise bracketing, and audit-based intervals. We isolate the contribution of each component through a fixed-history leave-one-out diagnostic: one interval is removed at a time while the realized operating history is held fixed. The exercise therefore measures how each information source changes retention uncertainty and the next audit decision, rather than comparing closed-loop policies that generate different data.

Figure 6 shows that the three intervals play distinct roles across retention regimes. At $\mu_\rho = 0.85$, removing the audit-based interval raises the counterfactual audit-trigger rate from 0.33% to 64.97%, while removing the bracketing interval also produces a material increase. Exact audits and observation-wise brackets therefore provide most of the protection against redundant additional counts when shrinkage is pronounced. The ranking changes sharply as retention rises: at $\mu_\rho = 0.98$, removing Kaplan–Meier raises the trigger rate from essentially zero to 63.76% and produces the widest confidence set. In this regime, Kaplan–Meier extracts risk-set information from ordinary censored observations that observation-wise bracketing largely discards. The three bounds are therefore complementary rather than redundant. Section D.2 reports the complete protocol and cell-level results.

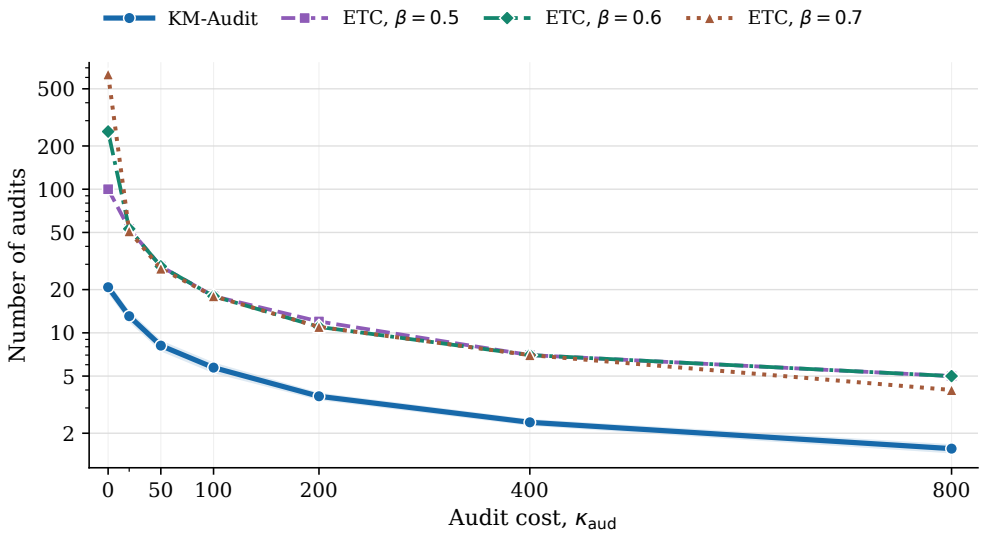


Figure 7 **Total number of physical audits as a function of the audit cost. KM–Audit points are mean realized counts, with shaded pointwise 95% confidence intervals. ETC counts are deterministic conditional on the experimental inputs. The vertical axis is logarithmic.**

6.2. Experiment 2: Sensitivity to Operating and Audit Costs

Our second experiment varies the cost of a physical audit and the relative costs of excess inventory and unmet demand. Unless otherwise stated, we fix $T = 10{,}000$ and $\mu_\rho = 0.95$ and compare KM–Audit with ETC under $\beta \in \{0.5, 0.6, 0.7\}$. Each policy receives the relevant cost parameters through its prescribed decision rule, without additional ex post calibration.

Impact of audit cost. We next vary the direct cost of a physical count, κ_{aud}, from 0 to 800, holding all other experimental inputs fixed. Figure 7 reports the total number of audits over the horizon. For ETC, these are the budgets selected before operations begin; for KM–Audit, they are the mean realized counts generated by its adaptive decisions.

KM–Audit responds strongly to the price of information: its mean audit count falls from 20.78 when physical counts carry no direct fee to 1.56 at $\kappa_{\text{aud}} = 800$, including the mandatory initialization count. The ETC budgets also decline, but at every positive audit cost they remain approximately 2.6 to 4.1 times as large as the realized KM–Audit count. KM–Audit thus adapts its information purchases to realized informativeness rather than committing the entire budget ex ante. Section D.3 reports the complete audit-count values.

Impact of holding and lost-sales costs. We next examine the two costs governing the inventory trade-off. In the left panel of Figure 8, we hold $b = 40$ fixed and vary the unit holding cost over $h \in \{0.67, 1, 1.5\}$. In the right panel, we hold $h = 1$ fixed and vary the lost-sales penalty over $b \in \{26.7, 40, 60\}$. These values are labeled Low, Medium, and High in the figures. To isolate the operating-cost channel, each policy's information-acquisition protocol is held fixed across the cost environments.

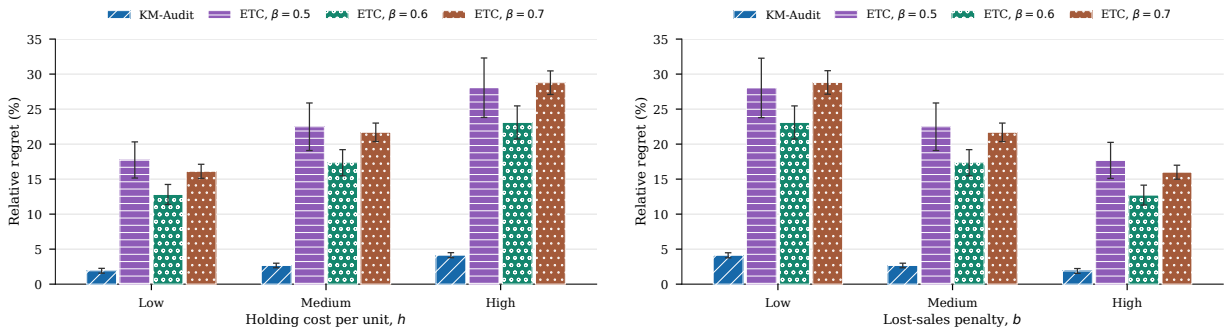


Figure 8 **Relative regret under alternative holding-cost levels (left) and lost-sales-penalty levels (right). In every environment, the clairvoyant EI and BI families are optimized separately over the common base-stock grid. Error bars are pointwise 95% confidence intervals across 500 simulation paths, conditional on the estimated reference costs.**

As holding inventory becomes more expensive, KM–Audit operates more leanly and errors that produce excess stock become increasingly costly. Its relative regret rises from 1.91% at $h = 0.67$ to 4.14% at $h = 1.5$. The best ETC specification is more sensitive to the same change: its regret rises from 12.81% to 23.09%, consistent with the increasing cost of its high-stock learning phase and of an overly aggressive commitment decision.

Increasing the lost-sales penalty produces the complementary response. KM–Audit carries more inventory, prevents more shortages, and its relative regret falls from 4.12% at $b = 26.7$ to 1.88% at $b = 60$. This decrease is relative to the reoptimized benchmark; absolute cost still rises because each unmet unit is more expensive. ETC moves in the same direction but remains substantially farther from the benchmark. Section D.3 reports the associated inventory targets, stockout rates, and absolute costs.

Viewed jointly, the two panels exhibit a near symmetry governed by the service-to-inventory cost ratio b/h. The low-holding-cost environment, with $b/h \approx 60$, produces almost the same relative regrets as the high-lost-sales-cost environment, where $b/h = 60$. Similarly, the high-holding-cost and low-lost-sales-cost environments both have $b/h \approx 26.7$ and yield nearly identical outcomes. Because these matched environments are approximately proportional rescalings of the same inventory trade-off, this correspondence is best interpreted as an internal scaling check: within the range considered here, b/h is the primary determinant of the desired inventory position.

KM–Audit has the lowest mean regret in all six cost environments, and its pointwise confidence intervals do not overlap those of any ETC policy. Among the ETC alternatives, $\beta = 0.6$ is uniformly best, consistent with the trade-off between committing too early with imprecise estimates and exploring too long at a costly high-stock level.

6.3. Experiment 3: Evaluation Using Retailer Data

For the retail partner introduced in Section 1, this experiment tests whether KM–Audit's information-acquisition logic remains useful under intermittent, heterogeneous, and nonstationary demand. It com-

bines actual dated order histories and SKU-level economics with a counterfactual shrinkage process calibrated from external food-loss data. Demand chronology and product heterogeneity are therefore empirical, whereas inventory trajectories, physical losses, stockouts, and audit outcomes are simulated.

Retailer data. We use the retailer's data for 2023 and for January 1 through November 14, 2024. After matching paid, nonrefunded cart lines to product records, the data contain 27,741 retained transaction lines representing 40,916 ordered units in 2023 and 23,023 lines representing 34,991 units in 2024. Daily SKU-level quantities are treated as a paid-order proxy for potential demand. The final cohort contains 101 SKUs from nine categories; the figures focus on meat (21 SKUs), prepared meals (16), and produce (12).

Demand calibration and chronological replay. The 2023 observations initialize the demand model and determine its preprocessing parameters. Each fixed 2024 demand path is then processed chronologically for $T = 319$ days, without shuffling or resampling, once for every simulated shrinkage realization. All policies use the same calendar-adjusted online demand model. For demand estimation, they observe only past sales and stockout indicators. A rolling Kaplan–Meier estimator combines exact demand observations with right-censored stockout observations; no policy observes current or future demand before ordering.

Construction of the shrinkage environment. The retailer data contain no historical physical-inventory records, audit outcomes, or observed shrinkage realizations. We therefore construct a counterfactual environment using the Food and Agriculture Organization of the United Nations' *Food Loss and Waste Database*.[3] Its category-level targets are translated into intermittent daily retention processes and frozen before the 2024 replay. For meat, prepared meals, and produce, the calibrated one-period mean retentions are 0.99940, 0.99743, and 0.99693. Because inventory carried across days is repeatedly exposed, these daily laws generate validation loss-to-inbound ratios of 2.13%, 7.26%, and 10.33%, close to the corresponding targets of 2.15%, 7.18%, and 10.30%. The simulator keeps the retention realizations hidden; policies receive an exact observation only through a simulated physical count or an eligible reconciled stockout.

Economic inputs and policies. For SKU i, the selling price P_i is the median valid unit price in its 2023 paid transactions, and c_i is given for each product. We set

$$h_i = \frac{0.25 c_i}{365}, \qquad b_i = P_i - c_i, \qquad \kappa_{\mathrm{aud}} = 22\frac{6+4}{60} = \$3.67.$$

Thus, holding cost corresponds to an assumed annual carrying rate of 25%, the lost-sales penalty equals the foregone gross-margin proxy, and an audit assigns six minutes of setup plus four minutes of counting at a labor rate of \$22 per hour. These inputs are declared economic assumptions rather than measured retailer accounting costs.

We compare KM–Audit with ETC under $\beta \in \{0.10, 0.20, 0.30\}$. The $\beta = 0.20$ specification is selected from the candidate grid using the 2023 chronology, its two neighboring specifications show the cost of shorter and longer front-loaded exploration.

[3] https://www.fao.org/platform-food-loss-waste/flw-data

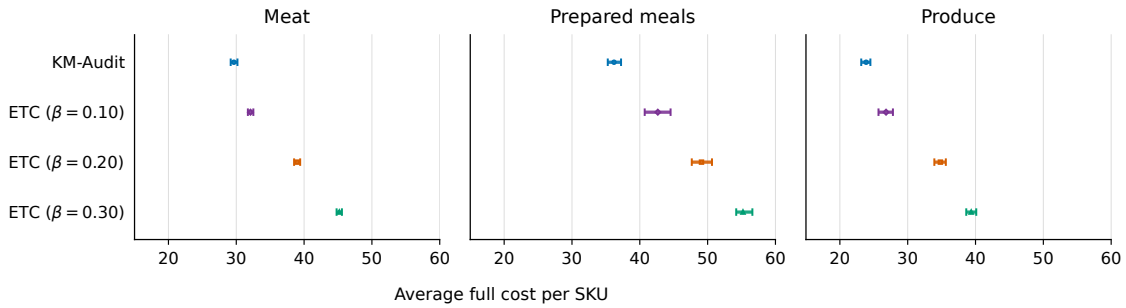


Figure 9 **Average full cost per SKU over the 319-day retailer-data replay. Full cost includes holding, lost-sales, and physical-audit costs and does not use a clairvoyant denominator. Error bars are pointwise 95% paired bootstrap intervals conditional on the fixed SKU cohort and demand histories.**

Performance measures and uncertainty. We report full cost, lost sales as a percentage of potential demand, and units scrapped per 100 units received. Full cost includes holding, lost-sales, and physical-audit costs; scrapped units are reported separately rather than priced again in the monetary objective. Each SKU is evaluated on 40 common shrinkage paths with its 2024 demand chronology held fixed. Error bars are pointwise 95% paired bootstrap intervals conditional on the displayed SKUs and demand histories. Section D.4 of the Electronic Companion gives the complete cohort construction, calibration, implementation, metric, and resampling details.

Full-cost performance. Figure 9 shows that KM–Audit has the lowest average full cost in all three displayed categories. Its average 319-day cost per SKU is 29.65 for meat, 36.20 for prepared meals, and 23.87 for produce. The corresponding costs under the least exploratory ETC specification, $\beta = 0.10$, are 32.09, 42.65, and 26.81. KM–Audit therefore reduces cost by 7.6%, 15.1%, and 11.0%, respectively, even relative to the ex post cheapest displayed ETC specification.

The comparison with the 2023-selected specification, ETC with $\beta = 0.20$, is more pronounced. Relative to its costs of 38.97, 49.09, and 34.82, KM–Audit reduces average cost by 23.9% in meat, 26.3% in prepared meals, and 31.5% in produce.

The cost difference is closely related to the amount and timing of information acquisition. Across the complete 101-SKU cohort, KM–Audit conducts an average of 1.42 audits per SKU. ETC averages 2.00, 3.97, and 5.54 audits under $\beta = 0.10$, 0.20, and 0.30, respectively. KM–Audit averages only 1.31 counts in meat, 1.84 in prepared meals, and 1.26 in produce. Its advantage therefore comes from preserving the option to count while avoiding redundant observations, not from counting more often.

Service and physical waste. Figures 10 and 11 reveal that the monetary advantage of adaptive auditing does not arise from a single uniform inventory response. Instead, KM–Audit uses its information differently across product environments.

For meat, KM–Audit and the 2023-selected ETC policy deliver essentially the same service: their lost-sales rates are 6.79% and 6.83%, respectively. KM–Audit scraps 2.41 units per 100 units received, compared

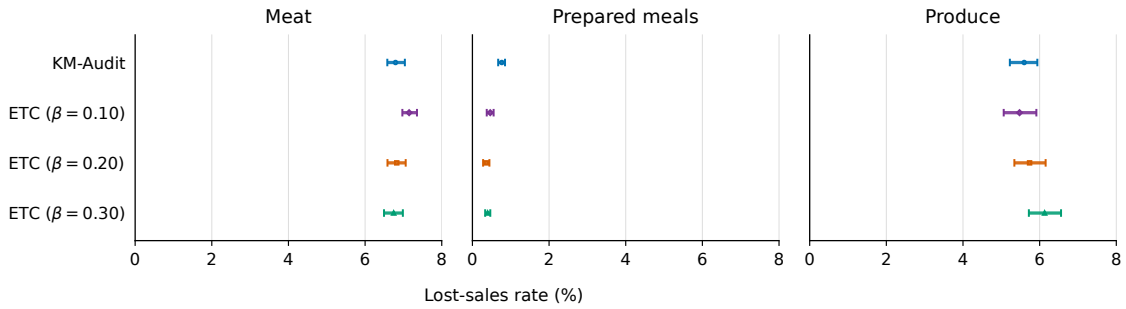


Figure 10 **Lost sales as a percentage of the 2024 paid-order sequence treated as potential demand. All policies process the same dated demand sequences and the same shrinkage paths. Error bars are pointwise 95% paired bootstrap intervals.**

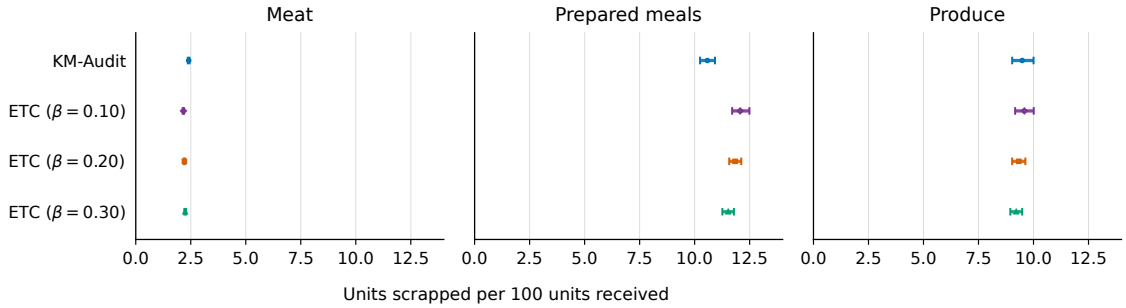


Figure 11 **Units scrapped through physical shrinkage per 100 units received. The denominator is inbound replenishment rather than units sold. Error bars are pointwise 95% paired bootstrap intervals.**

with 2.21 under ETC. The much lower full cost is therefore achieved while preserving service, with a modest increase in physical product loss.

Prepared meals exhibit a different trade-off. KM–Audit scraps 10.58 units per 100 units received, compared with 11.84 under the selected ETC policy and 11.52 even under the lowest-waste displayed ETC specification. Its lost-sales rate is higher, 0.77% rather than 0.36%, but remains below one percent. In this category, adaptive replenishment materially reduces both monetary cost and physical waste by operating somewhat more leanly while retaining a high service level.

For produce, the physical outcomes are close. KM–Audit has a lost-sales rate of 5.60% and scraps 9.49 units per 100 units received, compared with 5.74% and 9.33, respectively, under the selected ETC specification. The large full-cost difference is consequently driven mainly by the cost of front-loaded exploration and counting rather than by a substantial difference in the steady-state service–waste position.

Inventory-record choice and audit cost. The preceding comparison evaluates complete replenishment policies. Figure 12 instead isolates inventory-record choice by comparing EI-only and BI-only policies over audit prices ranging from zero to eight times the baseline fee. The restriction applies only to the record convention: both policies continue to reoptimize the base-stock level, and physical counts remain available.

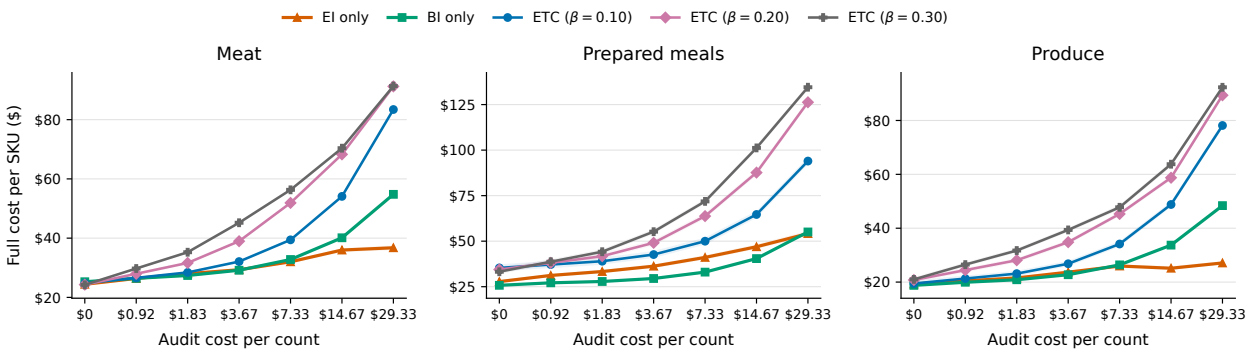


Figure 12 **Average full cost per SKU under alternative inventory-record conventions and physical-count prices. EI only uses the expected-inventory convention. BI only fixes the inventory-record coefficient at one. Each ETC policy recomputes its cost-balanced count budget at every price.**

Every audit-price cell is replayed from the beginning. Further design details are given in Section D.5 of the Electronic Companion.

At every positive audit price, EI has lower mean cost than all three ETC policies. Among the ETC alternatives, $\beta = 0.1$ is uniformly least costly once the count price is positive, as longer front-loaded exploration becomes difficult to recover over a 319-day horizon. Because EI also continues to revise its replenishment decisions whereas ETC commits once, this comparison captures the joint value of adaptive auditing and continued policy revision rather than audit timing alone.

Turning to inventory-record choice, at the baseline count price of $3.67, BI has the lower mean cost in all three categories, although its advantage is only 0.5% in meat. The advantage is larger in prepared meals (18.6%) and produce (4.0%). In these categories, the unadjusted book record acts as a lean-inventory control. BI does not fully replace estimated shrinkage, thereby reducing inventory exposure when the corresponding service benefit is insufficient to justify the additional stock. A more accurate shrinkage adjustment therefore need not minimize total operating cost.

This ranking changes with the price of physical information. In produce, EI overtakes BI between count prices of $3.67 and $7.33; in prepared meals, the reversal occurs between $14.67 and $29.33. Meat exhibits no uniform ranking: EI is slightly better at the two lowest prices, BI is better at $1.83 and $3.67, and EI regains the advantage from $7.33 onward. The high-price reversals partly reflect BI's required initialization count, while EI can decline a count whose estimated decision value does not cover its fee. Thus, the inventory-record convention and the information-acquisition protocol should be chosen jointly.

7. Conclusion

Cloud e-commerce management platforms have made it inexpensive for small retailers to sell online, but they have not given these retailers the warehouse or enterprise systems needed to track physical stock. Such retailers replenish from the book inventory that their platform simulates from orders and sales, and

unrecorded shrinkage makes that record drift above the sellable stock. Unobserved shrinkage turns replenishment into both a control and an information problem. It also raises a basic operational question: should replenishment use the EI update to account for average shrinkage or the BI update based on recorded transactions, that is, should the inventory planner discount the CEMP's book inventory or order against it as recorded? KM-Audit combines routine sales and reliable stockout signals with occasional targeted counts to learn the base-stock level, choose between EI and stockout-reconciled BI, and decide when a targeted physical count is worth its cost.

The structural contribution is an observable cycle-level performance representation that separates learning the shrinkage process from comparing candidate EI and BI policies. KM-Audit pools naturally occurring exact and right-censored retention observations from decision cycles with exact observations purchased through audit episodes. Each audit begins at a known-zero stockout boundary, places the fixed audit-start order, and ends with a physical count of the focal SKU. The audit rule works in two stages: a cube-root target margin of error widens the gaps between audits as the horizon grows, and an audit is executed only when the realized confidence radius, which also reflects routine sales after stockouts, still exceeds that target. We establish a $\widetilde{O}(T^{2/3})$ regret upper bound against the better non-auditing policy across the two continuous target classes, whose actual long-run average operating cost is $g^{\star}_{\cup}$, and an $\Omega(\sqrt{T})$ lower bound showing that shrinkage uncertainty alone imposes a nontrivial learning cost. The guarantee is therefore relative to the better optimized policy across the two operational rule classes.

Operational requirements. KM-Audit is designed for settings without continuous item-level inventory visibility. Its operational inputs are order quantities, realized sales, reliable stockout indicators, and occasional end-of-period physical counts of the focal SKU. The first three are already recorded by a CEMP; only the counts require additional effort. These observations suffice to update the mean-retention confidence set and cycle-level performance estimates. The confidence set supports cost evaluation for both modes and implementation of EI, whereas BI continues to use the book balance between reliable stockout resets. Once those quantities are updated, the audit trigger is operationally simple: conduct another audit when the confidence radius $r_{\mu}(m)$ exceeds the uncertainty threshold a_{m+1}. The method does not reconstruct the exact inventory state between reconciliation points or replace ERP, RFID, or warehouse-management systems. Rather, it supports replenishment when continuous physical visibility is unavailable or too costly, as it is for retailers, such as our partner, whose storefront has outgrown the back-end systems they can afford.

Three implications follow. First, a more shrinkage-adjusted inventory record is not automatically a better operating policy. EI can prevent the service failures created by an overstated book balance, but the resulting replenishment can also increase holding exposure and physical waste. BI can therefore be the lower-cost rule when its lean-inventory discipline is worth more than the service protection forgone. The inventory-record convention and stock target should be chosen jointly rather than treating record correction as an end in itself.

Second, the value of adaptive auditing lies in preserving the option to buy an exact observation, not in counting more often. KM–Audit compares the cost of a complete audit episode with the operating consequences of unresolved retention uncertainty. Informative stockout cycles can narrow that uncertainty and make a planned count redundant, whereas weak routine information leaves the option to count later. Audit intensity should therefore respond jointly to the count cost, the decision sensitivity to retention error, and the informativeness of ordinary operation.

Third, better shrinkage information moves the retailer along a cost–service–waste frontier rather than improving every outcome simultaneously. Correcting an overstated record can reduce lost sales by supporting a higher service posture, but the additional inventory may also increase exposure to subsequent loss. Conversely, a leaner policy can reduce scrappage at the price of more shortages. The preferred response is therefore category- and cost-dependent, which is precisely why the audit, tracking rule, and base-stock target must be learned together.

Several extensions would broaden the operational scope of the model: covariate-dependent retention, positive replenishment lead times, and multi-item counts that bundle audit costs across SKUs or storage locations. Each extension changes either the regenerative structure or the value of exact information and therefore requires more than a mechanical modification of KM–Audit.

More broadly, limited traceability need not be all or nothing. Targeted counts can be used selectively, with routine operating data determining when exact visibility is worth its cost and whether replenishment should adjust for shrinkage.

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Electronic Companion to "Learning to Replenish and Audit under Limited Inventory Traceability"

Appendix A: Supplementary Model and Algorithm Details

Audit-start order. The exact-recovery identity in (1) holds for every positive s_{aud}. We therefore treat this quantity as an exogenous operational design parameter rather than an additional learning decision. A smaller audit-start order limits possible carryover inventory but can increase lost sales in the audit period; a larger order can reduce audit-period shortage costs but may leave a longer recovery tail. The analysis uses only the uniform bound $s_{\mathrm{aud}} \le \bar{S}$, so the regret rate does not depend on its precise value.

Grouped Kaplan–Meier quantities. For the pooled sample $\mathscr{D}_\rho(m) = \{(U_\ell, \Delta_\ell) : 1 \le \ell \le m\}$, let $0 \le r_{1,m} < \cdots < r_{J_m,m} \le 1$ be the distinct uncensored values. The grouped event count and risk set at $r_{j,m}$ are

$$d_{j,m} := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell = r_{j,m},\ \Delta_\ell = 1\}, \qquad n_{j,m} := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \ge r_{j,m}\}.$$

For a truncation level $a \in \mathcal{A}_T$, define

$$R_m(a) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \ge a\}, \qquad \widehat{\mathsf{H}}_m(a) := \sum_{j: r_{j,m} \le a} \frac{d_{j,m}}{n_{j,m}}.$$

These quantities enter the explicit restricted-mean radius in (72) below.

Appendix B: Proof of Proposition 1

Proof. Fix $S \in [0, \bar{S}]$ and $\nu \in (0, 1]$, and suppress the dependence on (S, ν) in the period-level notation. All expectations are taken under the true dynamics induced by the fixed policy π_S^ν. Assumption 3 gives the stockout time a geometric tail. Together with bounded orders and demand, this implies the integrability needed below. In particular,

$$0 < \mathbb{E}[\Lambda_1^{S,\nu}] < \infty, \qquad \mathbb{E}[V_1^{S,\nu} + W_1^{S,\nu}] < \infty,$$

so that the renewal-reward theorem applies.

Let

$$B_t := X_t + q_t$$

denote the pre-shrinkage inventory, and define the physical shrinkage in period t by

$$R_t^{\mathrm{sh}} := (1 - \rho_t) B_t.$$

Then

$$A_t = \rho_t B_t, \qquad X_{t+1} = B_t - R_t^{\mathrm{sh}} - Y_t.$$

Since $X_{t+1} = A_t - Y_t$ and $L_t = D_t - Y_t$, the one-period cost can be written as

$$C_t = hX_{t+1} + bL_t = h(A_t - Y_t) + b(D_t - Y_t) = bD_t + hA_t - (h+b)Y_t. \tag{46}$$

We next relate the expected usable inventory A_t to the expected physical shrinkage R_t^{sh}. Use the full, possibly latent, pre-outcome filtration $\mathcal{F}_t^-$ defined in Assumption 3. The quantity $B_t = X_t + q_t$ is $\mathcal{F}_t^-$-measurable, whereas

ρ_t is independent of $\mathcal{F}_t^-$ and has mean μ_ρ. Moreover, for a cycle started from a stockout state, the event $\{t \le \Lambda_1^{S,\nu}\}$ depends only on periods before t, and is therefore $\mathcal{F}_t^-$-measurable. Hence

$$\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}A_t\right] = \mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}\rho_t B_t\right] = \mu_\rho\,\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}B_t\right],$$

while

$$\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}R_t^{\mathrm{sh}}\right] = \mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}(1-\rho_t)B_t\right] = (1-\mu_\rho)\,\mathbb{E}\left[\mathbf{1}_{\{t\le\Lambda_1^{S,\nu}\}}B_t\right].$$

Since $\mu_\rho \in (0,1)$, summing over $t \ge 1$ gives

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] = \frac{\mu_\rho}{1-\mu_\rho}\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} R_t^{\mathrm{sh}}\right]. \tag{47}$$

Now use the inventory balance over a completed cycle. Because stockout periods reset the true inventory to zero, a completed cycle starts and ends with zero true inventory:

$$X_{\tau_0+1} = 0, \qquad X_{\tau_1+1} = 0.$$

Summing

$$X_{t+1} = X_t + q_t - R_t^{\mathrm{sh}} - Y_t$$

over $t \in \mathcal{C}_1$ yields

$$0 = Q_1^{S,\nu} - \sum_{t\in\mathcal{C}_1} R_t^{\mathrm{sh}} - V_1^{S,\nu}.$$

Therefore

$$\sum_{t\in\mathcal{C}_1} R_t^{\mathrm{sh}} = Q_1^{S,\nu} - V_1^{S,\nu} = W_1^{S,\nu}. \tag{48}$$

Combining (47) and (48),

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] = \frac{\mu_\rho}{1-\mu_\rho}\,\mathbb{E}[W_1^{S,\nu}]. \tag{49}$$

Next, $\Lambda_1^{S,\nu}$ is an integrable stopping time for the joint demand–retention filtration. Wald's identity and independence of the i.i.d. demand increments therefore give

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} D_t\right] = \mathbb{E}[D_t]\,\mathbb{E}[\Lambda_1^{S,\nu}]. \tag{50}$$

Using (46), (49), and (50), the expected total true cost in one completed cycle is

$$\begin{aligned}\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} C_t\right] &= b\,\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} D_t\right] + h\,\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} A_t\right] - (h+b)\,\mathbb{E}[V_1^{S,\nu}] \\ &= b\,\mathbb{E}[D_t]\,\mathbb{E}[\Lambda_1^{S,\nu}] + \frac{h\mu_\rho}{1-\mu_\rho}\,\mathbb{E}[W_1^{S,\nu}] - (h+b)\,\mathbb{E}[V_1^{S,\nu}].\end{aligned}$$

Since

$$\alpha(\mu_\rho) := \frac{h\mu_\rho}{1-\mu_\rho},$$

this becomes

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} C_t\right] = b\,\mathbb{E}[D_t]\,\mathbb{E}[\Lambda_1^{S,\nu}] + \alpha(\mu_\rho)\,\mathbb{E}[W_1^{S,\nu}] - (h+b)\,\mathbb{E}[V_1^{S,\nu}]. \tag{51}$$

By the renewal-reward theorem,

$$g(S;\nu) = \frac{\mathbb{E}\left[\sum_{t\in\mathcal{C}_1} C_t\right]}{\mathbb{E}[\Lambda_1^{S,\nu}]}.$$

Dividing (51) by $\mathbb{E}[\Lambda_1^{S,\nu}]$ gives

$$g(S;\nu) = b\,\mathbb{E}[D_t] + \alpha(\mu_\rho)\,\frac{\mathbb{E}[W_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]} - (h+b)\,\frac{\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}. \tag{52}$$

Because the stockout cycles are regenerative and identically distributed under the fixed policy π_S^ν,

$$w(S;\nu) = \frac{\mathbb{E}[W_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}, \qquad v(S;\nu) = \frac{\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]}.$$

Substituting these identities into (52) proves

$$g(S;\nu) = b\,\mathbb{E}[D_t] + \alpha(\mu_\rho)\,w(S;\nu) - (h+b)\,v(S;\nu).$$

It remains to verify the equivalent pseudo-cost representation. Under the fixed policy π_S^ν, the cycle pseudo-cost random variable is

$$\widetilde{K}_k^{S,\nu}(\mu_\rho) = \alpha(\mu_\rho) W_k^{S,\nu} - (h+b) V_k^{S,\nu}.$$

Since the completed cycles are identically distributed under the fixed policy,

$$\begin{aligned}\gamma(S;\nu) &= \frac{\mathbb{E}[\widetilde{K}_1^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda_1^{S,\nu}]} \\ &= \frac{\alpha(\mu_\rho)\mathbb{E}[W_1^{S,\nu}] - (h+b)\mathbb{E}[V_1^{S,\nu}]}{\mathbb{E}[\Lambda_1^{S,\nu}]} \\ &= \alpha(\mu_\rho) w(S;\nu) - (h+b) v(S;\nu).\end{aligned}$$

Therefore

$$g(S;\nu) = b\,\mathbb{E}[D_t] + \gamma(S;\nu).$$

The additive term $b\,\mathbb{E}[D_t]$ is independent of both the base-stock target and the inventory-tracking mode. In particular, with

$$\nu_{\mathrm{EI}} := \mu_\rho, \qquad \nu_{\mathrm{BI}} := 1,$$

minimizing $g(S;\nu_c)$ over $S \in [0,\bar{S}]$ and $c \in \mathcal{C}_{\mathrm{tr}}$ is equivalent to minimizing $\gamma(S;\nu_c)$ over the same set, including the book-inventory endpoint $\nu = 1$. This proves the proposition.

□

Appendix C: Proofs for the regret analysis

C.1. Auxiliary episode bounds

We first record a few deterministic and stochastic bounds used repeatedly below. Throughout this subsection, the length of an episode that crosses the calendar horizon is understood as its full length under the analysis-only continuation of that episode until the next stockout. This continuation is used only in the proof and does not provide the algorithm with post-horizon observations.

Lemma 1 *Under Assumptions 1 and 3, consider any policy used by the algorithm or by the benchmark. In particular, every decision and recovery order-up-to level belongs to $[0, \bar{S}]$, and $s_{\mathrm{aud}} \le \bar{S}$. Then, for every calendar period t,*

$$q_t \le \bar{S}, \qquad X_t \le (t-1)\bar{S}, \qquad C_t \le ht\bar{S} + b\overline{D}.$$

Consequently,

$$\sum_{t=1}^{T} C_t \le \frac{h\bar{S}}{2} T(T+1) + b\overline{D}\, T,$$

and

$$\sum_{t=1}^{T} C_t^{\mathrm{alg}} \le \frac{h\bar{S}}{2} T(T+1) + (b\overline{D} + \kappa_{\mathrm{aud}})T.$$

Moreover, let episode ℓ be either a decision episode or an audit episode, including any post-audit recovery tail, and suppose that it starts from a stockout state. Let $\mathcal{G}_{\ell-1}$ denote the episode-start history and let $\mathcal{C}_\ell$ and Λ_ℓ denote its full period set and length. Then

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_\ell} C_t \,\middle|\, \mathcal{G}_{\ell-1}\right] \le \bar{C}_{\mathrm{op}}, \qquad \bar{C}_{\mathrm{op}} := \frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0}. \tag{53}$$

In particular, the total algorithmic episode cost satisfies

$$\mathbb{E}\left[\sum_{t\in\mathcal{C}_\ell} C_t^{\mathrm{alg}} \,\middle|\, \mathcal{G}_{\ell-1}\right] \le \begin{cases} \bar{C}_{\mathrm{op}}, & \text{if episode } \ell \text{ is a decision episode,} \\ \bar{C}_{\mathrm{aud}}, & \text{if episode } \ell \text{ is an audit episode,} \end{cases} \tag{54}$$

where

$$\bar{C}_{\mathrm{aud}} := \kappa_{\mathrm{aud}} + \bar{C}_{\mathrm{op}} = \kappa_{\mathrm{aud}} + \frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0}.$$

Proof. For a decision or recovery period, the order rule gives

$$q_t = (S - \hat{X}_t)^+ \le S \le \bar{S}.$$

In the first period of an audit episode,

$$q_t = s_{\mathrm{aud}} \le \bar{S}.$$

Hence $q_t \le \bar{S}$ under every policy considered in the paper.

Since $\rho_t \le 1$ and $D_t \ge 0$,

$$X_{t+1} = \big(\rho_t(X_t + q_t) - D_t\big)^+ \le X_t + q_t \le X_t + \bar{S}.$$

Starting from $X_1 = 0$, induction gives

$$X_t \le (t-1)\bar{S}.$$

Therefore

$$C_t = hX_{t+1} + b(D_t - Y_t) \le ht\bar{S} + b\overline{D}.$$

Summing over $t = 1, \dots, T$ proves the operating-cost horizon bound. Since $J_t^{\mathrm{aud}} \le 1$,

$$C_t^{\mathrm{alg}} = C_t + \kappa_{\mathrm{aud}} J_t^{\mathrm{aud}} \le C_t + \kappa_{\mathrm{aud}},$$

which proves the algorithmic-cost horizon bound.

We next prove the episode-level bound. Suppose episode ℓ begins at calendar time t_ℓ, so

$$X_{t_\ell} = 0.$$

Use $r = 1, 2, \ldots$ as the local period index within the episode. Repeating the preceding induction from the episode-start state gives

$$X_{t_\ell+r-1} \le (r-1)\bar{S}, \qquad X_{t_\ell+r} \le r\bar{S}.$$

Consequently, on the event $\{\Lambda_\ell \ge r\}$,

$$C_{t_\ell+r-1} \le hr\bar{S} + b\overline{D}.$$

By Lemma 2,

$$\mathbb{P}(\Lambda_\ell \ge r \mid \mathcal{G}_{\ell-1}) \le (1-p_0)^{r-1}.$$

Therefore

$$\begin{aligned}
\mathbb{E}\left[\sum_{t\in\mathcal{C}_\ell} C_t \,\middle|\, \mathcal{G}_{\ell-1}\right] &= \sum_{r\ge1} \mathbb{E}[C_{t_\ell+r-1}\mathbf{1}\{\Lambda_\ell \ge r\} \mid \mathcal{G}_{\ell-1}] \\
&\le \sum_{r\ge1} (hr\bar{S} + b\overline{D})\mathbb{P}(\Lambda_\ell \ge r \mid \mathcal{G}_{\ell-1}) \\
&\le \sum_{r\ge1} (hr\bar{S} + b\overline{D})(1-p_0)^{r-1} \\
&= \frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0}.
\end{aligned}$$

This proves (53).

A decision episode incurs no audit fee. An audit episode incurs exactly one direct fee κ_{aud}, in its first period, regardless of the length of its recovery tail. Adding this one-time fee proves (54).

□

Lemma 2 *Under Assumption 3, every decision episode and every audit episode operated under the bounded recovery rule has, conditional on its episode-start history, a length stochastically dominated by a geometric random variable with success probability p_0. More precisely, for every episode ℓ and every integer $r \ge 1$,*

$$\mathbb{P}(\Lambda_\ell \ge r \mid \mathcal{G}_{\ell-1}) \le (1-p_0)^{r-1}. \tag{55}$$

In particular,

$$\mathbb{E}[\Lambda_\ell \mid \mathcal{G}_{\ell-1}] \le \frac{1}{p_0}.$$

The same geometric domination holds conditionally from any full pre-outcome history at which the episode remains active. Moreover, with probability at least $1 - T^{-4}$, every episode initiated by calendar time T has full length at most $\bar{\Lambda}_T$.

Proof. Assumption 3 states that, conditional on the full pre-outcome history of any active period, the episode ends in that period with probability at least p_0. Thus, conditional on the episode having survived through its first $r-1$ periods, its probability of surviving period r is at most $1-p_0$. Iterating this bound proves (55). Summing the tail gives

$$\mathbb{E}[\Lambda_\ell \mid \mathcal{G}_{\ell-1}] = \sum_{r\ge1} \mathbb{P}(\Lambda_\ell \ge r \mid \mathcal{G}_{\ell-1}) \le \sum_{r\ge1} (1-p_0)^{r-1} = \frac{1}{p_0}.$$

Starting the same argument from any active within-episode period proves the conditional residual-length claim.

At most T episodes can be initiated by calendar time T, because every episode consumes at least one period. For definiteness, set $t_\ell = \infty$ if episode ℓ is not initiated. Using the conditional tail bound,

$$\begin{aligned}
&\mathbb{P}\big(\exists \ell : t_\ell \le T,\ \Lambda_\ell > \bar{\Lambda}_T\big) \\
&\quad \le \sum_{\ell=1}^{T} \mathbb{E}\big[\mathbf{1}\{t_\ell \le T\}\mathbb{P}\big(\Lambda_\ell > \bar{\Lambda}_T \mid \mathcal{G}_{\ell-1}\big)\big] \\
&\quad \le T(1-p_0)^{\bar{\Lambda}_T} \\
&\quad \le T\exp(-p_0\bar{\Lambda}_T) \\
&\quad \le T^{-4},
\end{aligned}$$

by the definition of $\bar{\Lambda}_T$.

□

C.2. Confidence events for shrinkage learning

The following lemma establishes simultaneous coverage of the bracketing confidence interval.

Lemma 3 *Under Assumption 2, for every* $1 \le m \le T$,

$$\mathbb{P}(\mu_\rho \notin \mathcal{C}_{\mathrm{br}}(m)) \le \frac{\delta_m}{3}.$$

Consequently, under the schedule

$$\delta_m := \frac{1}{64T^4(m+1)^2}, \qquad m \ge 0,$$

we have

$$\mathbb{P}(\forall\, 1 \le m \le T : \mu_\rho \in \mathcal{C}_{\mathrm{br}}(m)) \ge 1 - T^{-4}.$$

Proof. For every episode ℓ, the episode-start observation satisfies

$$L_\ell = U_\ell \le \rho_{t_\ell} \le H_\ell \qquad \text{almost surely.}$$

Let $\mathcal{G}_{\ell-1}$ contain the complete history through the end of episode $\ell-1$, together with the randomization and selected action for episode ℓ, but not the current demand and retention realizations. Thus the episode type and episode-start order are $\mathcal{G}_{\ell-1}$-measurable, while ρ_{t_ℓ} is fresh and independent of $\mathcal{G}_{\ell-1}$. Therefore

$$\mathbb{E}[\rho_{t_\ell} \mid \mathcal{G}_{\ell-1}] = \mu_\rho.$$

The pathwise brackets consequently imply

$$\mathbb{E}[L_\ell - \mu_\rho \mid \mathcal{G}_{\ell-1}] \le 0,$$

and

$$\mathbb{E}[\mu_\rho - H_\ell \mid \mathcal{G}_{\ell-1}] \le 0.$$

Thus

$$S_m^L := \sum_{\ell=1}^{m} (L_\ell - \mu_\rho)$$

and

$$S_m^H := \sum_{\ell=1}^{m} (\mu_\rho - H_\ell)$$

are supermartingales.

Moreover, their increments have range length at most one:

$$L_\ell - \mu_\rho \in [-\mu_\rho, 1 - \mu_\rho],$$

and

$$\mu_\rho - H_\ell \in [\mu_\rho - 1, \mu_\rho].$$

The one-sided Hoeffding–Azuma inequality for supermartingales therefore gives, for every $r > 0$,

$$\mathbb{P}\big(\bar{L}_m - \mu_\rho > r\big) = \mathbb{P}(S_m^L > mr) \le \exp(-2mr^2),$$

and

$$\mathbb{P}\big(\mu_\rho - \bar{H}_m > r\big) = \mathbb{P}(S_m^H > mr) \le \exp(-2mr^2).$$

With

$$r_{\mathrm{br}}(m) := \sqrt{\frac{\log(6/\delta_m)}{2m}},$$

we obtain

$$\mathbb{P}\big(\bar{L}_m - \mu_\rho > r_{\mathrm{br}}(m)\big) \le \frac{\delta_m}{6},$$

and

$$\mathbb{P}\big(\mu_\rho - \bar{H}_m > r_{\mathrm{br}}(m)\big) \le \frac{\delta_m}{6}.$$

Assumption 2 gives

$$\mu_\rho \in [\underline{\mu}, \bar{\mu}].$$

Hence intersecting the raw bracketing interval with $[\underline{\mu}, \bar{\mu}]$ cannot remove μ_ρ. Therefore

$$\begin{aligned}
\mathbb{P}(\mu_\rho \notin \mathcal{C}_{\mathrm{br}}(m)) &\le \mathbb{P}\big(\bar{L}_m - \mu_\rho > r_{\mathrm{br}}(m)\big) \\
&\quad + \mathbb{P}\big(\mu_\rho - \bar{H}_m > r_{\mathrm{br}}(m)\big) \\
&\le \frac{\delta_m}{3}.
\end{aligned}$$

Finally, a union bound gives

$$\begin{aligned}
&\mathbb{P}(\exists\, 1 \le m \le T : \mu_\rho \notin \mathcal{C}_{\mathrm{br}}(m)) \\
&\quad \le \sum_{m=1}^{T} \frac{\delta_m}{3} \\
&\quad \le \frac{1}{192T^4} \sum_{m=1}^{\infty} \frac{1}{(m+1)^2} \\
&\quad \le T^{-4}.
\end{aligned}$$

This proves the simultaneous statement.

□

The following lemma establishes simultaneous coverage of the audit confidence interval.

Lemma 4 *Suppose that*

$$\delta_m := \frac{1}{64T^4(m+1)^2}, \qquad m \geq 0,$$

and that every audit is selected at a stockout boundary before the current demand and retention realizations are observed. Under Assumption 2, with probability at least $1 - T^{-4}$*,*

$$\mu_\rho \in \mathcal{C}_{\mathrm{aud}}(m) \qquad \textit{simultaneously for every } 0 \leq m \leq T.$$

Proof. Let $\mathcal{G}_{\ell-1}$ contain the complete history through the end of episode $\ell - 1$, together with the randomization and selected action for episode ℓ, but not the current demand and retention realizations, and define the predictable audit indicator

$$\mathsf{A}_\ell := \mathbf{1}\{\text{episode } \ell \text{ is audited}\}.$$

Thus A_ℓ is $\mathcal{G}_{\ell-1}$-measurable.

Let

$$T_\ell := \rho_{t_\ell}$$

be the fresh retention realization in the first period of episode ℓ. Since every audit episode starts from a stockout boundary,

$$X_{t_\ell} = 0.$$

On the event $\{\mathsf{A}_\ell = 1\}$, the audit-start order $s_{\mathrm{aud}} > 0$ therefore satisfies

$$T_\ell s_{\mathrm{aud}} = Y_{t_\ell} + X_{t_\ell+1}.$$

Hence the physical count reveals

$$U_\ell = \frac{Y_{t_\ell} + X_{t_\ell+1}}{s_{\mathrm{aud}}} = T_\ell.$$

Thus every selected audit contributes an exact observation of the fresh retention draw, whether or not the audited period stocks out. Define

$$N_{\mathrm{aud}}(m) := \sum_{\ell=1}^{m} \mathsf{A}_\ell$$

and

$$S_m := \sum_{\ell=1}^{m} \mathsf{A}_\ell (T_\ell - \mu_\rho).$$

Equivalently, because $U_\ell = T_\ell$ whenever $\mathsf{A}_\ell = 1$,

$$S_m = \sum_{\ell=1}^{m} \mathsf{A}_\ell (U_\ell - \mu_\rho).$$

Because $T_\ell \in [0,1]$ is fresh and satisfies

$$\mathbb{E}[T_\ell \mid \mathcal{G}_{\ell-1}] = \mu_\rho,$$

Hoeffding's lemma gives, for every $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\exp\{\lambda(T_\ell - \mu_\rho)\} \mid \mathcal{G}_{\ell-1}] \leq \exp\{\lambda^2/8\}.$$

Since A_ℓ is $\mathcal{G}_{\ell-1}$-measurable,

$$\begin{aligned}
&\mathbb{E}\left[\exp\left\{\lambda \mathsf{A}_\ell(T_\ell-\mu_\rho)-\frac{\lambda^2}{8}\mathsf{A}_\ell\right\}\middle|\mathcal{G}_{\ell-1}\right]\\
&\quad=1-\mathsf{A}_\ell+\mathsf{A}_\ell\mathbb{E}\left[\exp\left\{\lambda(T_\ell-\mu_\rho)-\frac{\lambda^2}{8}\right\}\middle|\mathcal{G}_{\ell-1}\right]\\
&\quad\le 1.
\end{aligned}$$

Consequently,

$$\mathcal{M}_m(\lambda):=\exp\left\{\lambda S_m-\frac{\lambda^2}{8}N_{\mathrm{aud}}(m)\right\},\qquad m\ge 0,$$

is a nonnegative supermartingale with

$$\mathcal{M}_0(\lambda)=1.$$

For $n\ge 1$, define

$$\sigma_n:=\inf\{m\ge 1: N_{\mathrm{aud}}(m)=n\},$$

with $\sigma_n=\infty$ if the nth audit does not occur. On $\{\sigma_n<\infty\}$, define the nth selected audit draw by

$$\rho_n^{\mathrm{aud}}:=T_{\sigma_n}=U_{\sigma_n},$$

and let

$$\bar\rho_n^{\mathrm{aud}}:=\frac{1}{n}\sum_{j=1}^{n}\rho_j^{\mathrm{aud}}.$$

On $\{\sigma_n<\infty\}$, we then have

$$S_{\sigma_n}=n\big(\bar\rho_n^{\mathrm{aud}}-\mu_\rho\big).$$

Fix $r>0$. On the event

$$\left\{\sigma_n\le T,\ \bar\rho_n^{\mathrm{aud}}-\mu_\rho>r\right\},$$

we have

$$S_{\sigma_n}>nr,\qquad N_{\mathrm{aud}}(\sigma_n)=n.$$

Choosing $\lambda=4r$ gives

$$\mathcal{M}_{\sigma_n}(4r)>\exp\{2nr^2\}.$$

Therefore

$$\begin{aligned}
&\mathbb{P}\big(\sigma_n\le T,\ \bar\rho_n^{\mathrm{aud}}-\mu_\rho>r\big)\\
&\quad\le\mathbb{P}\left(\max_{0\le m\le T}\mathcal{M}_m(4r)>\exp\{2nr^2\}\right)\\
&\quad\le\exp(-2nr^2),
\end{aligned}$$

where the last inequality is the maximal inequality for nonnegative supermartingales.

Applying the same argument to $\mathcal{M}_m(-4r)$ gives

$$\mathbb{P}\big(\sigma_n\le T,\ \big|\bar\rho_n^{\mathrm{aud}}-\mu_\rho\big|>r\big)\le 2\exp(-2nr^2).$$

For $n\ge 1$, define

$$\widetilde{r}_{\mathrm{aud}}(n):=\sqrt{\frac{\log(6/\delta_n)}{2n}}.$$

Then

$$\mathbb{P}\big(\sigma_n \le T,\ \big|\bar{\rho}_n^{\mathrm{aud}} - \mu_\rho\big| > \widetilde{r}_{\mathrm{aud}}(n)\big) \le \frac{\delta_n}{3}.$$

A union bound over $n = 1, \dots, T$ yields

$$\begin{aligned}
&\mathbb{P}\big(\exists\, n \le T : \sigma_n \le T,\ \big|\bar{\rho}_n^{\mathrm{aud}} - \mu_\rho\big| > \widetilde{r}_{\mathrm{aud}}(n)\big) \\
&\quad \le \sum_{n=1}^{T} \frac{\delta_n}{3} \\
&\quad \le \frac{1}{192T^4} \sum_{n=1}^{\infty} \frac{1}{(n+1)^2} \\
&\quad \le T^{-4}.
\end{aligned}$$

Work on the complementary event. Fix any completed-episode count $0 \le m \le T$, and set

$$n := N_{\mathrm{aud}}(m).$$

If $n = 0$, then

$$\mathcal{C}_{\mathrm{aud}}(m) = [\underline{\mu}, \bar{\mu}],$$

and the result follows from Assumption 2.

Suppose $n \ge 1$. Then

$$\sigma_n \le m \le T$$

and, by the definition of the audit sample mean,

$$\widehat{\mu}_{\mathrm{aud}}(m) = \bar{\rho}_n^{\mathrm{aud}}.$$

Therefore

$$|\widehat{\mu}_{\mathrm{aud}}(m) - \mu_\rho| \le \widetilde{r}_{\mathrm{aud}}(n).$$

Because $n \le m$ and the schedule $(\delta_m)_{m \ge 0}$ is nonincreasing,

$$\delta_m \le \delta_n.$$

Hence

$$\begin{aligned}
\widetilde{r}_{\mathrm{aud}}(n) &= \sqrt{\frac{\log(6/\delta_n)}{2n}} \\
&\le \sqrt{\frac{\log(6/\delta_m)}{2n}} \\
&= r_{\mathrm{aud}}(m).
\end{aligned}$$

Thus

$$|\widehat{\mu}_{\mathrm{aud}}(m) - \mu_\rho| \le r_{\mathrm{aud}}(m).$$

The audit confidence interval is centered at the projected estimator

$$\widetilde{\mu}_{\mathrm{aud}}(m) = \Pi_{[\underline{\mu}, \bar{\mu}]}\big(\widehat{\mu}_{\mathrm{aud}}(m)\big).$$

Since

$$\mu_\rho \in [\underline{\mu}, \bar{\mu}],$$

we have

$$\Pi_{[\underline{\mu},\bar{\mu}]}(\mu_\rho) = \mu_\rho.$$

Nonexpansiveness of Euclidean projection therefore gives

$$\begin{aligned} |\widetilde{\mu}_{\mathrm{aud}}(m) - \mu_\rho| &= \left|\Pi_{[\underline{\mu},\bar{\mu}]}\big(\widehat{\mu}_{\mathrm{aud}}(m)\big) - \Pi_{[\underline{\mu},\bar{\mu}]}(\mu_\rho)\right| \\ &\le |\widehat{\mu}_{\mathrm{aud}}(m) - \mu_\rho| \\ &\le r_{\mathrm{aud}}(m). \end{aligned}$$

Consequently,

$$\mu_\rho \in \left[\widetilde{\mu}_{\mathrm{aud}}(m) - r_{\mathrm{aud}}(m),\, \widetilde{\mu}_{\mathrm{aud}}(m) + r_{\mathrm{aud}}(m)\right].$$

Because $\mu_\rho \in [\underline{\mu}, \bar{\mu}]$, the final intersection with that known interval does not remove μ_ρ. Hence

$$\mu_\rho \in \mathcal{C}_{\mathrm{aud}}(m)$$

simultaneously for all $0 \le m \le T$.

□

Lemma 5 *Let H and $\widehat{H}$ be finite, càdlàg, nondecreasing cumulative hazard functions on $[0, a]$, normalized by*

$$H(0-) = \widehat{H}(0-) = 0,$$

and satisfying

$$0 \le \Delta H(u) \le 1, \qquad 0 \le \Delta \widehat{H}(u) \le 1 \qquad \textit{for all } u \in [0, a].$$

Let

$$S(x) := \prod_{u \le x} (1 - dH(u)), \qquad \widehat{S}(x) := \prod_{u \le x} (1 - d\widehat{H}(u))$$

be the associated product-integral survival functions. If

$$\sup_{0 \le x \le a} |\widehat{H}(x) - H(x)| \le \zeta,$$

then

$$\sup_{0 \le x \le a} |\widehat{S}(x) - S(x)| \le \zeta \exp\{\widehat{H}(a)\}.$$

Proof. By the forward Volterra equation for product integrals (Gill and Johansen, 1990, Proposition 5, Eq. (40)), applied to the additive measures $-H$ and $-\widehat{H}$, respectively, the product-integral survival functions satisfy

$$S(x) = 1 - \int_{[0,x]} S(u-)\, dH(u),$$

and

$$\widehat{S}(x) = 1 - \int_{[0,x]} \widehat{S}(u-)\, d\widehat{H}(u).$$

Set

$$D(x) := \widehat{S}(x) - S(x), \qquad G(x) := \widehat{H}(x) - H(x).$$

Subtracting the two equations gives

$$D(x) = -\int_{[0,x]} D(u-)\, d\widehat{H}(u) - \int_{[0,x]} S(u-)\, dG(u). \tag{56}$$

We first control the second term. Since G and S are functions of bounded variation, integration by parts gives, for every $y \le a$,

$$\int_{[0,y]} S(u-)\, dG(u) = S(y)G(y) - \int_{[0,y]} G(u)\, dS(u),$$

where the initial boundary term vanishes because $G(0-) = 0$. Since S is nonincreasing, $0 \le S \le 1$, and $-dS$ is a positive measure with total mass

$$\int_{[0,y]} (-dS(u)) = S(0-) - S(y) = 1 - S(y),$$

we obtain

$$\begin{aligned}\left| \int_{[0,y]} S(u-)\, dG(u) \right| &\le S(y)|G(y)| + \int_{[0,y]} |G(u)|\, (-dS(u)) \\ &\le \zeta S(y) + \zeta\{1 - S(y)\} = \zeta. \end{aligned} \tag{57}$$

Define

$$M(x) := \sup_{0 \le y \le x} |D(y)|.$$

Using (56) and (57), we obtain

$$M(x) \le \zeta + \int_{[0,x]} M(u-)\, d\widehat{H}(u).$$

The Stieltjes version of Gronwall's inequality therefore yields

$$M(x) \le \zeta\, \mathcal{E}_{\widehat{H}}(x),$$

where

$$\mathcal{E}_{\widehat{H}}(x) := \exp\{\widehat{H}^c(x)\} \prod_{u \le x} \big(1 + \Delta\widehat{H}(u)\big)$$

and $\widehat{H}^c$ denotes the continuous part of $\widehat{H}$. Since $\log(1+z) \le z$ for $z \ge 0$,

$$\mathcal{E}_{\widehat{H}}(x) \le \exp\left\{ \widehat{H}^c(x) + \sum_{u \le x} \Delta\widehat{H}(u) \right\} = \exp\{\widehat{H}(x)\}.$$

Consequently,

$$M(x) \le \zeta \exp\{\widehat{H}(x)\} \le \zeta \exp\{\widehat{H}(a)\}.$$

Taking $x = a$ proves the result.

□

For the remaining shrinkage-confidence arguments, define

$$\bar{F}_\rho(x) := \mathbb{P}(\rho_t > x), \qquad \bar{F}_\rho^-(x) := \mathbb{P}(\rho_t \geq x),$$

and define the shrinkage hazard measure by

$$d\mathsf{H}_\rho(x) := \frac{\mathbb{P}(\rho_t \in dx)}{\bar{F}_\rho^-(x)}.$$

The ratio is set to zero wherever $\bar{F}_\rho^-(x) = 0$, which is immaterial because the retention distribution assigns no mass there. The cumulative hazard may be extended-valued after the upper support endpoint; all perturbation arguments below are restricted to levels a with a nonempty empirical risk set, for which $\bar{F}_\rho^-(a) > 0$ almost surely and $\mathsf{H}_\rho(a) < \infty$. At an atom x,

$$\Delta\mathsf{H}_\rho(x) = \mathbb{P}(\rho_t = x \mid \rho_t \geq x).$$

The survival function is the corresponding product integral:

$$\bar{F}_\rho(x) = \prod_{u \leq x} \big(1 - d\mathsf{H}_\rho(u)\big).$$

Lemma 6 *Under Assumptions 1 and 4, define, for episode ℓ,*

$$\mathsf{R}_\ell(u) := \mathbf{1}\{U_\ell \geq u\}, \qquad N_\ell(x) := \mathbf{1}\{U_\ell \leq x,\ \Delta_\ell = 1\}.$$

Let

$$R_m(u) := \sum_{\ell=1}^{m} \mathsf{R}_\ell(u).$$

For every Borel interval $B \subseteq [0,1]$, define

$$N_m(B) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \in B,\ \Delta_\ell = 1\},$$

and

$$\mathbb{M}_m(B) := N_m(B) - \int_B R_m(u)\, d\mathsf{H}_\rho(u),$$

$$\mathbb{A}_m(B) := \int_B R_m(u)\, d\mathsf{H}_\rho(u).$$

Set

$$\lambda_m := 4\log\left(\frac{32(m+1)^4|\mathcal{A}_T|}{\delta_m}\right). \tag{58}$$

Then, with probability at least $1 - T^{-4}$, the following holds simultaneously for every $1 \leq m \leq T$, every $a \in \mathcal{A}_T$ satisfying $R_m(a) \geq 1$, and every Borel interval $B \subseteq [0,a]$:

$$|\mathbb{M}_m(B)| \leq 2\sqrt{\big(\mathbb{A}_m(B) + 1\big)\lambda_m} + 14\lambda_m. \tag{59}$$

Proof. Write

$$K_T := |\mathcal{A}_T|.$$

For a Borel set $B \subseteq [0,1]$, use the notation

$$\mathsf{H}_\rho(B) := \int_B d\mathsf{H}_\rho(u).$$

We first establish the martingale concentration bound for a fixed deterministic interval.

Fix an episode ℓ. Let $\mathcal{G}_{\ell-1}$ be the sigma-field generated by all information available through the end of episode $\ell-1$, together with any independent algorithmic randomization used to select the action for episode ℓ, but before the current demand and shrinkage variables are realized. Thus the episode type and episode-start order are $\mathcal{G}_{\ell-1}$-measurable.

Let

$$T_\ell := \rho_{t_\ell},$$

and define the effective censoring threshold

$$C_\ell^\dagger := \begin{cases} +\infty, & \text{if episode } \ell \text{ is audited,} \\ D_{t_\ell}/\xi_\ell, & \text{if episode } \ell \text{ is a decision episode.} \end{cases}$$

Conditional on $\mathcal{G}_{\ell-1}$, the variable T_ℓ has the retention distribution and is independent of $C_\ell^\dagger$. By the definition of the episode-start observation,

$$U_\ell = \min\{T_\ell, C_\ell^\dagger\}, \qquad \Delta_\ell = \mathbf{1}\{T_\ell < C_\ell^\dagger\}.$$

For an audited episode, $C_\ell^\dagger = +\infty$, so $U_\ell = T_\ell$ and $\Delta_\ell = 1$.

Define the censoring-counting process

$$N_\ell^{\mathrm{c}}(x) := \mathbf{1}\{U_\ell \le x,\ \Delta_\ell = 0\}.$$

For $u \in [0,1]$, let

$$\mathscr{F}_{\ell,u} := \mathcal{G}_{\ell-1} \vee \sigma(N_\ell(v), N_\ell^{\mathrm{c}}(v) : 0 \le v \le u)\,,$$

completed in the usual way, and define

$$\mathscr{F}_{\ell,u-} := \bigvee_{v<u} \mathscr{F}_{\ell,v}, \qquad \mathscr{F}_{\ell,0-} := \mathcal{G}_{\ell-1}.$$

The at-risk indicator

$$\mathsf{R}_\ell(u) = \mathbf{1}\{U_\ell \ge u\}$$

is $\mathscr{F}_{\ell,u-}$-measurable.

On the event $\{\mathsf{R}_\ell(u) = 1\}$, we have

$$T_\ell \ge u, \qquad C_\ell^\dagger \ge u.$$

Conditional independence therefore gives, in compensator notation,

$$\mathbb{E}[dN_\ell(u) \mid \mathscr{F}_{\ell,u-}] = \mathsf{R}_\ell(u)\, d\mathsf{H}_\rho(u).$$

In particular, at an atom u,

$$\mathbb{P}(T_\ell = u \mid \mathscr{F}_{\ell,u-}, \mathsf{R}_\ell(u) = 1) = \Delta\mathsf{H}_\rho(u).$$

Assumption 4 rules out $T_\ell = C_\ell^\dagger$ in a decision episode, while no finite censoring tie is possible in an audited episode.

It follows that

$$M_\ell(x) := N_\ell(x) - \int_{[0,x]} \mathsf{R}_\ell(u)\, d\mathsf{H}_\rho(u)$$

is a martingale with respect to $(\mathscr{F}_{\ell,x})_{x\in[0,1]}$.

Concatenate these within-episode filtrations in chronological order. After the first-period censored observation of episode ℓ has been revealed in the level variable, reveal the remainder of the episode, including any post-audit recovery tail, before selecting the action for episode $\ell+1$. These additional observations carry zero increment in the concatenated martingale. Consequently, no future episode action is revealed before the observations on which that action depends.

For every fixed deterministic Borel interval B, integrating $\mathbf{1}\{u \in B\}$ against the concatenated martingales yields a martingale, denoted by $\mathbb{M}^B$, whose terminal value after m episodes is

$$\mathbb{M}_m(B) = N_m(B) - \int_B R_m(u)\, d\mathsf{H}_\rho(u).$$

Because episodes are processed one at a time, simultaneous events at a common hazard atom are also revealed one at a time. Every martingale jump therefore has absolute value at most one.

Let H_ρ^c denote the continuous part of the hazard measure. The predictable quadratic variation satisfies

$$\begin{aligned}\left\langle \mathbb{M}^B \right\rangle_m &= \sum_{\ell=1}^{m} \left[\int_B \mathsf{R}_\ell(u)\, d\mathsf{H}_\rho^c(u) + \sum_{u\in B} \mathsf{R}_\ell(u)\Delta\mathsf{H}_\rho(u)\{1-\Delta\mathsf{H}_\rho(u)\} \right] \\ &\le \sum_{\ell=1}^{m} \int_B \mathsf{R}_\ell(u)\, d\mathsf{H}_\rho(u) = \mathbb{A}_m(B). \end{aligned} \tag{60}$$

For deterministic $v>0$ and $y>0$, the bounded-jump Bernstein–Freedman inequality for local martingales (Dzhaparidze and Van Zanten, 2001, Corollary 3.4(iii)) gives

$$\mathbb{P}\left(|\mathbb{M}_m(B)| > \sqrt{2vy} + \frac{2y}{3},\ \left\langle \mathbb{M}^B \right\rangle_m \le v \right) \le 2e^{-y}. \tag{61}$$

Suppose that

$$\mathsf{H}_\rho(B) \le L.$$

Since $R_m(u) \le m$,

$$0 \le \mathbb{A}_m(B) \le m\mathsf{H}_\rho(B) \le mL.$$

Set

$$q^\star := \lceil \log_2(mL \vee 1) \rceil.$$

Partition the possible values of $\mathbb{A}_m(B)$ into

$$E_0 := \{\mathbb{A}_m(B) \le 1\},$$

and

$$E_q := \{2^{q-1} < \mathbb{A}_m(B) \le 2^q\}, \qquad q = 1, \dots, q^\star.$$

On E_0, use $v=1$ in (61). On E_q, use $v=2^q$. By (60), the corresponding predictable quadratic-variation condition holds on each bin. Moreover, on E_q,

$$2^q \le 2\mathbb{A}_m(B).$$

Thus, on every variance bin, the threshold

$$2\sqrt{(\mathbb{A}_m(B)+1)y} + 2y$$

dominates the corresponding Freedman threshold

$$\sqrt{2vy}+\frac{2y}{3}.$$

A union bound over the at most

$$Q_m(L) := 2+\lceil \log_2(mL+1)\rceil$$

variance bins therefore gives

$$\mathbb{P}\left(|\mathbb{M}_m(B)|>2\sqrt{\big(\mathbb{A}_m(B)+1\big)y}+2y\right)\le 2Q_m(L)e^{-y}. \tag{62}$$

We next construct a deterministic hazard partition that permits a uniform union bound. To do so, we first rule out a nonempty empirical risk set at a level having arbitrarily large true cumulative hazard. Let

$$\bar{F}_\rho^-(a) := \mathbb{P}(T_\ell \ge a).$$

The product-integral representation gives

$$\bar{F}_\rho^-(a) = \exp\{-\mathsf{H}_\rho^c(a-)\}\prod_{u<a}\big(1-\Delta\mathsf{H}_\rho(u)\big).$$

Using $1-z\le e^{-z}$,

$$\bar{F}_\rho^-(a)\le \exp\{-\mathsf{H}_\rho(a-)\}.$$

Since

$$\mathsf{H}_\rho(a)=\mathsf{H}_\rho(a-)+\Delta\mathsf{H}_\rho(a),\qquad \Delta\mathsf{H}_\rho(a)\le 1,$$

we obtain

$$\bar{F}_\rho^-(a)\le \exp\{1-\mathsf{H}_\rho(a)\}.$$

Moreover,

$$U_\ell\ge a \quad\Longrightarrow\quad T_\ell\ge a.$$

Because T_ℓ is fresh conditional on $\mathcal{G}_{\ell-1}$,

$$\mathbb{E}[\mathbf{1}\{U_\ell\ge a\}\mid \mathcal{G}_{\ell-1}]\le \mathbb{P}(T_\ell\ge a)=\bar{F}_\rho^-(a).$$

Therefore

$$\mathbb{E}[R_m(a)]=\sum_{\ell=1}^m \mathbb{E}[\mathbf{1}\{U_\ell\ge a\}]\le m\bar{F}_\rho^-(a),$$

and hence

$$\mathbb{P}(R_m(a)\ge 1)\le \mathbb{E}[R_m(a)]\le m\exp\{1-\mathsf{H}_\rho(a)\}.$$

Set

$$L_m := 1+\log\left(\frac{8mK_T}{\delta_m}\right).$$

A union bound over $a\in\mathcal{A}_T$ gives

$$\mathbb{P}(\exists a\in\mathcal{A}_T: R_m(a)\ge 1,\ \mathsf{H}_\rho(a)>L_m)\le\frac{\delta_m}{8}. \tag{63}$$

Because $0 \in \mathcal{A}_T$ and

$$\mathsf{H}_\rho(0) = \mathbb{P}(\rho_t = 0) \le 1 < L_m,$$

the finite set

$$\{a \in \mathcal{A}_T : \mathsf{H}_\rho(a) \le L_m\}$$

is nonempty. Define

$$a_m^\star := \max\{a \in \mathcal{A}_T : \mathsf{H}_\rho(a) \le L_m\}.$$

On the complement of the event in (63), every $a \in \mathcal{A}_T$ satisfying $R_m(a) \ge 1$ belongs to $[0, a_m^\star]$.

Set

$$\varepsilon_m := (m+1)^{-4}.$$

Construct a deterministic ordered partition $\mathcal{P}_m$ of $[0, a_m^\star]$ into disjoint Borel intervals with the following properties:

1. every point of $\mathcal{A}_T \cap [0, a_m^\star]$ is a partition boundary;
2. every atom u satisfying
$$\Delta\mathsf{H}_\rho(u) > \varepsilon_m$$
is isolated as a singleton partition cell;
3. every remaining nonsingleton partition cell C satisfies
$$\mathsf{H}_\rho(C) \le 2\varepsilon_m.$$

To construct such a partition, first insert all points of $\mathcal{A}_T \cap [0, a_m^\star]$ as boundaries and isolate every atom larger than ε_m. On each remaining component, every atom has mass at most ε_m. Starting from the left endpoint, close a cell when its accumulated hazard mass first reaches ε_m. Any overshoot is caused by an atom of mass at most ε_m, so each completed cell has hazard mass at most $2\varepsilon_m$. The final cell in each component has hazard mass at most ε_m.

Let B_m^{atom} denote the number of atoms larger than ε_m. Since

$$\mathsf{H}_\rho([0, a_m^\star]) = \mathsf{H}_\rho(a_m^\star) \le L_m,$$

we have

$$B_m^{\mathrm{atom}} \le \frac{L_m}{\varepsilon_m}.$$

After inserting the K_T truncation-grid boundaries and isolating the large atoms, there are at most

$$K_T + 2B_m^{\mathrm{atom}} + 1$$

remaining components. The greedy construction produces at most

$$\frac{L_m}{\varepsilon_m} + K_T + 2B_m^{\mathrm{atom}} + 1$$

nonsingleton cells. Adding the singleton cells gives, with

$$G_m := |\mathcal{P}_m|,$$

the bound

$$G_m \le \frac{L_m}{\varepsilon_m} + K_T + 3B_m^{\mathrm{atom}} + 1 \le K_T + 4L_m\varepsilon_m^{-1} + 1.$$

In particular,

$$G_m \le 4K_T + 8L_m\varepsilon_m^{-1} + 8. \tag{64}$$

Call a union of consecutive cells of $\mathcal{P}_m$ a *partition interval*. There are at most G_m^2 partition intervals. Every such interval B satisfies

$$\mathsf{H}_\rho(B) \le L_m.$$

We now verify the union-bound bookkeeping. Since $m \ge 1$, $K_T \ge 1$, and $0 < \delta_m \le 1$,

$$L_m = 1 + \log\left(\frac{8mK_T}{\delta_m}\right) \le \frac{8mK_T}{\delta_m}.$$

Using $\varepsilon_m^{-1} = (m+1)^4$ in (64),

$$G_m \le 76(m+1)^5 K_T \delta_m^{-1}.$$

Moreover,

$$Q_m(L_m) = 2 + \lceil \log_2(mL_m + 1) \rceil \le 12(m+1)^2 K_T \delta_m^{-1}.$$

Apply (62) with $L = L_m$ and $y = \lambda_m$ to every partition interval. Since

$$e^{-\lambda_m} = \frac{\delta_m^4}{2^{20}(m+1)^{16}K_T^4},$$

a union bound gives

$$\begin{aligned} 2Q_m(L_m)G_m^2 e^{-\lambda_m} &\le \frac{2 \cdot 12 \cdot 76^2}{2^{20}} \frac{\delta_m}{(m+1)^4 K_T} \\ &\le \frac{\delta_m}{8}, \end{aligned}$$

where the last inequality uses $m \ge 1$ and $K_T \ge 1$.

Consequently, outside an event of probability at most $\delta_m/8$,

$$|\mathbb{M}_m(B)| \le 2\sqrt{\big(\mathbb{A}_m(B) + 1\big)\lambda_m} + 2\lambda_m \tag{65}$$

simultaneously for every partition interval B.

It remains to extend the bound from partition intervals to arbitrary Borel intervals. Let B be an arbitrary Borel interval in $[0, a_m^\star]$. Let B° be the union of all cells of $\mathcal{P}_m$ that are contained entirely in B, with $B^\circ = \emptyset$ if there is no such cell. Because B is an interval, B° is either empty or a partition interval.

The remainder $B \setminus B^\circ$ consists of at most two interval fragments, each lying inside one nonsingleton boundary cell. A singleton cell corresponding to an isolated large atom is either entirely contained in B or entirely excluded from B, and therefore produces no boundary approximation error.

Let C be one of the possible nonsingleton boundary cells. By the construction of $\mathcal{P}_m$,

$$\mathsf{H}_\rho(C) \le 2\varepsilon_m.$$

Hence

$$\mathbb{A}_m(C) \le m\mathsf{H}_\rho(C) \le 2m\varepsilon_m \le \frac{1}{4}.$$

Since $\lambda_m \geq 1$, applying (65) to the single-cell partition interval C gives

$$|\mathbb{M}_m(C)| \leq 2\sqrt{\frac{5}{4}\lambda_m} + 2\lambda_m \leq 5\lambda_m.$$

Also,

$$N_m(C) = \mathbb{M}_m(C) + \mathbb{A}_m(C) \leq |\mathbb{M}_m(C)| + \mathbb{A}_m(C).$$

Therefore, for every Borel interval $J \subseteq C$,

$$\begin{aligned} |\mathbb{M}_m(J)| &\leq N_m(J) + \mathbb{A}_m(J) \\ &\leq N_m(C) + \mathbb{A}_m(C) \\ &\leq |\mathbb{M}_m(C)| + 2\mathbb{A}_m(C) \\ &\leq 6\lambda_m. \end{aligned}$$

If $B^\circ \neq \emptyset$, then (65) and $\mathbb{A}_m(B^\circ) \leq \mathbb{A}_m(B)$ give

$$|\mathbb{M}_m(B^\circ)| \leq 2\sqrt{\big(\mathbb{A}_m(B)+1\big)\lambda_m} + 2\lambda_m.$$

The same inequality is trivially true when $B^\circ = \emptyset$. Adding the contributions of the at most two boundary fragments yields

$$|\mathbb{M}_m(B)| \leq 2\sqrt{\big(\mathbb{A}_m(B)+1\big)\lambda_m} + 14\lambda_m.$$

This is (59).

For a fixed m, the failure probabilities in (63) and (65) sum to at most

$$\frac{\delta_m}{4}.$$

On their complement, every $a \in \mathcal{A}_T$ satisfying $R_m(a) \geq 1$ lies in $[0, a_m^\star]$, so the preceding bound holds for every Borel interval $B \subseteq [0, a]$.

Finally, under

$$\delta_m = \frac{1}{64T^4(m+1)^2},$$

a union bound over $m = 1, \ldots, T$ gives

$$\sum_{m=1}^{T} \frac{\delta_m}{4} \leq \frac{1}{256T^4} \sum_{m=1}^{\infty} \frac{1}{(m+1)^2} \leq T^{-4}.$$

This proves the simultaneous statement.

□

Lemma 7 *Consider the pooled episode-start observations*

$$\{(U_\ell, \Delta_\ell) : \ell \geq 1\}.$$

Suppose that every episode type is selected predictably from the preceding history, before the current demand and shrinkage variables are realized. Define

$$R_m(x) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \geq x\},$$

$$N_m(x) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \le x,\ \Delta_\ell = 1\},$$

and

$$\widehat{\mathsf{H}}_m(x) := \int_{[0,x]} \frac{dN_m(u)}{R_m(u)}.$$

Under Assumptions 1 and 4, with probability at least $1 - T^{-4}$*, the following holds simultaneously for every* $1 \le m \le T$ *and every* $a \in \mathcal{A}_T$ *satisfying* $R_m(a) \ge 1$*:*

$$\sup_{0 \le x \le a} \left| \widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x) \right| \le 4\sqrt{\frac{\mathsf{H}_\rho(a)\lambda_m}{R_m(a)}} + 32\frac{\lambda_m}{R_m(a)}. \tag{66}$$

Proof. Fix $m \le T$ and $a \in \mathcal{A}_T$ such that

$$r := R_m(a) \ge 1.$$

On $[0, a]$, define the signed counting-process residual measure

$$d\mathbb{M}_m(u) := dN_m(u) - R_m(u)\, d\mathsf{H}_\rho(u).$$

Since $R_m(u) \ge R_m(a) = r$ for every $u \le a$,

$$\widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x) = \int_{[0,x]} \frac{d\mathbb{M}_m(u)}{R_m(u)}, \qquad 0 \le x \le a.$$

The proof controls this algebraic identity by partitioning $[0, a]$ according to the realized risk-set size. Let

$$K := \left\lfloor \log_2 \left(\frac{m}{r} \right) \right\rfloor, \qquad q_k := 2^k r, \qquad k = 0, \dots, K.$$

Define

$$I_k := \{u \in [0, a] : q_k \le R_m(u) < 2q_k\}.$$

Since $r \le R_m(u) \le m$ on $[0, a]$, these sets are disjoint Borel intervals, possibly empty, and their nonempty members partition $[0, a]$. Let

$$h_k := \int_{I_k} d\mathsf{H}_\rho(u).$$

Then

$$\sum_{k=0}^{K} h_k = \mathsf{H}_\rho(a).$$

We first record a deterministic Abel-summation inequality. Let I be any nonempty Borel interval on which

$$q \le R_m(u) < 2q.$$

Fix $x \in I$, and set

$$J_x := I \cap [0, x].$$

If $J_x = \emptyset$, the desired inequality is trivial. Otherwise, the function

$$w(u) := \frac{1}{R_m(u)}$$

is a nondecreasing finite-step function on J_x. Partition J_x into consecutive Borel intervals

$$J_{x,1}, \dots, J_{x,p}$$

on which w is constant, with corresponding values

$$0 \le w_1 \le \cdots \le w_p \le \frac{1}{q}.$$

Write

$$\nu_j := \mathbb{M}_m(J_{x,j}), \qquad S_j := \sum_{i=j}^{p} \nu_i = \mathbb{M}_m\left(\bigcup_{i=j}^{p} J_{x,i}\right).$$

Each union on the right is a Borel interval contained in I. Abel summation gives

$$\sum_{j=1}^{p} w_j \nu_j = w_1 S_1 + \sum_{j=2}^{p} (w_j - w_{j-1}) S_j.$$

Therefore

$$\begin{aligned}
\left| \int_{I \cap [0,x]} \frac{d\mathbb{M}_m(u)}{R_m(u)} \right| &\le \left[w_1 + \sum_{j=2}^{p} (w_j - w_{j-1}) \right] \sup_{\substack{B \subseteq I \\ B \text{ a Borel interval}}} |\mathbb{M}_m(B)| \\
&= w_p \sup_{\substack{B \subseteq I \\ B \text{ a Borel interval}}} |\mathbb{M}_m(B)| \\
&\le \frac{1}{q} \sup_{\substack{B \subseteq I \\ B \text{ a Borel interval}}} |\mathbb{M}_m(B)|. \qquad (67)
\end{aligned}$$

Apply Lemma 6. For every interval $B \subseteq I_k$,

$$\mathbb{A}_m(B) = \int_B R_m(u)\, d\mathsf{H}_\rho(u) \le 2q_k h_k.$$

Hence

$$\sup_{\substack{B \subseteq I_k \\ B \text{ an interval}}} |\mathbb{M}_m(B)| \le 2\sqrt{(2q_k h_k + 1)\lambda_m} + 14\lambda_m.$$

Using (67),

$$\begin{aligned}
\sup_{x \in I_k} \left| \int_{I_k \cap [0,x]} \frac{d\mathbb{M}_m(u)}{R_m(u)} \right| &\le \frac{2}{q_k} \sqrt{(2q_k h_k + 1)\lambda_m} + \frac{14\lambda_m}{q_k} \\
&\le 2\sqrt{\frac{2h_k \lambda_m}{q_k}} + 16\frac{\lambda_m}{q_k}, \qquad (68)
\end{aligned}$$

where we used

$$\sqrt{2q_k h_k + 1} \le \sqrt{2q_k h_k} + 1$$

and $\lambda_m \ge 1$.

Summing (68) over the nonempty strata gives

$$\sup_{0 \le x \le a} \left| \widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x) \right| \le 2\sqrt{2\lambda_m} \sum_{k=0}^{K} \sqrt{\frac{h_k}{2^k r}} + \frac{16\lambda_m}{r} \sum_{k=0}^{K} 2^{-k}. \qquad (69)$$

By Cauchy–Schwarz,

$$\sum_{k=0}^{K} \sqrt{h_k 2^{-k}} \le \sqrt{\left(\sum_{k=0}^{K} h_k \right) \left(\sum_{k=0}^{K} 2^{-k} \right)} \le \sqrt{2\mathsf{H}_\rho(a)}.$$

Also,

$$\sum_{k=0}^{K} 2^{-k} \le 2.$$

Therefore

$$\sup_{0 \le x \le a} \left| \widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x) \right| \le 4\sqrt{\frac{\mathsf{H}_\rho(a)\lambda_m}{R_m(a)}} + 32\frac{\lambda_m}{R_m(a)}. \qquad (70)$$

Equation (70) is (66).

Kaplan–Meier restricted-mean bound. For every $m \geq 1$ and $a \in \mathcal{A}_T$ satisfying $R_m(a) \geq 1$, define

$$\zeta_m(a) := 40\frac{\lambda_m}{R_m(a)} + 4\sqrt{\frac{\lambda_m \widehat{\mathsf{H}}_m(a)}{R_m(a)} + 36\left(\frac{\lambda_m}{R_m(a)}\right)^2}. \tag{71}$$

Define the restricted-mean radius by

$$e_{\mathrm{KM}}(m;a) := \begin{cases} a, & R_m(a) = 0, \\ a \min\left\{1, \exp\{\widehat{\mathsf{H}}_m(a)\}\zeta_m(a)\right\}, & R_m(a) \geq 1. \end{cases} \tag{72}$$

The truncated Kaplan–Meier confidence interval is

$$\mathcal{C}_{\mathrm{KM}}(m;a) := [\widehat{\mu}_{\mathrm{KM}}(m;a) - e_{\mathrm{KM}}(m;a),\, \widehat{\mu}_{\mathrm{KM}}(m;a) + e_{\mathrm{KM}}(m;a) + (1-a)] \cap [\underline{\mu}, \bar{\mu}]. \tag{73}$$

Define the nonempty-candidate set

$$\mathcal{A}_T^{\mathrm{KM}}(m) := \{a \in \mathcal{A}_T : \mathcal{C}_{\mathrm{KM}}(m;a) \neq \varnothing\}.$$

It is nonempty because $a = 0$ yields $[\underline{\mu}, \bar{\mu}]$. The algorithm selects

$$a_m^{\mathrm{KM}} \in \operatorname*{arg\,min}_{a \in \mathcal{A}_T^{\mathrm{KM}}(m)} \operatorname{diam}\big(\mathcal{C}_{\mathrm{KM}}(m;a)\big), \qquad \mathcal{C}_{\mathrm{KM}}(m) := \mathcal{C}_{\mathrm{KM}}\big(m; a_m^{\mathrm{KM}}\big). \tag{74}$$

Lemma 8 *Under the conditions of Lemma 7 and Assumptions 1–4, with probability at least $1 - T^{-4}$, the following holds simultaneously for every $1 \leq m \leq T$ and every $a \in \mathcal{A}_T$:*

$$\left|\widehat{\mu}_{\mathrm{KM}}(m;a) - \int_0^a \mathbb{P}(\rho_t > x)\, dx\right| \leq e_{\mathrm{KM}}(m;a).$$

Consequently,

$$\mu_\rho \in \mathcal{C}_{\mathrm{KM}}(m) \qquad \textit{for every } 1 \leq m \leq T.$$

Proof. The proof uses only the pooled episode-start observations

$$\mathscr{D}_\rho(m) = \{(U_\ell, \Delta_\ell) : 1 \leq \ell \leq m\}.$$

For episode ℓ, let

$$T_\ell := \rho_{t_\ell},$$

and define the effective censoring threshold

$$C_\ell^\dagger := \begin{cases} +\infty, & \text{if episode } \ell \text{ is audited,} \\ D_{t_\ell}/\xi_\ell, & \text{if episode } \ell \text{ is a decision episode.} \end{cases}$$

By construction,

$$U_\ell = \min\{T_\ell, C_\ell^\dagger\}, \qquad \Delta_\ell = \mathbf{1}\{T_\ell < C_\ell^\dagger\}.$$

For an audited episode,

$$C_\ell^\dagger = +\infty, \qquad U_\ell = T_\ell, \qquad \Delta_\ell = 1.$$

For a decision episode, the episode-start order is predictable, while D_{t_ℓ} and ρ_{t_ℓ} are fresh, mutually independent, and independent of the preceding history. Assumption 4 rules out the censoring tie

$$T_\ell = C_\ell^\dagger$$

almost surely.

Define the pooled risk and event-counting processes by

$$R_m(x) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \geq x\},$$

and

$$N_m(x) := \sum_{\ell=1}^{m} \mathbf{1}\{U_\ell \leq x,\ \Delta_\ell = 1\}.$$

The empirical cumulative hazard is

$$\widehat{\mathsf{H}}_m(x) := \int_{[0,x]} \frac{dN_m(u)}{R_m(u)} = \sum_{j: r_{j,m} \leq x} \frac{d_{j,m}}{n_{j,m}},$$

where the integrand is defined to be zero whenever $R_m(u) = 0$. Its product-integral survival function is the Kaplan–Meier estimator:

$$\widehat{\bar{F}}_{\rho,m}(x) = \prod_{u \leq x} \big(1 - d\widehat{\mathsf{H}}_m(u)\big).$$

Work on the simultaneous event supplied by Lemma 7. This event has probability at least $1 - T^{-4}$.

Fix $m \leq T$ and $a \in \mathcal{A}_T$. If

$$R_m(a) = 0,$$

then both

$$\widehat{\mu}_{\mathrm{KM}}(m; a) \quad \text{and} \quad \int_0^a \mathbb{P}(\rho_t > x)\, dx$$

belong to $[0, a]$. Therefore

$$\left| \widehat{\mu}_{\mathrm{KM}}(m; a) - \int_0^a \mathbb{P}(\rho_t > x)\, dx \right| \leq a = e_{\mathrm{KM}}(m; a).$$

Hence suppose

$$R_m(a) \geq 1.$$

Because T and $\mathcal{A}_T$ are finite, we may intersect the simultaneous Nelson–Aalen event with a probability-one event on which, for every $m \leq T$ and $a \in \mathcal{A}_T$,

$$R_m(a) \geq 1 \quad \Longrightarrow \quad \bar{F}_\rho^-(a) > 0.$$

Indeed, if

$$\bar{F}_\rho^-(a) = \mathbb{P}(T_\ell \geq a) = 0,$$

then

$$U_\ell \geq a \quad \Longrightarrow \quad T_\ell \geq a$$

implies

$$R_m(a) = 0 \qquad \text{almost surely.}$$

Whenever $\bar{F}_\rho^-(a) > 0$, monotonicity of $\bar{F}_\rho^-$ gives

$$\bar{F}_\rho^-(u) \geq \bar{F}_\rho^-(a) \qquad \text{for all } u \leq a.$$

Consequently,

$$\mathsf{H}_\rho(a) = \int_{[0,a]} \frac{\mathbb{P}(\rho_t \in du)}{\bar{F}_\rho^-(u)} \le \frac{\mathbb{P}(\rho_t \le a)}{\bar{F}_\rho^-(a)} \le \frac{1}{\bar{F}_\rho^-(a)} < \infty.$$

Set

$$\eta_{m,a} := \sup_{0 \le x \le a} \left| \widehat{\mathsf{H}}_m(x) - \mathsf{H}_\rho(x) \right|.$$

By Lemma 7,

$$\eta_{m,a} \le 4\sqrt{\frac{\mathsf{H}_\rho(a)\lambda_m}{R_m(a)}} + 32\frac{\lambda_m}{R_m(a)}. \tag{75}$$

Since

$$\mathsf{H}_\rho(a) \le \widehat{\mathsf{H}}_m(a) + \eta_{m,a},$$

define

$$x_{m,a} := \frac{\lambda_m}{R_m(a)}, \qquad \widehat{h}_{m,a} := \widehat{\mathsf{H}}_m(a).$$

Then (75) implies

$$\eta_{m,a} \le 4\sqrt{x_{m,a}\left(\widehat{h}_{m,a} + \eta_{m,a}\right)} + 32x_{m,a}. \tag{76}$$

If

$$\eta_{m,a} \le 32x_{m,a},$$

then immediately $\eta_{m,a} \le \zeta_m(a)$. Otherwise,

$$\eta_{m,a} - 32x_{m,a} > 0,$$

so subtracting $32x_{m,a}$ from both sides of (76) and squaring gives

$$\left(\eta_{m,a} - 32x_{m,a}\right)^2 \le 16x_{m,a}\left(\widehat{h}_{m,a} + \eta_{m,a}\right).$$

Equivalently,

$$\eta_{m,a}^2 - 80x_{m,a}\eta_{m,a} + 1024x_{m,a}^2 - 16x_{m,a}\widehat{h}_{m,a} \le 0.$$

The upper root of this quadratic is

$$40x_{m,a} + 4\sqrt{x_{m,a}\widehat{h}_{m,a} + 36x_{m,a}^2}.$$

Thus, in both cases,

$$\eta_{m,a} \le 40\frac{\lambda_m}{R_m(a)} + 4\sqrt{\frac{\lambda_m \widehat{\mathsf{H}}_m(a)}{R_m(a)} + 36\left(\frac{\lambda_m}{R_m(a)}\right)^2} = \zeta_m(a). \tag{77}$$

Moreover, because

$$R_m(u) \ge R_m(a) \ge 1 \qquad \text{for all } u \le a,$$

we have

$$\widehat{\mathsf{H}}_m(a) = \int_{[0,a]} \frac{dN_m(u)}{R_m(u)} \le \frac{N_m(a)}{R_m(a)} \le m.$$

Thus $\widehat{\mathsf{H}}_m$ is finite on $[0,a]$.

The cumulative hazard functions H_ρ and $\widehat{\mathsf{H}}_m$ are therefore finite, càdlàg, and nondecreasing on $[0,a]$, are normalized by

$$\mathsf{H}_\rho(0-) = \widehat{\mathsf{H}}_m(0-) = 0,$$

and have jumps in $[0,1]$. Indeed,

$$\Delta \mathsf{H}_\rho(u) = \mathbb{P}(\rho_t = u \mid \rho_t \geq u) \in [0,1],$$

and

$$\Delta \widehat{\mathsf{H}}_m(u) = \frac{\Delta N_m(u)}{R_m(u)} \in [0,1].$$

Lemma 5, applied with

$$H = \mathsf{H}_\rho, \qquad \widehat{H} = \widehat{\mathsf{H}}_m, \qquad \zeta = \zeta_m(a),$$

gives

$$\sup_{0 \leq x \leq a} \left| \widehat{\bar{F}}_{\rho,m}(x) - \bar{F}_\rho(x) \right| \leq \exp\{\widehat{\mathsf{H}}_m(a)\}\zeta_m(a).$$

Since both survival functions take values in $[0,1]$,

$$\sup_{0 \leq x \leq a} \left| \widehat{\bar{F}}_{\rho,m}(x) - \bar{F}_\rho(x) \right| \leq \min\left\{1,\, \exp\{\widehat{\mathsf{H}}_m(a)\}\zeta_m(a)\right\}.$$

Integrating over $[0,a]$ gives

$$\begin{aligned}
\left| \widehat{\mu}_{\mathrm{KM}}(m;a) - \int_0^a \mathbb{P}(\rho_t > x)\, dx \right| &\leq \int_0^a \left| \widehat{\bar{F}}_{\rho,m}(x) - \bar{F}_\rho(x) \right| dx \\
&\leq a \min\left\{1,\, \exp\{\widehat{\mathsf{H}}_m(a)\}\zeta_m(a)\right\} \\
&= e_{\mathrm{KM}}(m;a).
\end{aligned}$$

Finally,

$$\mu_\rho = \int_0^1 \mathbb{P}(\rho_t > x)\, dx = \int_0^a \mathbb{P}(\rho_t > x)\, dx + \int_a^1 \mathbb{P}(\rho_t > x)\, dx,$$

and

$$0 \leq \int_a^1 \mathbb{P}(\rho_t > x)\, dx \leq 1 - a.$$

Therefore

$$\mu_\rho \in \left[\widehat{\mu}_{\mathrm{KM}}(m;a) - e_{\mathrm{KM}}(m;a),\, \widehat{\mu}_{\mathrm{KM}}(m;a) + e_{\mathrm{KM}}(m;a) + (1-a)\right].$$

Assumption 2 also gives

$$\mu_\rho \in [\underline{\mu}, \bar{\mu}].$$

Hence

$$\mu_\rho \in \mathcal{C}_{\mathrm{KM}}(m;a)$$

simultaneously for every $m \leq T$ and every $a \in \mathcal{A}_T$.

In particular, the result holds for the data-dependent minimizer a_m^{KM}. Therefore

$$\mu_\rho \in \mathcal{C}_{\mathrm{KM}}(m) \qquad \text{for every } 1 \leq m \leq T.$$

□

C.3. Cycle concentration and tracker-parameter stability

Lemma 9 *With probability at least $1 - T^{-4}$, the following holds simultaneously for all policy arms $(i,c) \in \{1,\dots,M\} \times \mathcal{C}_{\mathrm{tr}}$ and all $m \le T$ with $N_{i,c}(m) \ge 1$:*

$$\left| \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \left(\widetilde{K}_{\ell}^{\mathrm{obs}}(\mu_\rho) - \gamma(s_i; \nu_{c,\ell-1}) \Lambda_\ell \right)}{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell} \right| \le B_{i,c}^{\mathrm{stat}}(m).$$

Proof. Fix a policy arm (i,c), and enumerate its completed decision episodes, whenever they occur, by

$$\ell_{i,c,1} < \ell_{i,c,2} < \cdots .$$

For the jth such episode, write

$$\Lambda_{i,c,j} := \Lambda_{\ell_{i,c,j}}, \qquad \widetilde{K}_{i,c,j}^{\mathrm{obs}}(\mu) := \widetilde{K}_{\ell_{i,c,j}}^{\mathrm{obs}}(\mu), \qquad \nu_{i,c,j} := \nu_{c,\ell_{i,c,j}-1},$$

and define

$$X_{i,c,j} := \widetilde{K}_{i,c,j}^{\mathrm{obs}}(\mu_\rho) - \gamma(s_i; \nu_{i,c,j}) \Lambda_{i,c,j}.$$

Let $\mathcal{H}_{i,c,j}$ denote the complete history available immediately before episode $\ell_{i,c,j}$ begins. Conditional on $\mathcal{H}_{i,c,j}$, the episode starts from the stockout state, the pair $(s_i, \nu_{i,c,j})$ is fixed, and the future demand and shrinkage sequence is fresh. Therefore

$$\mathbb{E}[X_{i,c,j} \mid \mathcal{H}_{i,c,j}] = 0.$$

Since the history through the end of the $(j-1)$st completed episode of policy arm (i,c) is contained in $\mathcal{H}_{i,c,j}$, the centered cycle increments along the policy arm's realized occurrences form a martingale-difference sequence in the policy-arm-specific chronological filtration.

Let

$$\bar{c}_{\mathrm{per}} := \alpha(\bar{\mu}) \bar{S} + \big(\alpha(\bar{\mu}) + h + b\big) \overline{D}.$$

For every completed decision episode,

$$\left| \widetilde{K}_{i,c,j}^{\mathrm{obs}}(\mu_\rho) \right| \le \bar{c}_{\mathrm{per}} \Lambda_{i,c,j}.$$

Moreover,

$$|\gamma(s_i; \nu_{i,c,j})| = \frac{\left| \mathbb{E}[\widetilde{K}_1^{s_i, \nu_{i,c,j}}(\mu_\rho)] \right|}{\mathbb{E}[\Lambda_1^{s_i, \nu_{i,c,j}}]} \le \bar{c}_{\mathrm{per}}.$$

Hence

$$|X_{i,c,j}| \le 2 \bar{c}_{\mathrm{per}} \Lambda_{i,c,j}. \tag{78}$$

The positive-stockout-probability assumption gives, conditional on $\mathcal{H}_{i,c,j}$,

$$\mathbb{P}(\Lambda_{i,c,j} \ge t \mid \mathcal{H}_{i,c,j}) \le (1 - p_0)^{t-1}, \qquad t \ge 1. \tag{79}$$

We next convert this geometric tail into a conditional Bernstein moment bound. For every integer $k \ge 1$, the tail-integral identity and (79) give

$$\begin{aligned} \mathbb{E}[\Lambda_{i,c,j}^k \mid \mathcal{H}_{i,c,j}] &= \int_0^\infty k x^{k-1} \mathbb{P}(\Lambda_{i,c,j} > x \mid \mathcal{H}_{i,c,j}) \, dx \\ &\le 1 + e \int_1^\infty k x^{k-1} e^{-p_0 x} \, dx \\ &\le 1 + \frac{e k!}{p_0^k} \le \frac{4 k!}{p_0^k}. \end{aligned}$$

Combining this estimate with (78), for every integer $k \geq 2$,

$$\mathbb{E}[|X_{i,c,j}|^k \mid \mathcal{H}_{i,c,j}] \leq 4k! \left(\frac{2\bar{c}_{\mathrm{per}}}{p_0}\right)^k. \tag{80}$$

Set

$$v := \frac{32\bar{c}_{\mathrm{per}}^2}{p_0^2}, \qquad c := \frac{4\bar{c}_{\mathrm{per}}}{p_0}.$$

Then (80) implies the conditional Bernstein condition

$$\mathbb{E}[|X_{i,c,j}|^k \mid \mathcal{H}_{i,c,j}] \leq \frac{k!}{2} v c^{k-2}, \qquad k \geq 2.$$

Since $\mathbb{E}[X_{i,c,j} \mid \mathcal{H}_{i,c,j}] = 0$, for every $|\lambda| < 1/c$,

$$\mathbb{E}[\exp\{\lambda X_{i,c,j}\} \mid \mathcal{H}_{i,c,j}] \leq \exp\left\{\frac{\lambda^2 v}{2(1 - c|\lambda|)}\right\}. \tag{81}$$

For each $n \geq 1$, let $\ell_{i,c,n} = \infty$ if the nth completed decision episode of policy arm (i,c) never occurs. Iterating (81) in the policy-arm-specific filtration and stopping the resulting exponential supermartingale at the nth policy-arm occurrence yields, for every fixed $n \geq 1$, every $x > 0$, and $S_{i,c,n} := \sum_{j=1}^{n} X_{i,c,j}$ on $\{\ell_{i,c,n} < \infty\}$,

$$\mathbb{P}(\ell_{i,c,n} \leq T,\ S_{i,c,n} \geq x) \leq \exp\left\{-\frac{x^2}{2(nv + cx)}\right\}.$$

The same stopped-supermartingale bound applied to $-S_{i,c,n}$ gives

$$\mathbb{P}\Big(\ell_{i,c,n} \leq T,\ |S_{i,c,n}| > \sqrt{2nvL} + 2cL\Big) \leq 2e^{-L} \tag{82}$$

for every $L > 0$. Taking

$$L = L_{2M,T} := \log(128MT^5)$$

in (82) gives, except on an event of probability at most $2e^{-L_{2M,T}}$,

$$|S_{i,c,n}| \leq \frac{8\bar{c}_{\mathrm{per}}}{p_0}\left(\sqrt{nL_{2M,T}} + L_{2M,T}\right) \tag{83}$$

whenever $\ell_{i,c,n} \leq T$.

A union bound over the $2M$ policy arms and $n = 1, \ldots, T$ gives

$$4MTe^{-L_{2M,T}} = \frac{1}{32T^4} \leq T^{-4}.$$

Work on the resulting simultaneous event. Fix $m \leq T$, and let $n = N_{i,c}(m) \geq 1$. The set $\mathcal{I}_{i,c}(m)$ consists of the first n completed decision episodes of policy arm (i,c). Since every cycle lasts at least one period,

$$\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell = \sum_{j=1}^{n} \Lambda_{i,c,j} \geq n.$$

Therefore (83) implies

$$\begin{aligned}
&\left|\frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \left(\widetilde{K}_\ell^{\mathrm{obs}}(\mu_\rho) - \gamma(s_i; \nu_{c,\ell-1})\Lambda_\ell\right)}{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell}\right| \\
&\quad \leq \frac{8\bar{c}_{\mathrm{per}}}{p_0}\left(\sqrt{\frac{L_{2M,T}}{n}} + \frac{L_{2M,T}}{n}\right) \\
&\quad \leq \frac{8\bar{c}_{\mathrm{per}}}{p_0}\left(1 + \sqrt{L_{2M,T}}\right)\sqrt{\frac{L_{2M,T}}{n}} \\
&\quad = B_{i,c}^{\mathrm{stat}}(m),
\end{aligned}$$

where the last equality uses

$$c_{\mathrm{stat},T} := \frac{8\bar{c}_{\mathrm{per}}}{p_0}\left(1+\sqrt{L_{2M,T}}\right).$$

This proves the simultaneous claim.

□

Lemma 10 *Under Assumptions 1, 2, 3, and 4, there exists a finite constant L_{impl}, depending only on the model primitives, such that for every $S \in [0,\bar{S}]$ and every $\nu,\nu' \in [\underline{\mu},\bar{\mu}]$,*

$$|g(S;\nu) - g(S;\nu')| \le L_{\mathrm{impl}}|\nu-\nu'|.$$

Equivalently,

$$|\gamma(S;\nu) - \gamma(S;\nu')| \le L_{\mathrm{impl}}|\nu-\nu'|.$$

Proof. Fix $S \in [0,\bar{S}]$ and $\nu,\nu' \in [\underline{\mu},\bar{\mu}]$. The comparison is made over one generic regenerative cycle. Couple the two systems with the same exogenous sequence $(D_t,\rho_t)_{t\ge1}$, and start both systems from a stockout state:

$$X_1^{\nu} = X_1^{\nu'} = 0, \qquad \hat{X}_1^{\nu} = \hat{X}_1^{\nu'} = 0.$$

All time indices in this proof are local to this fresh cycle. Let

$$\Delta_\nu := |\nu-\nu'|.$$

Define the two cycle lengths by

$$\Lambda^{\nu} := \inf\{t\ge1 : Z_t^{\nu}=1\}, \qquad \Lambda^{\nu'} := \inf\{t\ge1 : Z_t^{\nu'}=1\}.$$

Define the first stockout-pattern mismatch by

$$M := \inf\left\{t\ge1 : \Lambda^{\nu}\ge t,\ \Lambda^{\nu'}\ge t,\ Z_t^{\nu}\ne Z_t^{\nu'}\right\},$$

with $M=\infty$ if no such period occurs. For $t\ge1$, define the event that both coupled cycles are alive at the start of local period t:

$$\mathcal{H}_t := \{\Lambda^{\nu}\ge t,\ \Lambda^{\nu'}\ge t\}. \tag{84}$$

On $\mathcal{H}_t$, neither system has stocked out in periods $1,\dots,t-1$.

We first record pathwise sensitivity bounds. For every $r\le t$, on $\mathcal{H}_t$,

$$|\hat{X}_r^{\nu} - \hat{X}_r^{\nu'}| \le S\Delta_\nu, \qquad |q_r^{\nu} - q_r^{\nu'}| \le S\Delta_\nu, \tag{85}$$

and

$$|X_r^{\nu} - X_r^{\nu'}| \le (r-1)S\Delta_\nu, \qquad |A_r^{\nu} - A_r^{\nu'}| \le rS\Delta_\nu. \tag{86}$$

Indeed, within either cycle the tracker remains in $[0,S]$, so

$$q_r = S - \hat{X}_r.$$

On $\mathcal{H}_t$, neither system stocks out in periods $1,\dots,t-1$. Therefore, for every $r<t$, sales equal demand in both systems and

$$\hat{X}_{r+1}^{\nu} = (\nu S - D_r)^+, \qquad \hat{X}_{r+1}^{\nu'} = (\nu' S - D_r)^+.$$

Since $x \mapsto x^+$ is one-Lipschitz,

$$|\hat{X}^{\nu}_{r+1} - \hat{X}^{\nu'}_{r+1}| \le S\Delta_\nu.$$

The order bound in (85) follows. Moreover, for every $r < t$,

$$X^{\nu}_{r+1} = \rho_r(X^{\nu}_r + q^{\nu}_r) - D_r, \qquad X^{\nu'}_{r+1} = \rho_r(X^{\nu'}_r + q^{\nu'}_r) - D_r.$$

Using $\rho_r \le 1$,

$$|X^{\nu}_{r+1} - X^{\nu'}_{r+1}| \le |X^{\nu}_r - X^{\nu'}_r| + |q^{\nu}_r - q^{\nu'}_r|.$$

Starting from $X^{\nu}_1 = X^{\nu'}_1 = 0$, induction gives the true-inventory bound in (86). Finally,

$$A^{\nu}_r = \rho_r(X^{\nu}_r + q^{\nu}_r), \qquad A^{\nu'}_r = \rho_r(X^{\nu'}_r + q^{\nu'}_r),$$

so the usable-inventory bound follows from the true-inventory and order bounds.

If a stockout-pattern mismatch occurs in period t, then both cycles are alive at the start of that period, and the realized demand lies between the two usable-inventory thresholds. Let $\mathcal{F}^-_t$ denote the full pre-outcome filtration of this coupled construction. Using Assumption 4 and (86),

$$\begin{aligned}\mathbb{P}(M = t) &\le \mathbb{E}\Big[\mathbf{1}_{\mathcal{H}_t}\mathbb{P}\Big(D_t \in [A^{\nu}_t \wedge A^{\nu'}_t,\, A^{\nu}_t \vee A^{\nu'}_t] \,|\, \mathcal{F}^-_t, \rho_t\Big)\Big] \\ &\le \bar{f}_D\, tS\Delta_\nu\, \mathbb{P}(\mathcal{H}_t). \end{aligned} \tag{87}$$

Because $\mathcal{H}_t \subseteq \{\Lambda^\nu \ge t\}$, the geometric episode tail in Lemma 2 gives

$$\mathbb{P}(\mathcal{H}_t) \le (1 - p_0)^{t-1}. \tag{88}$$

Combining (87) and (88),

$$\mathbb{P}(M = t) \le \bar{f}_D\, tS\Delta_\nu (1 - p_0)^{t-1}. \tag{89}$$

Consequently,

$$\mathbb{P}(M < \infty) \le \frac{\bar{f}_D S}{p_0^2}\Delta_\nu \le \frac{\bar{f}_D \bar{S}}{p_0^2}\Delta_\nu. \tag{90}$$

Let

$$\alpha_{\max} := \alpha(\bar{\mu}), \qquad \bar{c}_{\text{per}} := \alpha_{\max}\bar{S} + (\alpha_{\max} + h + b)\overline{D},$$

and define the one-period pseudo-cost increments

$$k^{\nu}_t := \alpha(\mu_\rho)(q^{\nu}_t - Y^{\nu}_t) - (h + b)Y^{\nu}_t,$$

$$k^{\nu'}_t := \alpha(\mu_\rho)(q^{\nu'}_t - Y^{\nu'}_t) - (h + b)Y^{\nu'}_t.$$

The corresponding cycle pseudo-costs are

$$\widetilde{K}^{S,\nu}(\mu_\rho) := \sum_{t=1}^{\Lambda^\nu} k^{\nu}_t, \qquad \widetilde{K}^{S,\nu'}(\mu_\rho) := \sum_{t=1}^{\Lambda^{\nu'}} k^{\nu'}_t.$$

The sales map is one-Lipschitz in usable inventory. For every common demand realization D_t,

$$|Y^{\nu}_t - Y^{\nu'}_t| \le |A^{\nu}_t - A^{\nu'}_t|. \tag{91}$$

Combining (91) with (85) and (86), on $\mathcal{H}_t$,

$$\begin{aligned}|k_t^\nu - k_t^{\nu'}| &\le \alpha_{\max}|q_t^\nu - q_t^{\nu'}| + (\alpha_{\max}+h+b)|Y_t^\nu - Y_t^{\nu'}| \\ &\le \alpha_{\max} S\Delta_\nu + (\alpha_{\max}+h+b)tS\Delta_\nu \\ &\le c_1 t\Delta_\nu, \end{aligned} \tag{92}$$

where

$$c_1 := (2\alpha_{\max}+h+b)\bar{S}.$$

Therefore

$$\begin{aligned}\mathbb{E}\left[\sum_{t\ge1}|k_t^\nu - k_t^{\nu'}|\mathbf{1}_{\mathcal{H}_t}\right] &\le \sum_{t\ge1} c_1 t\Delta_\nu(1-p_0)^{t-1} \\ &= \frac{c_1}{p_0^2}\Delta_\nu. \end{aligned} \tag{93}$$

To control the portion after a mismatch, observe the pathwise inequality

$$\begin{aligned}&\left|\widetilde{K}^{S,\nu}(\mu_\rho) - \widetilde{K}^{S,\nu'}(\mu_\rho)\right| \\ &\le \sum_{t\ge1}|k_t^\nu - k_t^{\nu'}|\mathbf{1}_{\mathcal{H}_t} + \mathbf{1}_{\{M<\infty\}}\left(\sum_{t=M+1}^{\Lambda^\nu}|k_t^\nu| + \sum_{t=M+1}^{\Lambda^{\nu'}}|k_t^{\nu'}|\right),\end{aligned}$$

where an empty sum is interpreted as zero. Each one-period pseudo-cost is bounded in absolute value by $\bar{c}_{\mathrm{per}}$. Conditional on the mismatch history, the expected number of additional periods in either still-alive cycle is at most $1/p_0$. Hence, by (90),

$$\mathbb{E}[\text{post-mismatch pseudo-cost contribution}] \le \frac{2\bar{c}_{\mathrm{per}}}{p_0}\mathbb{P}(M<\infty) \le \frac{2\bar{c}_{\mathrm{per}}\bar{f}_D\bar{S}}{p_0^3}\Delta_\nu. \tag{94}$$

Combining (93) and (94),

$$\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)] - \mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \le L_K\Delta_\nu, \tag{95}$$

where one may take

$$L_K := \frac{c_1}{p_0^2} + \frac{2\bar{c}_{\mathrm{per}}\bar{f}_D\bar{S}}{p_0^3}. \tag{96}$$

The cycle lengths can differ only if a mismatch occurs. Pathwise,

$$|\Lambda^\nu - \Lambda^{\nu'}| \le \mathbf{1}_{\{M<\infty\}}\left[(\Lambda^\nu - M)^+ + (\Lambda^{\nu'} - M)^+\right].$$

Conditional on the mismatch history, each residual term has conditional expectation at most $1/p_0$. Therefore

$$\left|\mathbb{E}[\Lambda^\nu] - \mathbb{E}[\Lambda^{\nu'}]\right| \le \frac{2}{p_0}\mathbb{P}(M<\infty) \le \frac{2\bar{f}_D\bar{S}}{p_0^3}\Delta_\nu =: L_\Lambda\Delta_\nu. \tag{97}$$

The pseudo-cost rates are

$$\gamma(S;\nu) = \frac{\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda^\nu]}, \qquad \gamma(S;\nu') = \frac{\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]}{\mathbb{E}[\Lambda^{\nu'}]}.$$

Every cycle lasts at least one period, so

$$\mathbb{E}[\Lambda^\nu] \ge 1, \qquad \mathbb{E}[\Lambda^{\nu'}] \ge 1.$$

Moreover,

$$\left|\widetilde{K}^{S,\nu}(\mu_\rho)\right| \le \bar{c}_{\mathrm{per}}\Lambda^{\nu}, \qquad \left|\widetilde{K}^{S,\nu'}(\mu_\rho)\right| \le \bar{c}_{\mathrm{per}}\Lambda^{\nu'},$$

and geometric domination gives

$$\mathbb{E}[\Lambda^{\nu}] \le \frac{1}{p_0}, \qquad \mathbb{E}[\Lambda^{\nu'}] \le \frac{1}{p_0}.$$

Consequently,

$$\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]\right| \le \frac{\bar{c}_{\mathrm{per}}}{p_0}, \qquad \left|\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \le \frac{\bar{c}_{\mathrm{per}}}{p_0}. \tag{98}$$

Using (95)–(98),

$$\begin{aligned}|\gamma(S;\nu)-\gamma(S;\nu')| &\le \left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]-\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right| \\ &\quad + \left|\mathbb{E}[\widetilde{K}^{S,\nu'}(\mu_\rho)]\right|\left|\frac{1}{\mathbb{E}[\Lambda^{\nu}]}-\frac{1}{\mathbb{E}[\Lambda^{\nu'}]}\right| \\ &\le L_K\Delta_\nu + \frac{\bar{c}_{\mathrm{per}}}{p_0}L_\Lambda\Delta_\nu. \end{aligned} \tag{99}$$

Thus $\gamma(S;\nu)$ is Lipschitz in ν. Since

$$g(S;\nu) = b\mathbb{E}[D_t] + \gamma(S;\nu),$$

the same Lipschitz bound holds for $g(S;\nu)$. Taking

$$L_{\mathrm{impl}} := L_K + \frac{\bar{c}_{\mathrm{per}}}{p_0}L_\Lambda$$

completes the proof.

□

C.4. Policy-arm confidence and optimism

Lemma 11 *For a fixed grid, let*

$$(i^\star, c^\star) \in \operatorname*{arg\,min}_{\substack{1\le i\le M\\ c\in\bar{\mathcal{C}}_{\mathrm{tr}}}} g(s_i;\nu_c), \qquad g^\star_{\cup,M} := g(s_{i^\star};\nu_{c^\star}).$$

Set, by convention,

$$\mathcal{C}_\mu(0) := [\underline{\mu}, \bar{\mu}], \qquad r_\mu(0) := \frac{\bar{\mu}-\underline{\mu}}{2}.$$

There exists an event $\mathcal{E}_T$ with probability at least $1-6T^{-4}$ such that, on $\mathcal{E}_T$, the following statements hold.

(i) For every $0 \le m \le T$, $\mu_\rho \in \mathcal{C}_\mu(m)$.

(ii) For every $m \le T$ and every policy arm (i,c) with $N_{i,c}(m) \ge 1$,

$$|\hat{\gamma}_{i,c}(m) - \gamma(s_i;\nu_c)| \le B_{i,c}(m).$$

(iii) The grid-optimal policy arm $(i^\star, c^\star)$ is never eliminated.

(iv) Whenever episode $m+1$ is a decision episode and the chosen policy arm (i_{m+1}, c_{m+1}) has already been sampled,

$$g(s_{i_{m+1}};\nu_{c_{m+1}}) - g^\star_{\cup,M} \le 2B_{i_{m+1},c_{m+1}}(m).$$

Proof. Let $\mathcal{E}_T$ be the intersection of the Kaplan–Meier, bracketing, audit, and policy-arm cycle-concentration events in Lemmas 8, 3, 4, and 9. The displayed failure bounds in those lemmas and a union bound give

$$\mathbb{P}(\mathcal{E}_T^c) \le 6T^{-4}.$$

On this event the three mean-retention confidence intervals contain μ_ρ, so

$$|\hat{\mu}_m - \mu_\rho| \le r_\mu(m), \qquad 0 \le m \le T.$$

This proves part (i).

Fix $m \le T$ and a policy arm (i,c) with $N_{i,c}(m) \ge 1$. Write

$$\gamma_{i,c} := \gamma(s_i;\nu_c), \qquad \mathcal{T}_{i,c}(m) := \sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell.$$

The exact error decomposition is

$$\hat{\gamma}_{i,c}(m) - \gamma_{i,c} = A_{i,c}^{\mathrm{stat}}(m) + A_{i,c}^{\mu,\mathrm{eval}}(m) + A_{i,c}^{\mu,\mathrm{impl}}(m),$$

where

$$A_{i,c}^{\mathrm{stat}}(m) := \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \left\{ \widetilde{K}_\ell^{\mathrm{obs}}(\mu_\rho) - \gamma(s_i;\nu_{c,\ell-1})\Lambda_\ell \right\}}{\mathcal{T}_{i,c}(m)},$$

$$A_{i,c}^{\mu,\mathrm{eval}}(m) := \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} [\alpha(\hat{\mu}_m) - \alpha(\mu_\rho)] W_\ell}{\mathcal{T}_{i,c}(m)},$$

and

$$A_{i,c}^{\mu,\mathrm{impl}}(m) := \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} [\gamma(s_i;\nu_{c,\ell-1}) - \gamma(s_i;\nu_c)] \Lambda_\ell}{\mathcal{T}_{i,c}(m)}.$$

Lemma 9 gives

$$|A_{i,c}^{\mathrm{stat}}(m)| \le B_{i,c}^{\mathrm{stat}}(m).$$

Since $0 \le W_\ell \le Q_\ell \le \bar{S}\Lambda_\ell$,

$$|A_{i,c}^{\mu,\mathrm{eval}}(m)| \le L_\alpha \bar{S} r_\mu(m) = B_{i,c}^{\mu,\mathrm{eval}}(m).$$

For BI, $\nu_{\mathrm{BI},\ell-1} = \nu_{\mathrm{BI}} = 1$, and hence $A_{i,\mathrm{BI}}^{\mu,\mathrm{impl}}(m) = 0$. For EI, Lemma 10 gives

$$|\gamma(s_i;\nu_{\mathrm{EI},\ell-1}) - \gamma(s_i;\nu_{\mathrm{EI}})| \le L_{\mathrm{impl}} |\hat{\mu}_{\ell-1} - \mu_\rho| \le L_{\mathrm{impl}} r_\mu(\ell-1).$$

Consequently, for either class,

$$|A_{i,c}^{\mu,\mathrm{impl}}(m)| \le \mathbf{1}\{c = \mathrm{EI}\} L_{\mathrm{impl}} \frac{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell r_\mu(\ell-1)}{\sum_{\ell \in \mathcal{I}_{i,c}(m)} \Lambda_\ell} = B_{i,c}^{\mu,\mathrm{impl}}(m).$$

Adding the three bounds proves part (ii).

If $N_{i^\star,c^\star}(m) = 0$, then $\mathrm{LCB}_{i^\star,c^\star}(m) = -\infty$, so the benchmark optimizer cannot be eliminated. Otherwise, part (ii) gives

$$\mathrm{LCB}_{i^\star,c^\star}(m) \le \gamma(s_{i^\star};\nu_{c^\star}).$$

For every active policy arm (j,d), either it is unsampled and $\mathrm{UCB}_{j,d}(m) = +\infty$, or part (ii) gives

$$\mathrm{UCB}_{j,d}(m) \ge \gamma(s_j;\nu_d).$$

The pseudo-cost identity and optimality of $(i^\star, c^\star)$ imply

$$\gamma(s_{i^\star};\nu_{c^\star}) \le \gamma(s_j;\nu_d).$$

Thus the elimination rule never removes $(i^\star, c^\star)$, proving part (iii).

Finally, suppose (i_{m+1}, c_{m+1}) has already been sampled. The initialization rule then implies that every active policy arm has been sampled; in particular, $N_{i^\star,c^\star}(m) \ge 1$. Optimism and part (ii) give

$$\begin{aligned}\gamma(s_{i_{m+1}};\nu_{c_{m+1}}) &\le \mathrm{UCB}_{i_{m+1},c_{m+1}}(m)\\ &= \mathrm{LCB}_{i_{m+1},c_{m+1}}(m) + 2B_{i_{m+1},c_{m+1}}(m)\\ &\le \mathrm{LCB}_{i^\star,c^\star}(m) + 2B_{i_{m+1},c_{m+1}}(m)\\ &\le \gamma(s_{i^\star};\nu_{c^\star}) + 2B_{i_{m+1},c_{m+1}}(m).\end{aligned}$$

Since $b\mathbb{E}[D_t]$ is common to every policy arm, this is exactly the claim in part (iv).

□

C.5. Base-stock sensitivity and discretization

Lemma 12 *Under Assumptions 1, 3, and 4, there exists a finite constant L_S, depending only on the model primitives, such that for every tracker parameter $\nu \in [\underline{\mu}, \bar{\mu}] \cup \{1\}$ and every $S, S' \in [0, \bar{S}]$,*

$$|g(S;\nu) - g(S';\nu)| \le L_S |S - S'|.$$

In particular, for every inventory-tracking mode $c \in \mathcal{C}_{\mathrm{tr}}$,

$$|g(S;\nu_c) - g(S';\nu_c)| \le L_S |S - S'|.$$

Proof. Fix $\nu \in [\underline{\mu}, \bar{\mu}] \cup \{1\}$ and $S, S' \in [0, \bar{S}]$. Couple the two systems with the same exogenous sequence $(D_t, \rho_t)_{t\ge1}$, and start both cycles from

$$X_1^S = X_1^{S'} = 0, \qquad \hat{X}_1^S = \hat{X}_1^{S'} = 0.$$

Write

$$\Delta_S := |S - S'|.$$

Let

$$\Lambda^{S,\nu} := \inf\{t \ge 1 : Z_t^S = 1\}, \qquad \Lambda^{S',\nu} := \inf\{t \ge 1 : Z_t^{S'} = 1\}$$

be the two coupled cycle lengths, and let

$$M := \inf\left\{t \ge 1 : \Lambda^{S,\nu} \ge t,\ \Lambda^{S',\nu} \ge t,\ Z_t^S \ne Z_t^{S'}\right\}$$

be the first period in which the two coupled systems disagree on the stockout indicator. If no such disagreement occurs before both cycles end, set $M = \infty$. For $t \ge 1$, define

$$\mathcal{H}_t := \{\Lambda^{S,\nu} \ge t,\ \Lambda^{S',\nu} \ge t,\ M \ge t\}. \tag{100}$$

Thus, on $\mathcal{H}_t$, both systems are alive at the start of period t, and their stockout histories agree through period $t-1$.

We first record the pathwise sensitivity of the coupled systems on $\mathcal{H}_t$. For every $r \le t$, on $\mathcal{H}_t$,

$$|\hat{X}_r^S - \hat{X}_r^{S'}| \le \Delta_S, \qquad |q_r^S - q_r^{S'}| \le 2\Delta_S, \tag{101}$$

and

$$|X_r^S - X_r^{S'}| \le 2(r-1)\Delta_S, \qquad |A_r^S - A_r^{S'}| \le 2r\Delta_S. \tag{102}$$

Indeed, the tracker always satisfies $\hat{X}_r^S \le S$ and $\hat{X}_r^{S'} \le S'$ within a cycle, so

$$q_r^S = S - \hat{X}_r^S, \qquad q_r^{S'} = S' - \hat{X}_r^{S'}.$$

On $\mathcal{H}_t$, the stockout histories coincide before period t. Hence in every preceding non-stockout period $r < t$,

$$\hat{X}_{r+1}^S = (\nu S - D_r)^+, \qquad \hat{X}_{r+1}^{S'} = (\nu S' - D_r)^+.$$

Since $x \mapsto x^+$ is one-Lipschitz and $\nu \le 1$,

$$|\hat{X}_{r+1}^S - \hat{X}_{r+1}^{S'}| \le \nu|S - S'| \le \Delta_S.$$

The order bound in (101) follows from $q_r = S - \hat{X}_r$. For the true inventories, whenever both coupled cycles are alive and no stockout occurs,

$$X_{r+1}^S = \rho_r(X_r^S + q_r^S) - D_r, \qquad X_{r+1}^{S'} = \rho_r(X_r^{S'} + q_r^{S'}) - D_r.$$

Therefore, using $\rho_r \le 1$,

$$|X_{r+1}^S - X_{r+1}^{S'}| \le |X_r^S - X_r^{S'}| + |q_r^S - q_r^{S'}|.$$

Starting from $X_1^S = X_1^{S'} = 0$ and using (101), induction gives the true-inventory bound in (102). Since

$$A_r^S = \rho_r(X_r^S + q_r^S), \qquad A_r^{S'} = \rho_r(X_r^{S'} + q_r^{S'}),$$

the bound on $|A_r^S - A_r^{S'}|$ follows as well.

A stockout-pattern mismatch in period t can occur only if both systems are alive at the start of period t, their stockout histories agree before that period, and D_t falls between the two usable-inventory thresholds. Let $\mathcal{F}_t^-$ denote the full pre-outcome filtration of this coupled construction. Hence, using Assumption 4 and (102),

$$\begin{aligned}\mathbb{P}(M = t) &\le \mathbb{E}\Big[\mathbf{1}_{\mathcal{H}_t}\mathbb{P}\Big(D_t \in [A_t^S \wedge A_t^{S'},\, A_t^S \vee A_t^{S'}] \mid \mathcal{F}_t^-, \rho_t\Big)\Big] \\ &\le 2\bar{f}_D\, t\Delta_S\, \mathbb{P}(\mathcal{H}_t).\end{aligned} \tag{103}$$

Because $\mathcal{H}_t \subseteq \{\Lambda^{S,\nu} \ge t\}$, the geometric episode tail in Lemma 2 gives

$$\mathbb{P}(\mathcal{H}_t) \le (1 - p_0)^{t-1}. \tag{104}$$

Combining (103) and (104) gives

$$\mathbb{P}(M = t) \le 2\bar{f}_D\, t\Delta_S(1 - p_0)^{t-1}. \tag{105}$$

Therefore

$$\mathbb{P}(M < \infty) \le \sum_{t \ge 1} 2\bar{f}_D\, t\Delta_S(1 - p_0)^{t-1} = \frac{2\bar{f}_D}{p_0^2}\Delta_S. \tag{106}$$

Let

$$\alpha_{\max} := \alpha(\bar{\mu}), \qquad \bar{c}_{\mathrm{per}} := \alpha_{\max}\bar{S} + (\alpha_{\max} + h + b)\overline{D},$$

and define the one-period pseudo-cost increment under base-stock level S by

$$k_t^S := \alpha(\mu_\rho)(q_t^S - Y_t^S) - (h+b)Y_t^S, \qquad k_t^{S'} := \alpha(\mu_\rho)(q_t^{S'} - Y_t^{S'}) - (h+b)Y_t^{S'}.$$

Let

$$\widetilde{K}^{S,\nu}(\mu_\rho) := \sum_{t=1}^{\Lambda^{S,\nu}} k_t^S, \qquad \widetilde{K}^{S',\nu}(\mu_\rho) := \sum_{t=1}^{\Lambda^{S',\nu}} k_t^{S'}$$

be the two cycle pseudo-costs under tracker parameter ν. Set

$$c_1 := 2(2\alpha_{\max} + h + b).$$

On the event $\mathcal{H}_t$, both coupled cycles are alive at the start of period t. Since the sales map is one-Lipschitz in usable inventory,

$$|Y_t^S - Y_t^{S'}| \le |A_t^S - A_t^{S'}|.$$

Using (101) and (102), we obtain

$$\begin{aligned} |k_t^S - k_t^{S'}| &\le \alpha_{\max}|q_t^S - q_t^{S'}| + (\alpha_{\max} + h + b)|Y_t^S - Y_t^{S'}| \\ &\le 2\alpha_{\max}\Delta_S + 2(\alpha_{\max} + h + b)t\Delta_S \\ &\le c_1 t\Delta_S. \end{aligned} \tag{107}$$

The expected contribution through the first mismatch is therefore bounded by

$$\begin{aligned} \mathbb{E}\left[\sum_{t\ge 1} |k_t^S - k_t^{S'}|\,\mathbf{1}_{\mathcal{H}_t}\right] &\le \sum_{t\ge 1} c_1 t\Delta_S (1-p_0)^{t-1} \\ &= \frac{c_1}{p_0^2}\Delta_S. \end{aligned} \tag{108}$$

After the first mismatch, each one-period pseudo-cost is bounded in absolute value by $\bar{c}_{\mathrm{per}}$, and conditional on a mismatch the expected remaining length of each coupled cycle is at most $1/p_0$. Hence

$$\mathbb{E}[\text{post-mismatch pseudo-cost difference}] \le \frac{2\bar{c}_{\mathrm{per}}}{p_0}\mathbb{P}(M < \infty) \le \frac{4\bar{c}_{\mathrm{per}}\bar{f}_D}{p_0^3}\Delta_S. \tag{109}$$

Equations (108) and (109) imply

$$\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)] - \mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right| \le L_K^S\Delta_S, \tag{110}$$

where one may take

$$L_K^S := \frac{c_1}{p_0^2} + \frac{4\bar{c}_{\mathrm{per}}\bar{f}_D}{p_0^3}. \tag{111}$$

The expected cycle lengths are also Lipschitz. If no stockout-pattern mismatch occurs, then the two coupled cycles have the same length. Conditional on a mismatch, the expected remaining length of each coupled cycle is at most $1/p_0$. Thus (106) gives

$$\left|\mathbb{E}[\Lambda^{S,\nu}] - \mathbb{E}[\Lambda^{S',\nu}]\right| \le \frac{2}{p_0}\mathbb{P}(M < \infty) \le \frac{4\bar{f}_D}{p_0^3}\Delta_S =: L_\Lambda^S\Delta_S. \tag{112}$$

The pseudo-cost rates are

$$\gamma(S;\nu)=\frac{\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda^{S,\nu}]},\qquad \gamma(S';\nu)=\frac{\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]}{\mathbb{E}[\Lambda^{S',\nu}]}.$$

Since every cycle lasts at least one period,

$$\mathbb{E}[\Lambda^{S,\nu}]\ge 1,\qquad \mathbb{E}[\Lambda^{S',\nu}]\ge 1.$$

Moreover,

$$\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]\right|\le\frac{\bar{c}_{\mathrm{per}}}{p_0},\qquad \left|\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right|\le\frac{\bar{c}_{\mathrm{per}}}{p_0}. \tag{113}$$

Using (110)–(113),

$$\begin{aligned}|\gamma(S;\nu)-\gamma(S';\nu)|&\le\left|\mathbb{E}[\widetilde{K}^{S,\nu}(\mu_\rho)]-\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right|\\&\quad+\left|\mathbb{E}[\widetilde{K}^{S',\nu}(\mu_\rho)]\right|\left|\frac{1}{\mathbb{E}[\Lambda^{S,\nu}]}-\frac{1}{\mathbb{E}[\Lambda^{S',\nu}]}\right|\\&\le L_K^S\Delta_S+\frac{\bar{c}_{\mathrm{per}}}{p_0}L_\Lambda^S\Delta_S.\end{aligned} \tag{114}$$

Thus $\gamma(S;\nu)$ is Lipschitz in S, uniformly over $\nu\in[\underline{\mu},\bar{\mu}]\cup\{1\}$. Since

$$g(S;\nu)=b\mathbb{E}[D_t]+\gamma(S;\nu),$$

the same Lipschitz bound holds for $g(S;\nu)$. Taking

$$L_S:=L_K^S+\frac{\bar{c}_{\mathrm{per}}}{p_0}L_\Lambda^S$$

proves the first claim. Since $\nu_{\mathrm{EI}}=\mu_\rho\in[\underline{\mu},\bar{\mu}]$ and $\nu_{\mathrm{BI}}=1$, setting $\nu=\nu_c$ gives, for each $c\in\mathcal{C}_{\mathrm{tr}}$,

$$|g(S;\nu_c)-g(S';\nu_c)|\le L_S|S-S'|.$$

□

C.6. Pathwise summation lemmas

Audit-count and estimation-error terms. The mean-retention-estimation part of the regret proof tracks two pathwise quantities:

$$A_T^{\mathrm{init}}\qquad\text{and}\qquad \mathcal{U}_T^{\mu}=\sum_{\ell\in\mathcal{D}_T^{\mathrm{init}}}r_\mu(\ell-1).$$

The first is the number of audit episodes initiated by the horizon; it is multiplied later by the audit-episode regret coefficient $\bar{C}_{\mathrm{aud}}$. The second is the cumulative mean-retention confidence radius at decision-episode starts; it is multiplied by the aggregate estimation-sensitivity coefficient $C_{\mathrm{sh,T}}$ and controls pseudo-cost evaluation error and tracker-parameter error. The next two lemmas bound these pathwise quantities separately and link them to cumulative regret through $\bar{C}_{\mathrm{aud}}$ and $C_{\mathrm{sh,T}}$.

Lemma 13 *Suppose the algorithm uses the threshold schedule*

$$a_m:=\min\left\{1,\vartheta_T\left(\frac{L_T}{m}\right)^{1/3}\right\},\qquad m\ge 1.$$

On every sample path, the number A_T^{init} of audit episodes initiated by time T satisfies

$$A_T^{\mathrm{init}}\le 1+\mathbf{1}\{a_T<1\}\min\left\{T-1,\frac{L_T}{2a_T^2}\right\}\le 1+\frac{L_T^{1/3}T^{2/3}}{2\vartheta_T^2}. \tag{115}$$

Proof. The mandatory calibration audit accounts for the additive one. Because $\mathcal{C}_\mu(m) \subseteq [\underline{\mu}, \bar{\mu}]$ on every sample path,

$$r_\mu(m) \le r_{\max} := \frac{\bar{\mu} - \underline{\mu}}{2} < 1.$$

If $a_T = 1$, monotonicity gives $a_{m+1} \ge a_T = 1$ at every episode boundary reached by the horizon. The condition $r_\mu(m) > a_{m+1}$ can therefore never hold, so no additional audit is initiated.

Suppose instead that $a_T < 1$ and consider an additional audit launched at episode boundary m, after $p \ge 1$ audits have been completed. The fallback construction of the combined confidence interval guarantees

$$\mathcal{C}_\mu(m) \subseteq \mathcal{C}_{\mathrm{aud}}(m).$$

The audit-only radius and the definition of L_T therefore give

$$r_\mu(m) \le r_{\mathrm{aud}}(m) \le \sqrt{\frac{L_T}{2p}}.$$

An additional audit is initiated only if $r_\mu(m) > a_{m+1}$. Since $m+1 \le T$ and (a_m) is nonincreasing,

$$\sqrt{\frac{L_T}{2p}} > a_{m+1} \ge a_T,$$

which implies $p < L_T/(2a_T^2)$. Applying this inequality to the last additional audit and observing that at most $T-1$ audits can follow the mandatory first audit proves the first inequality in (115). Finally, when $a_T < 1$,

$$a_T = \vartheta_T \left(\frac{L_T}{T}\right)^{1/3}, \qquad \frac{L_T}{2a_T^2} = \frac{L_T^{1/3} T^{2/3}}{2\vartheta_T^2}.$$

When $a_T = 1$, the additional-audit term is zero. This proves the second inequality and the lemma.

Lemma 14 *Under the schedule in Lemma 13, every decision episode $\ell \in \mathcal{D}_T^{\mathrm{init}}$ satisfies*

$$r_\mu(\ell - 1) \le a_\ell.$$

Consequently, along every sample path,

$$\mathcal{U}_T^\mu = \sum_{\ell \in \mathcal{D}_T^{\mathrm{init}}} r_\mu(\ell - 1) \le \frac{3}{2} \vartheta_T L_T^{1/3} \big(T^{2/3} - 1\big) \le \frac{3}{2} \vartheta_T L_T^{1/3} T^{2/3}. \tag{116}$$

Proof. Episode ℓ is a decision episode only if the audit condition fails at its start, which is precisely $r_\mu(\ell - 1) \le a_\ell$. The mandatory calibration audit is episode one, and at most T episodes can be initiated by the period horizon. Hence

$$\mathcal{U}_T^\mu \le \sum_{\ell=2}^{T} a_\ell \le \vartheta_T L_T^{1/3} \sum_{\ell=2}^{T} \ell^{-1/3}.$$

Since $x^{-1/3}$ is decreasing,

$$\sum_{\ell=2}^{T} \ell^{-1/3} \le \int_1^T x^{-1/3}\, dx = \frac{3}{2}\big(T^{2/3} - 1\big).$$

Substitution proves (116).

Lemma 15 *Along every sample path,*

$$\sum_{\substack{m \in \mathcal{D}_T^{\mathrm{init}}: \\ N_{i_m, c_m}(m-1) \ge 1}} B^{\mathrm{stat}}_{i_m, c_m}(m-1) \le 2 c_{\mathrm{stat},T} \sqrt{2MTL_{2M,T}},$$

where (i_m, c_m) is the policy arm selected in decision episode m.

Proof. For each policy arm (i,c), let $N_{i,c}^T$ be the number of its decision episodes initiated by horizon T. For the nth initiated decision episode of (i,c), all of its $n-1$ predecessors are complete, so the policy arm has been sampled before exactly when $n \geq 2$, and its statistical radius is at most

$$c_{\mathrm{stat},T}\sqrt{\frac{L_{2M,T}}{n-1}}.$$

Therefore

$$\sum_{\substack{m\in\mathcal{D}_T^{\mathrm{init}}:(i_m,c_m)=(i,c),\\ N_{i_m,c_m}(m-1)\geq 1}} B^{\mathrm{stat}}_{i_m,c_m}(m-1) \leq c_{\mathrm{stat},T}\sqrt{L_{2M,T}}\sum_{r=1}^{N_{i,c}^T-1} r^{-1/2} \leq 2c_{\mathrm{stat},T}\sqrt{L_{2M,T}N_{i,c}^T},$$

where the sum is zero if $N_{i,c}^T \leq 1$. Summing over the $2M$ policy arms and using Cauchy–Schwarz gives

$$\sum_{i=1}^{M}\sum_{c\in\mathcal{C}_{\mathrm{tr}}}\sqrt{N_{i,c}^T} \leq \sqrt{2M\sum_{i=1}^{M}\sum_{c\in\mathcal{C}_{\mathrm{tr}}}N_{i,c}^T} \leq \sqrt{2MT},$$

which proves the bound.

Lemma 16 *On the event that all episodes initiated before horizon T have length at most $\bar{\Lambda}_T$,*

$$\sum_{\substack{m\in\mathcal{D}_T^{\mathrm{init}}:\\ N_{i_m,c_m}(m-1)\geq 1}} B^{\mu,\mathrm{impl}}_{i_m,c_m}(m-1) \leq L_{\mathrm{impl}}\bar{\Lambda}_T\log(eT)\sum_{m\in\mathcal{D}_T^{\mathrm{init}}} r_\mu(m-1).$$

Proof. The tracker-parameter radius is identically zero for every BI policy arm. Fix an EI policy arm (i,EI), and enumerate its decision episodes initiated by horizon T by

$$\ell_{i,\mathrm{EI},1} < \ell_{i,\mathrm{EI},2} < \cdots < \ell_{i,\mathrm{EI},N_{i,\mathrm{EI}}^T}.$$

For its nth initiated episode with $n \geq 2$, the preceding $n-1$ episodes are complete. The tracker-parameter radius computed before the nth episode therefore satisfies, using $\Lambda_{\ell_{i,\mathrm{EI},j}} \leq \bar{\Lambda}_T$ and $\Lambda_{\ell_{i,\mathrm{EI},j}} \geq 1$,

$$B^{\mu,\mathrm{impl}}_{i,\mathrm{EI}}(\ell_{i,\mathrm{EI},n}-1) \leq L_{\mathrm{impl}}\bar{\Lambda}_T\frac{\sum_{j=1}^{n-1} r_\mu(\ell_{i,\mathrm{EI},j}-1)}{n-1}.$$

Thus

$$\sum_{n=2}^{N_{i,\mathrm{EI}}^T} B^{\mu,\mathrm{impl}}_{i,\mathrm{EI}}(\ell_{i,\mathrm{EI},n}-1) \leq L_{\mathrm{impl}}\bar{\Lambda}_T\sum_{n=2}^{N_{i,\mathrm{EI}}^T}\frac{1}{n-1}\sum_{j=1}^{n-1} r_\mu(\ell_{i,\mathrm{EI},j}-1).$$

Interchanging sums gives

$$\sum_{n=2}^{N_{i,\mathrm{EI}}^T} B^{\mu,\mathrm{impl}}_{i,\mathrm{EI}}(\ell_{i,\mathrm{EI},n}-1) \leq L_{\mathrm{impl}}\bar{\Lambda}_T\log(eT)\sum_{j=1}^{N_{i,\mathrm{EI}}^T} r_\mu(\ell_{i,\mathrm{EI},j}-1).$$

Summing over the M EI policy arms and then enlarging the resulting nonnegative radius sum to all decision episodes yields the claimed bound.

C.7. Proof of Theorem 1

Proof. Let

$$(i^\star, c^\star) \in \operatorname*{arg\,min}_{\substack{1\le i\le M\\ c\in\bar{\mathcal{C}}_{\mathrm{tr}}}} g(s_i;\nu_c), \qquad g^\star_{\cup,M} := g(s_{i^\star};\nu_{c^\star}),$$

and define the finite-grid regret

$$R_T^{\mathrm{grid}} := \mathbb{E}\left[\sum_{t=1}^{T} C_t^{\mathrm{alg}}\right] - T g^\star_{\cup,M}.$$

We first control R_T^{grid} and then add the continuous-target discretization error.

To place the finite-horizon cost on complete regenerative cycles, index episodes by $\ell = 1, 2, \ldots$, with t_ℓ denoting the calendar period in which episode ℓ starts. For the analysis only, if the episode active at period T has not yet reached its terminal stockout, continue that episode under the policy already selected for it until the next stockout. No post-horizon observation is used by the algorithm. Let

$$\mathcal{J}_T := \{\ell : t_\ell \le T\}$$

be the set of episodes initiated by the horizon. For every $\ell \in \mathcal{J}_T$, let Λ_ℓ be its full length under this analysis completion, and let

$$\Gamma_\ell := \sum_{t=t_\ell}^{t_\ell+\Lambda_\ell-1} C_t$$

be its full operating cost. Define the corresponding full algorithmic episode cost by

$$\Gamma_\ell^{\mathrm{alg}} := \Gamma_\ell + \kappa_{\mathrm{aud}} \mathbf{1}\{E_\ell = \mathrm{aud}\}.$$

Set

$$H_T := \sum_{\ell\in\mathcal{J}_T} \Lambda_\ell, \qquad \Omega_T := H_T - T.$$

Thus H_T is the terminal stockout time of the episode containing the horizon and $\Omega_T \ge 0$ is the post-horizon overshoot.

All period costs are nonnegative, so

$$\begin{aligned} R_T^{\mathrm{grid}} &\le \mathbb{E}\left[\sum_{\ell\in\mathcal{J}_T} \Gamma_\ell^{\mathrm{alg}}\right] - T g^\star_{\cup,M} \\ &= \mathbb{E}\left[\sum_{\ell\in\mathcal{J}_T} \left(\Gamma_\ell^{\mathrm{alg}} - g^\star_{\cup,M}\Lambda_\ell\right)\right] + g^\star_{\cup,M}\mathbb{E}[\Omega_T]. \end{aligned} \tag{117}$$

Recall the uniform upper bound on expected operating cost,

$$\bar{C}_{\mathrm{op}} := \frac{h\bar{S}}{p_0^2} + \frac{b\overline{D}}{p_0},$$

and the upper bound on full audit-episode cost,

$$\bar{C}_{\mathrm{aud}} := \kappa_{\mathrm{aud}} + \bar{C}_{\mathrm{op}}.$$

The cycle representation of g and $\mathbb{E}[\Lambda] \ge 1$ imply

$$0 \le g^\star_{\cup,M} \le \bar{C}_{\mathrm{op}}.$$

Let $\mathcal{F}_T$ denote the full system history through period T. On $\{\Omega_T > 0\}$, the continuation order for period $T+1$ is predictable from this history. Conditioning first on the corresponding pre-outcome history $\mathcal{F}_{T+1}^-$, the residual-length bound in Lemma 2 and the tower property give

$$\mathbb{E}[\Omega_T \mid \mathcal{F}_T] \le \frac{1}{p_0}, \qquad \mathbb{E}[\Omega_T] \le \frac{1}{p_0}.$$

Therefore the final term in (117) is bounded by the primitive constant

$$C_{\mathrm{bd}} := \frac{\bar{C}_{\mathrm{op}}}{p_0}. \tag{118}$$

Let

$$\mathcal{D}_T^{\mathrm{init}} := \{\ell \in \mathcal{J}_T : E_\ell \in \{1, \dots, M\} \times \mathcal{C}_{\mathrm{tr}}\}$$

be the set of decision episodes initiated by the horizon, and let

$$A_T^{\mathrm{init}} := \sum_{\ell \in \mathcal{J}_T} \mathbf{1}\{E_\ell = \mathrm{aud}\}$$

be the number of initiated audit episodes. For an audit episode, $g^\star_{\cup,M} \ge 0$, and the bounded recovery rule gives

$$\mathbb{E}[\Gamma_\ell^{\mathrm{alg}} - g^\star_{\cup,M}\Lambda_\ell \mid \mathcal{G}_{\ell-1}] \le \mathbb{E}[\Gamma_\ell^{\mathrm{alg}} \mid \mathcal{G}_{\ell-1}] \le \bar{C}_{\mathrm{aud}}.$$

Because the event that episode ℓ is initiated by T, as well as its episode type, is measurable at its start, summing by conditional expectation gives

$$\mathbb{E}\left[\sum_{\ell \in \mathcal{J}_T : E_\ell = \mathrm{aud}} \left(\Gamma_\ell^{\mathrm{alg}} - g^\star_{\cup,M}\Lambda_\ell\right)\right] \le \bar{C}_{\mathrm{aud}} \mathbb{E}[A_T^{\mathrm{init}}]. \tag{119}$$

We next bound the contribution of the decision episodes initiated by period T. For $\ell \in \mathcal{D}_T^{\mathrm{init}}$, let (i_ℓ, c_ℓ) be the selected policy arm and write

$$r_\ell^- := r_\mu(\ell - 1).$$

The selected policy arm (i_ℓ, c_ℓ), its execution parameter $\nu_{c_\ell,\ell-1}$, the start time t_ℓ, and the event $\{t_\ell \le T\}$ are $\mathcal{G}_{\ell-1}$-measurable. Conditional on $\mathcal{G}_{\ell-1}$, the full episode is a fresh regenerative cycle of $\pi_{s_{i_\ell}}^{\nu_{c_\ell,\ell-1}}$. Hence

$$\begin{aligned}&\mathbb{E}\big[\Gamma_\ell - g^\star_{\cup,M}\Lambda_\ell \mid \mathcal{G}_{\ell-1}\big] \\ &\quad = \mathbb{E}[\Lambda_\ell \mid \mathcal{G}_{\ell-1}]\big(g(s_{i_\ell}; \nu_{c_\ell,\ell-1}) - g^\star_{\cup,M}\big).\end{aligned} \tag{120}$$

By Lemma 2,

$$\mathbb{E}[\Lambda_\ell \mid \mathcal{G}_{\ell-1}] \le \frac{1}{p_0}.$$

Let $\mathcal{E}_{\ell-1} \in \mathcal{G}_{\ell-1}$ be the prefix event that the mean-retention and policy-arm confidence conclusions used in Lemma 11 hold through completed episode $\ell - 1$. These prefix events are nested, and

$$\mathbb{P}(\mathcal{E}_{\ell-1}^c) \le 6T^{-4} \qquad \text{for every } \ell \le T.$$

The separate bounded-length event will be used only in the pathwise summation of tracker-parameter radii below.

On $\mathcal{E}_{\ell-1}$, BI has no current tracker-parameter error, while Lemma 10 controls the corresponding EI error. Thus

$$g(s_{i_\ell}; \nu_{c_\ell,\ell-1}) - g(s_{i_\ell}; \nu_{c_\ell}) \le \mathbf{1}\{c_\ell = \mathrm{EI}\} L_{\mathrm{impl}} r_\ell^-.$$

If policy arm (i_ℓ, c_ℓ) has not previously completed a decision cycle, then the pseudo-cost identity gives

$$g(s_{i_\ell};\nu_{c_\ell}) - g^\star_{\cup,M} = \gamma(s_{i_\ell};\nu_{c_\ell}) - \gamma(s_{i^\star};\nu_{c^\star}) \le 2\bar{c}_{\mathrm{per}}.$$

If the policy arm has previously been sampled, the optimism conclusion of Lemma 11 gives

$$g(s_{i_\ell};\nu_{c_\ell}) - g^\star_{\cup,M} \le 2B_{i_\ell,c_\ell}(\ell-1).$$

Combining these bounds with (120), on $\mathcal{E}_{\ell-1}$,

$$\begin{aligned}
&\mathbb{E}\big[\Gamma_\ell - g^\star_{\cup,M}\Lambda_\ell \,\big|\, \mathcal{G}_{\ell-1}\big] \\
&\qquad \le \frac{\mathbf{1}\{c_\ell = \mathrm{EI}\}L_{\mathrm{impl}}r_\ell^-}{p_0} + \begin{cases} \dfrac{2\bar{c}_{\mathrm{per}}}{p_0}, & N_{i_\ell,c_\ell}(\ell-1) = 0, \\ \dfrac{2B_{i_\ell,c_\ell}(\ell-1)}{p_0}, & N_{i_\ell,c_\ell}(\ell-1) \ge 1. \end{cases}
\end{aligned} \tag{121}$$

On $\mathcal{E}^c_{\ell-1}$, nonnegativity of $g^\star_{\cup,M}$ and the episode-cost bound implies the crude conditional bound

$$\mathbb{E}\big[\Gamma_\ell - g^\star_{\cup,M}\Lambda_\ell \,\big|\, \mathcal{G}_{\ell-1}\big] \le \bar{C}_{\mathrm{op}}. \tag{122}$$

Since an episode can be initiated only after the preceding episode has completed, the indicator that episode ℓ belongs to $\mathcal{D}_T^{\mathrm{init}}$ is measurable at its start. We may therefore multiply (121) and (122) by that indicator, take expectations, and sum over $\ell = 1, \dots, T$. Each policy arm can be selected at most once while unsampled. Together with (117) and (119), this yields

$$\begin{aligned}
R_T^{\mathrm{grid}} &\le C_{\mathrm{bd}} + \bar{C}_{\mathrm{aud}}\mathbb{E}[A_T^{\mathrm{init}}] + \frac{2\bar{c}_{\mathrm{per}}}{p_0}(2M) \\
&\quad + \frac{2}{p_0}\mathbb{E}\left[\sum_{\substack{\ell\in\mathcal{D}_T^{\mathrm{init}}:\\ N_{i_\ell,c_\ell}(\ell-1)\ge 1}} B_{i_\ell,c_\ell}(\ell-1)\right] \\
&\quad + \frac{L_{\mathrm{impl}}}{p_0}\mathbb{E}\left[\sum_{\ell\in\mathcal{D}_T^{\mathrm{init}}} \mathbf{1}\{c_\ell = \mathrm{EI}\}r_\ell^-\right] + \bar{C}_{\mathrm{op}}\sum_{\ell=1}^{T}\mathbb{P}(\mathcal{E}^c_{\ell-1}).
\end{aligned} \tag{123}$$

The last term is $O(T^{-3})$.

It remains to control the confidence-radius terms in (123). For every selected policy arm that has already been sampled, decompose

$$B_{i_\ell,c_\ell}(\ell-1) = B^{\mathrm{stat}}_{i_\ell,c_\ell}(\ell-1) + B^{\mu,\mathrm{eval}}_{i_\ell,c_\ell}(\ell-1) + B^{\mu,\mathrm{impl}}_{i_\ell,c_\ell}(\ell-1).$$

The pathwise statistical-radius summation lemma, stated for decision episodes initiated by T, gives

$$\sum_{\substack{\ell\in\mathcal{D}_T^{\mathrm{init}}:\\ N_{i_\ell,c_\ell}(\ell-1)\ge 1}} B^{\mathrm{stat}}_{i_\ell,c_\ell}(\ell-1) \le 2c_{\mathrm{stat},T}\sqrt{2MTL_{2M,T}}. \tag{124}$$

Also,

$$\sum_{\substack{\ell\in\mathcal{D}_T^{\mathrm{init}}:\\ N_{i_\ell,c_\ell}(\ell-1)\ge 1}} B^{\mu,\mathrm{eval}}_{i_\ell,c_\ell}(\ell-1) \le L_\alpha \bar{S}\sum_{\ell\in\mathcal{D}_T^{\mathrm{init}}} r_\ell^-. \tag{125}$$

Let $\mathcal{L}_T$ be the event that every episode initiated by the horizon has length at most $\bar{\Lambda}_T$. By Lemma 2,

$$\mathbb{P}(\mathcal{L}_T^c) \le T^{-4}.$$

On $\mathcal{L}_T$, the tracker-parameter-radius summation lemma gives

$$\sum_{\substack{\ell \in \mathcal{D}_T^{\mathrm{init}}: \\ N_{i_\ell, c_\ell}(\ell-1) \ge 1}} B_{i_\ell, c_\ell}^{\mu, \mathrm{impl}}(\ell - 1) \le L_{\mathrm{impl}} \bar{\Lambda}_T \log(eT) \sum_{\ell \in \mathcal{D}_T^{\mathrm{init}}} r_\ell^-. \tag{126}$$

On the complement, each tracker-parameter radius is at most L_{impl} and at most T episodes can be initiated. Therefore

$$\begin{aligned} &\mathbb{E}\left[\sum_{\substack{\ell \in \mathcal{D}_T^{\mathrm{init}}: \\ N_{i_\ell, c_\ell}(\ell-1) \ge 1}} B_{i_\ell, c_\ell}^{\mu, \mathrm{impl}}(\ell - 1) \right] \\ &\qquad \le L_{\mathrm{impl}} \bar{\Lambda}_T \log(eT) \mathbb{E}\left[\sum_{\ell \in \mathcal{D}_T^{\mathrm{init}}} r_\ell^- \right] + L_{\mathrm{impl}} T \mathbb{P}(\mathcal{L}_T^c). \end{aligned} \tag{127}$$

The second term is $O(T^{-3})$.

Define

$$\mathcal{W}_T^{\mathrm{stat}} := \sum_{\substack{\ell \in \mathcal{D}_T^{\mathrm{init}}: \\ N_{i_\ell, c_\ell}(\ell-1) \ge 1}} B_{i_\ell, c_\ell}^{\mathrm{stat}}(\ell - 1), \qquad \mathcal{U}_T^{\mu} := \sum_{\ell \in \mathcal{D}_T^{\mathrm{init}}} r_\ell^-.$$

Combining the master inequality with (125) and (127), and using $\mathbf{1}\{c_\ell = \mathrm{EI}\} \le 1$, yields the adaptive finite-grid bound

$$\begin{aligned} R_T^{\mathrm{grid}} \le\ & C_{\mathrm{bd}} + \bar{C}_{\mathrm{aud}} \mathbb{E}[A_T^{\mathrm{init}}] + \frac{2\bar{c}_{\mathrm{per}}}{p_0}(2M) + \frac{2}{p_0} \mathbb{E}[\mathcal{W}_T^{\mathrm{stat}}] \\ & + C_{\mathrm{sh,T}} \mathbb{E}[\mathcal{U}_T^{\mu}] + O(T^{-3}). \end{aligned} \tag{128}$$

By definition,

$$R_T = R_T^{\mathrm{grid}} + T\varepsilon_M, \qquad \varepsilon_M = g_{\cup, M}^\star - g_{\cup}^\star.$$

Thus (128) proves (42).

Substituting (124)– (127) into (123) gives

$$\begin{aligned} R_T^{\mathrm{grid}} \le\ & C_{\mathrm{bd}} + \bar{C}_{\mathrm{aud}} \mathbb{E}[A_T^{\mathrm{init}}] + \frac{4\bar{c}_{\mathrm{per}}}{p_0} M + \frac{4c_{\mathrm{stat},T}}{p_0} \sqrt{2MTL_{2M,T}} \\ & + C_{\mathrm{sh,T}} \mathbb{E}\left[\sum_{\ell \in \mathcal{D}_T^{\mathrm{init}}} r_\ell^- \right] + O(T^{-3}). \end{aligned} \tag{129}$$

The audit-count lemma, with audits counted when they are initiated, gives

$$A_T^{\mathrm{init}} \le 1 + \frac{L_T^{1/3} T^{2/3}}{2\vartheta_T^2}$$

pathwise. The cumulative-radius lemma, likewise stated for decision episodes initiated by T, gives

$$\sum_{\ell \in \mathcal{D}_T^{\mathrm{init}}} r_\ell^- \le \frac{3}{2} \vartheta_T L_T^{1/3} T^{2/3}$$

pathwise. Therefore

$$\begin{aligned}
&\bar{C}_{\mathrm{aud}}\mathbb{E}[A_T^{\mathrm{init}}] + C_{\mathrm{sh,T}}\mathbb{E}\left[\sum_{\ell\in\mathcal{D}_T^{\mathrm{init}}} r_\ell^-\right] \\
&\quad\le \bar{C}_{\mathrm{aud}} + \left(\frac{\bar{C}_{\mathrm{aud}}}{2\vartheta_T^2} + \frac{3}{2}C_{\mathrm{sh,T}}\vartheta_T\right) L_T^{1/3}T^{2/3} \\
&\quad= \bar{C}_{\mathrm{aud}} + \left(\frac{3}{2}\right)^{5/3} \bar{C}_{\mathrm{aud}}^{1/3} C_{\mathrm{sh,T}}^{2/3} L_T^{1/3} T^{2/3},
\end{aligned} \tag{130}$$

where the final equality uses (37). When $M \le T$, $\bar{\Lambda}_T$, L_T, $L_{2M,T}$, $C_{\mathrm{sh,T}}$, and $c_{\mathrm{stat},T}$ are polylogarithmic in T. Hence (129) implies

$$R_T^{\mathrm{grid}} = \widetilde{O}\left(M + \sqrt{MT} + T^{2/3}\right). \tag{131}$$

Finally, we pass from the finite-grid benchmark to the continuous benchmark. Let $(c_\cup^\star, S_\cup^\star)$ be an optimizer of the continuous benchmark $g_\cup^\star$. The uniform grid contains a point s_j within distance $\bar{S}/M$ of $S_\cup^\star$. Lemma 12, applied at the fixed target parameter $\nu_{c_\cup^\star}$, therefore gives

$$0 \le g_{\cup,M}^\star - g_\cup^\star \le g(s_j;\nu_{c_\cup^\star}) - g(S_\cup^\star;\nu_{c_\cup^\star}) \le L_S\frac{\bar{S}}{M}.$$

Therefore

$$\begin{aligned}
R_T &= R_T^{\mathrm{grid}} + T\left(g_{\cup,M}^\star - g_\cup^\star\right) \\
&\le \widetilde{O}\left(M + \sqrt{MT} + T^{2/3}\right) + L_S\bar{S}\frac{T}{M}.
\end{aligned}$$

With $M = M_T := \lceil T^{1/3}\rceil$, the terms M_T, $\sqrt{M_T T}$, and T/M_T are all $O(T^{2/3})$. Thus

$$R_T = \widetilde{O}(T^{2/3}),$$

which proves the theorem.

C.8. Proof of Theorem 2

Hard instance class. Set

$$h = 1, \qquad b = 10, \qquad \bar{S} = 3,$$

take the known mean-retention interval to be

$$[\underline{\mu}, \bar{\mu}] = \left[\frac{3}{16}, \frac{9}{40}\right],$$

initialize $X_1 = 0$, and fix arbitrary common values

$$s_{\mathrm{aud}} \in (0, 3], \qquad \kappa_{\mathrm{aud}} > 0.$$

The demand law is known to the learner and common to both environments:

$$D_t \sim \mathrm{Unif}\left[1, \frac{6}{5}\right].$$

For $0 < \Delta \le 1/32$, define the two retention laws

	0	1/10	3/5
P_Δ^+	1/2	$3/16 - \Delta$	$5/16 + \Delta$
P_Δ^-	1/2	$3/16 + \Delta$	$5/16 - \Delta$.

(132)

Their respective means are

$$\mu_{\pm} = \frac{33}{160} \pm \frac{\Delta}{2} \in \left[\frac{3}{16}, \frac{9}{40}\right].$$

The hard parameter lies entirely in the retention distribution. Both environments satisfy Assumptions 1– 4, with

$$p_0 = \frac{1}{2}, \qquad \bar{f}_D = 5.$$

Let

$$\mathfrak{M}_{\mathrm{sh}} := \left\{ M_{\Delta}^{\sigma} : \sigma \in \{+,-\},\ 0 < \Delta \le \frac{1}{32} \right\}$$

denote this fixed class, with all other primitive parameters held fixed.

Learner classes. The class $\Pi_{\cup}$ consists of nonanticipative tracker-based algorithms operating in stockout-delimited episodes. Each ordinary episode runs some π_S^{ν}, with $S \in [0,3]$ and $\nu \in (0,1]$, until the next stockout. An audit may start only at a stockout boundary, uses the common audit-start order s_{aud}, and, if necessary, is followed to the next stockout by a nonanticipative recovery rule with orders in $[0,3]$. These choices may be adaptive and randomized. The oracle class Π_{or} uses the same physical dynamics, costs, audit actions, recovery controls, and benchmark, but receives the additional information specified below.

We prove the lower bound for the oracle-aided class in two steps. First, we show that every bounded execution policy must place physical pre-shrinkage inventory on the costly side of a fixed threshold in at least one of the two environments. Second, we show that even under full oracle feedback the two environments cannot be reliably distinguished over T periods when $\Delta = O(T^{-1/2})$.

Let Ξ be the learner's complete independent random seed. Immediately before the period-t order, enlarge the learner's information to

$$\mathcal{F}_{t-1}^{\mathrm{or}} := \sigma\big(\Xi, X_t, (D_s, \rho_s, X_{s+1}, C_s, q_s, J_s^{\mathrm{aud}})_{s<t}\big). \tag{133}$$

The common demand law is revealed to the learner, but the environment index $\sigma \in \{+,-\}$, and hence the retention law and its mean, are not revealed in advance. Under Π_{or}, the order $q_t \in [0,3]$ is $\mathcal{F}_{t-1}^{\mathrm{or}}$-measurable and $(D_t, \rho_t, X_{t+1}, C_t)$ is revealed exactly after the period. Physical audits and arbitrary bounded recovery controls remain available. Because ordinary, audit-start, and recovery orders are all bounded by $\bar{S} = 3$, every algorithm in $\Pi_{\cup}$ belongs to Π_{or} by (43). Since audit fees are nonnegative, omitting them can only weaken the resulting lower bound.

Write

$$B_t := X_t + q_t$$

for physical pre-shrinkage inventory, and let

$$g_{\sigma}(S) := g_{M_{\Delta}^{\sigma}}(S; \mu_{\sigma}), \qquad g_{\sigma,\mathrm{EI}}^{\star} := \inf_{S \in [0,3]} g_{\sigma}(S), \qquad g_{\sigma,\cup}^{\star} := g_{\cup}^{\star}(M_{\Delta}^{\sigma}),$$

and define the auxiliary regret against the EI policy class

$$R_{T,\mathrm{EI}}^{\pi}(M_{\Delta}^{\sigma}) := \mathbb{E}_{\sigma}^{\pi}\left[\sum_{t=1}^{T} C_t^{\mathrm{alg}}\right] - T g_{\sigma,\mathrm{EI}}^{\star}. \tag{134}$$

Because the two-class comparator contains the EI policy class,

$$g_{\sigma,\cup}^{\star} \le g_{\sigma,\mathrm{EI}}^{\star}, \qquad R_T^{\pi}(M_{\Delta}^{\sigma}) \ge R_{T,\mathrm{EI}}^{\pi}(M_{\Delta}^{\sigma}). \tag{135}$$

Lemma 17 *For every policy in Π_{or} and every period,*

$$0 \le X_t \le 2, \qquad 0 \le B_t \le 5.$$

Consequently, the state $\rho_t = 1/10$ always produces a stockout.

Proof. Since $q_t \le 3$, $\rho_t \le 3/5$, and $D_t \ge 1$,

$$X_{t+1} \le \left(\frac{3}{5}(X_t+3)-1\right)^+ \le \frac{3}{5}X_t + \frac{4}{5}.$$

Starting from $X_1 = 0$, induction gives $X_t \le 2$, and hence $B_t \le 5$. Therefore

$$\frac{1}{10}B_t \le \frac{1}{2} < 1 \le D_t,$$

which proves the final claim.

□

Parameterize a generic retention law in the hard family by

$$\mathbb{P}(\rho = 0) = \frac{1}{2}, \qquad \mathbb{P}\left(\rho = \frac{1}{10}\right) = \frac{1}{2} - \lambda, \qquad \mathbb{P}\left(\rho = \frac{3}{5}\right) = \lambda.$$

For deterministic $B \in [0,5]$, define

$$\ell_\lambda(B) := \mathbb{E}\left[(\rho B - D)^+ + 10(D - \rho B)^+\right], \qquad \Psi(B) := \mathbb{E}\left[\left(D - \frac{3}{5}B\right)^+\right],$$

where the first expectation is taken under the retention law indexed by λ and the common demand law. Since B_t is predictable and the current primitives are fresh, every oracle policy satisfies

$$\mathbb{E}[C_t \mid \mathcal{F}^{\mathrm{or}}_{t-1}] = \ell_\lambda(B_t). \tag{136}$$

Lemma 18 *For every $B \in [0,5]$,*

$$\ell_\lambda(B) = C(\lambda) + a(\lambda)B + 11\lambda\Psi(B), \tag{137}$$

where

$$C(\lambda) = \{10(1-\lambda) - \lambda\}\mathbb{E}[D], \qquad a(\lambda) = \frac{8}{5}\left(\lambda - \frac{5}{16}\right).$$

Moreover,

$$\Psi(B) = 0 \qquad (B \ge 2),$$

and, for $B = 2 - x$ with $0 \le x \le 1/3$,

$$\Psi(2-x) = \frac{9}{10}x^2. \tag{138}$$

Proof. By Lemma 17, retention outcomes 0 and $1/10$ always stock out. Conditioning on ρ_t gives

$$\ell_\lambda(B) = \frac{1}{2}\,10\mathbb{E}[D] + \left(\frac{1}{2} - \lambda\right)10\left\{\mathbb{E}[D] - \frac{1}{10}B\right\} + \lambda\mathbb{E}\left[\left(\frac{3}{5}B - D\right)^+ + 10\left(D - \frac{3}{5}B\right)^+\right].$$

The identity

$$z^+ + 10(-z)^+ = z + 11(-z)^+$$

with $z = 3B/5 - D$ yields (137). If $B \geq 2$, then $3B/5 \geq 6/5$, so $\Psi(B) = 0$. If $B = 2 - x$ and $0 \leq x \leq 1/3$, then

$$\frac{6}{5} - \frac{3}{5}B = \frac{3}{5}x.$$

Demand has density five on $[1, 6/5]$, and direct integration gives

$$\Psi(2-x) = \frac{5}{2}\left(\frac{3}{5}x\right)^2 = \frac{9}{10}x^2.$$

□

Set

$$B_c := \frac{11}{4}.$$

Lemma 19 *Under M_Δ^+, every $B \in [0,5]$ satisfies, relative to the optimal cost within the EI policy class,*

$$\ell_+(B) - g^\star_{+,\mathrm{EI}} \geq \Delta\, \mathbf{1}\{B \geq B_c\} - \frac{2}{25}\Delta. \tag{139}$$

Proof. Write

$$\lambda_+ := \frac{5}{16} + \Delta, \qquad a_+ := \frac{8}{5}\Delta, \qquad \ell_+ := \ell_{\lambda_+}.$$

For $B \leq 5/3$, one has $\Psi(B) = \mathbb{E}[D] - 3B/5$, and hence

$$\ell'_+(B) = a_+ - \frac{33}{5}\lambda_+ < 0.$$

On $[5/3, 2]$, put $B = 2 - x$ and use (138):

$$\ell_+(2-x) = C(\lambda_+) + 2a_+ - a_+ x + \frac{99}{10}\lambda_+ x^2,$$

so

$$\ell'_+(B) = a_+ - \frac{99}{5}\lambda_+(2-B).$$

For $B \geq 2$, $\ell'_+(B) = a_+ > 0$. Since

$$0 < \frac{5a_+}{99\lambda_+} < \frac{1}{3},$$

the global minimizer is

$$B^\star_+ = 2 - \frac{5a_+}{99\lambda_+},$$

and the global conditional minimum equals

$$c^{\min}_+ := \min_{0 \leq B \leq 5} \ell_+(B) = C(\lambda_+) + 2a_+ - \frac{5a_+^2}{198\lambda_+}. \tag{140}$$

Now consider the admissible EI policy $\pi_2^{\mu_+}$. Its tracker remains zero by induction: after a stockout it resets, while after a nonstockout $Y_t = D_t \geq 1 > 2\mu_+$, so the tracker update is zero. Thus the policy orders two in every period. Under $\rho_t = 3/5$, $B_t \geq 2$ and the period is a nonstockout almost surely; either of the other retention outcomes resets the true inventory to zero. This regenerative chain has a unique stationary law, and its stationary mean pre-inventory is

$$\bar{B}_2(\lambda_+) = 2 + \frac{\lambda_+/10}{1 - 3\lambda_+/5}. \tag{141}$$

Indeed, stationarity and the reset under the other two retention outcomes give

$$\mathbb{E}[X] = \lambda_+ \left\{\frac{3}{5}\mathbb{E}[X] + \frac{1}{10}\right\}, \qquad \bar{B}_2 = 2 + \mathbb{E}[X].$$

Uniform regeneration implies that the zero-start long-run average equals the invariant expected one-period cost. Since $\Psi(B) = 0$ along this policy,

$$g_+(2) = C(\lambda_+) + a_+ \bar{B}_2(\lambda_+). \tag{142}$$

For $\lambda_+ \le 11/32$,

$$\bar{B}_2(\lambda_+) - 2 \le \frac{11}{254} < \frac{1}{20}.$$

For every policy in the EI class and every period,

$$\mathbb{E}[C_t \mid \mathcal{F}_{t-1}^{\mathrm{or}}] = \ell_+(B_t) \ge c_+^{\min}.$$

Taking time averages and then minimizing over the EI policy class gives

$$c_+^{\min} \le g_{+,\mathrm{EI}}^{\star}.$$

The admissibility of $S = 2$ and (142) give

$$g_{+,\mathrm{EI}}^{\star} \le g_+(2) = C(\lambda_+) + a_+ \bar{B}_2(\lambda_+).$$

Consequently,

$$\begin{aligned} 0 \le g_{+,\mathrm{EI}}^{\star} - c_+^{\min} &\le a_+ \{\bar{B}_2(\lambda_+) - 2\} + \frac{5a_+^2}{198\lambda_+} \\ &\le \left(\frac{44}{635} + \frac{16}{2475}\right)\Delta < \frac{2}{25}\Delta. \end{aligned} \tag{143}$$

Thus

$$\ell_+(B) - g_{+,\mathrm{EI}}^{\star} \ge -\frac{2}{25}\Delta \qquad \text{for every } B \in [0,5].$$

If $B \ge B_c$, then $\Psi(B) = 0$, and

$$\begin{aligned} \ell_+(B) - g_{+,\mathrm{EI}}^{\star} &\ge \ell_+(B) - g_+(2) \\ &\ge a_+ \left\{\frac{11}{4} - 2 - \frac{1}{20}\right\} \\ &= \frac{28}{25}\Delta \ge \Delta. \end{aligned}$$

Combining the global and high-inventory bounds proves (139). □

Lemma 20 *Under M_Δ^-, every oracle policy π satisfies the auxiliary regret bound against the EI policy class*

$$R_{T,\mathrm{EI}}^{\pi}(M_\Delta^-) \ge \frac{2}{5}\Delta \sum_{t=1}^{T} \mathbb{P}_-^{\pi}(B_t < B_c)\,. \tag{144}$$

Proof. Write

$$\lambda_- := \frac{5}{16} - \Delta, \qquad a_- := -\frac{8}{5}\Delta, \qquad \ell_- := \ell_{\lambda_-}.$$

Equation (137) shows that $\ell_-(B)$ is nonincreasing: its linear coefficient is negative and $\Psi(B)$ is nonincreasing.

Let $\pi^{\max}$ be the non-auditing policy that orders three in every period, starting from $X_1 = 0$. Couple π and $\pi^{\max}$ with the same primitive sequence. Since every order of π is at most three and the state recursion is monotone,

$$X_t^{\pi} \le X_t^{\max}, \qquad B_t^{\pi} = X_t^{\pi} + q_t^{\pi} \le X_t^{\max} + 3 = B_t^{\max}$$

pathwise. Conditional on the coupled past,

$$\mathbb{E}_-[C_t^\pi - C_t^{\max} \mid \text{past}] = \ell_-(B_t^\pi) - \ell_-(B_t^{\max}) \geq 0.$$

On $\{B_t^\pi < B_c\}$, one also has $B_t^{\max} \geq 3$, so

$$\begin{aligned}\ell_-(B_t^\pi) - \ell_-(B_t^{\max}) &\geq \ell_-(B_c) - \ell_-(3) \\ &= |a_-| \left(3 - \frac{11}{4}\right) \\ &= \frac{2}{5}\Delta,\end{aligned}$$

where both comparison points exceed two and hence have $\Psi = 0$. Summing gives

$$\mathbb{E}_-^\pi \left[\sum_{t=1}^T C_t\right] - \mathbb{E}_-^{\pi^{\max}} \left[\sum_{t=1}^T C_t\right] \geq \frac{2}{5}\Delta \sum_{t=1}^T \mathbb{P}_-^\pi(B_t < B_c). \tag{145}$$

It remains to identify the EI-class comparator exactly. For every policy $\pi_S^{\mu_-}$, the tracker remains zero by induction because

$$3\mu_- < 1 \leq D_t.$$

Hence $\pi_S^{\mu_-}$ is the constant-order policy $q_t = S$. Coupling two such policies shows that the larger value of S produces larger pre-inventory pathwise. Since ℓ_- is nonincreasing, $S = 3$ minimizes the long-run average cost within the EI policy class. Thus

$$\pi^{\max} = \pi_3^{\mu_-}, \qquad g_{-,\mathrm{EI}}^\star = g_-(3).$$

The mass $1/2$ at zero forces a reset and gives the $S = 3$ chain a unique invariant law. Uniform regeneration also implies convergence to that law, so its invariant one-period expected cost equals the zero-start long-run average $g_-(3) = g_{-,\mathrm{EI}}^\star$. Let $\widetilde{X}_1$ have the invariant law, and couple the stationary chain with the zero-start chain using the same primitives. Monotonicity gives

$$X_t^{\max} \leq \widetilde{X}_t \qquad \text{for all } t.$$

Since ℓ_- is nonincreasing,

$$\mathbb{E}_-\left[C_t^{\max} - \widetilde{C}_t \mid X_t^{\max}, \widetilde{X}_t\right] = \ell_-(X_t^{\max} + 3) - \ell_-(\widetilde{X}_t + 3) \geq 0.$$

The stationary chain incurs expected cost $g_{-,\mathrm{EI}}^\star$ in every period, so

$$\mathbb{E}_-^{\pi^{\max}} \left[\sum_{t=1}^T C_t\right] \geq T g_{-,\mathrm{EI}}^\star. \tag{146}$$

Combining (145) and (146), and adding the learner's nonnegative audit fees, proves (144).

□

We use the total-variation convention

$$\mathrm{TV}(P, Q) := \sup_A |P(A) - Q(A)|.$$

For $\pi \in \Pi_{\mathrm{or}}$, let $\mathbb{P}_\sigma^{\pi, t-1}$ denote the law of the oracle pre-period history $\mathcal{F}_{t-1}^{\mathrm{or}}$ under M_Δ^σ.

Lemma 21 *For every $\pi \in \Pi_{\mathrm{or}}$ and every $t \le T$,*

$$\mathrm{KL}\big(\mathbb{P}_+^{\pi,t-1}, \mathbb{P}_-^{\pi,t-1}\big) \le 40(t-1)\Delta^2. \tag{147}$$

Proof. The demand laws are identical. Conditional on the seed and primitive sequence, all states, actions, audits, and observations are deterministic. Data processing and the adaptive KL chain rule therefore give

$$\mathrm{KL}\big(\mathbb{P}_+^{\pi,t-1}, \mathbb{P}_-^{\pi,t-1}\big) \le (t-1)\,\mathrm{KL}(P_\Delta^{+,\rho}, P_\Delta^{-,\rho}).$$

Only the masses at $1/10$ and $3/5$ differ. The chi-square upper bound gives

$$\mathrm{KL}(P_\Delta^{+,\rho}, P_\Delta^{-,\rho}) \le \frac{4\Delta^2}{3/16+\Delta} + \frac{4\Delta^2}{5/16-\Delta} \le 40\Delta^2$$

for $\Delta \le 1/32$.

□

Proof of Theorem 2. Fix an arbitrary $\pi \in \Pi_{\mathrm{or}}$. Set

$$\Delta_T := \frac{1}{8\sqrt{40T}}. \tag{148}$$

This belongs to $(0, 1/32]$ for every $T \ge 1$. By Lemma 21 and Pinsker's inequality,

$$\mathrm{TV}\big(\mathbb{P}_+^{\pi,t-1}, \mathbb{P}_-^{\pi,t-1}\big) \le \frac{1}{8} \qquad (t \le T). \tag{149}$$

Let

$$H_t := \{B_t \ge B_c\}.$$

Because the complete random seed is included in $\mathcal{F}_{t-1}^{\mathrm{or}}$, the order q_t and the event H_t are $\mathcal{F}_{t-1}^{\mathrm{or}}$-measurable. Hence

$$\mathbb{P}_+(H_t) + \mathbb{P}_-(H_t^c) \ge 1 - \mathrm{TV}\big(\mathbb{P}_+^{\pi,t-1}, \mathbb{P}_-^{\pi,t-1}\big) \ge \frac{7}{8}. \tag{150}$$

Conditional independence of the current primitives, Lemma 19, and nonnegative audit fees give

$$R_{T,\mathrm{EI}}^{\pi}(M_{\Delta_T}^{+}) \ge \Delta_T \sum_{t=1}^{T} \mathbb{P}_+(H_t) - \frac{2}{25}\Delta_T T. \tag{151}$$

Lemma 20 gives

$$R_{T,\mathrm{EI}}^{\pi}(M_{\Delta_T}^{-}) \ge \frac{2}{5}\Delta_T \sum_{t=1}^{T} \mathbb{P}_-(H_t^c). \tag{152}$$

Adding the two displays, using the smaller coefficient $2/5$, and applying (150),

$$\begin{aligned} R_{T,\mathrm{EI}}^{\pi}(M_{\Delta_T}^{+}) + R_{T,\mathrm{EI}}^{\pi}(M_{\Delta_T}^{-}) &\ge \frac{2}{5}\Delta_T \sum_{t=1}^{T} \{\mathbb{P}_+(H_t) + \mathbb{P}_-(H_t^c)\} - \frac{2}{25}\Delta_T T \\ &\ge \left(\frac{2}{5}\frac{7}{8} - \frac{2}{25}\right)\Delta_T T \\ &= \frac{27}{100}\Delta_T T. \end{aligned}$$

Therefore

$$\max_{\sigma\in\{+,-\}} R_{T,\mathrm{EI}}^{\pi}(M_{\Delta_T}^{\sigma}) \ge \frac{1}{8}\Delta_T T = \frac{1}{64\sqrt{40}}\sqrt{T}.$$

By (135), this auxiliary lower bound transfers pointwise to the two-class regret:

$$\max_{\sigma\in\{+,-\}} R_T^{\pi}(M_{\Delta_T}^{\sigma}) \ge \max_{\sigma\in\{+,-\}} R_{T,\mathrm{EI}}^{\pi}(M_{\Delta_T}^{\sigma}) \ge \frac{1}{64\sqrt{40}}\sqrt{T}.$$

Both alternatives belong to the fixed class $\mathfrak{M}_{\mathrm{sh}}$. The preceding argument holds for every $\pi \in \Pi_{\mathrm{or}}$; taking the infimum proves the oracle-class inequality. Since $\Pi_{\cup} \subseteq \Pi_{\mathrm{or}}$ by (43), the same lower bound holds for $\Pi_{\cup}$. Because the argument is in calendar time, ordinary cycles and recovery tails crossing T are included without a renewal remainder.

□

Appendix D: Supplementary Numerical-Experiment Details

This section specifies the benchmark construction, diagnostic protocol, retailer-data processing, and simulation-based uncertainty calculations for the numerical experiments in Section 6.

D.1. Synthetic benchmark construction

Explore–then–commit (ETC) concentrates learning at the beginning of the horizon and then uses one replenishment policy for all remaining periods. For an exploration exponent β, it operates at a prespecified high-stock level S_{learn}, chosen to reveal demand without censoring, for

$$H_\beta = \min\{T, \lceil T^\beta \rceil\}$$

periods. ETC chooses its physical-count budget separately from the exploration duration. Before operations begin, it selects

$$n_\beta^{\mathrm{aud}} \in \arg\min_{2 \le n \le H_\beta} \left\{ \kappa_{\mathrm{aud}} n + \frac{(T - H_\beta) c_{\mathrm{op}}}{2\sqrt{n}} \right\},$$

where c_{op} is a prespecified scale for the operating consequences of retention-estimation error. In the audit-count experiment,

$$c_{\mathrm{op}} = |g^\star_{\mathrm{BI}} - g^\star_{\mathrm{EI}}| = 3.15148,$$

computed on an independent selection stream at $\mu_\rho = 0.95$. This value is used for all three values of β and all audit fees. The first n_β^{aud} exploration periods are audited consecutively. ETC continues to operate at S_{learn} without auditing during the remaining $H_\beta - n_\beta^{\mathrm{aud}}$ periods.

At the end of exploration, ETC forms an empirical demand distribution from the high-stock demand observations and an empirical retention distribution from the exact count observations, including complete-loss realizations. It then optimizes the base-stock level separately over the common S-grid for the expected-inventory rule, $\nu = \widehat{\mu}_\rho$, and the book-inventory rule, $\nu = 1$. The two optimized (S, mode) pairs are evaluated under the same estimated dynamic inventory model, and the lower-cost pair is selected. ETC uses that pair for the remainder of the horizon without further audits, estimation updates, or policy revisions. Holding, lost-sales, and audit costs incurred during exploration are included in its reported performance.

The planning-horizon comparison uses 128 common demand–retention paths per cell, the mean-retention comparison uses 20, and each holding- and lost-sales-cost cell uses 500. In the audit-cost experiment, the reported KM–Audit means use 128 fee-specific paths; the ETC curves are deterministic planned budgets conditional on the displayed inputs.

D.2. Retention-bound diagnostic

Figure 6 isolates the role of the three confidence intervals in the audit gate. It evaluates trigger decisions on exogenously generated information histories; inventory evolution and operating costs are outside its scope. For each $\mu_\rho \in \{0.85, 0.95, 0.98\}$, we generate 100 independent information histories of 1,000 episodes. Retention follows the zero-inflated Beta family used in the synthetic design, with complete-loss probability 0.01 and conditional-Beta precision 4, and an ordinary episode has an independent censoring threshold distributed uniformly on $[0, 1]$. Each path begins with the mandatory exact initialization audit, leaving 999 eligible post-initialization episode starts.

The complete three-interval gate generates a common sequence of exact and censored observations. At every eligible start, each leave-one-out gate is evaluated on this same history after omitting the indicated interval. This common-history design isolates the direct contribution of each interval to the audit decision. The full intersection is nonempty on every simulated history. Trigger rates exclude the mandatory initialization audit and are computed over the 999 eligible post-initialization starts. Pointwise 95% intervals are Student-t intervals across the 100 independent paths.

At $\mu_\rho = 0.85$, the full intersection has a mean radius of 20.33 percentage points and a trigger rate of 0.33%. Removing the audit-only interval raises these values to 23.36 percentage points and 64.97%; removing bracketing yields 21.47 percentage points and 26.18%; and removing Kaplan–Meier raises the trigger rate to 8.01%. At $\mu_\rho = 0.95$, removing Kaplan–Meier increases the mean radius from 20.09 to 24.35 percentage points and the trigger rate from essentially zero to 64.22%. At $\mu_\rho = 0.98$, the corresponding radius rises from 18.98 to 24.01 percentage points and the trigger rate to 63.76%. Removing either of the other two intervals has a much smaller effect in these high-retention regimes.

D.3. Additional cost-sensitivity results

In the audit-cost experiment, KM–Audit conducts an average of 20.78, 8.15, 3.62, and 1.56 counts when κ_{aud} equals 0, 50, 200, and 800, respectively. These totals include the mandatory initialization count. At $\kappa_{\mathrm{aud}} = 50$, ETC schedules 29, 29, and 28 counts for $\beta = 0.5$, 0.6, and 0.7; at $\kappa_{\mathrm{aud}} = 200$, the respective budgets are 12, 11, and 11; and at $\kappa_{\mathrm{aud}} = 800$, they are 5, 5, and 4. When counts have no direct fee, the three ETC policies use 100, 252, and 631 counts. The similarity of the positive-cost ETC budgets reflects the separation between the duration of high-stock demand learning, which is governed by β, and the cost-adjusted retention-sample budget.

The operating-cost experiment also records policy adjustments not displayed in the main text. As h rises from 0.67 to 1.5 with $b = 40$, KM–Audit's average selected inventory target falls from 54.05 to 50.87 and its stockout rate rises from 4.50% to 5.79%. Conversely, as b rises from 26.7 to 60 with $h = 1$, the selected target rises from 50.87 to 54.07 and the stockout rate falls from 5.79% to 4.49%. Its average cost per period nevertheless rises from 41.48 to 63.69 because each unmet unit carries a larger penalty.

D.4. Retailer-data construction and chronological replay

Transaction processing and cohort construction. We use database exports for 2023 and for January 1 through November 14, 2024. The tables identify payment dates, paid and refunded status, ordered quantities, product identities and categories, transaction prices, and product-master cost fields. We match cart lines to cart headers and retain positive quantities from carts marked as paid and not refunded.

For each SKU, retained quantities are aggregated by payment date and expanded to a dense daily series, with zero assigned to dates having no retained order. Bulk orders and unusually large quantities are preserved. These series are a paid-order proxy for potential demand. The exports contain neither historical inventory availability nor stockout indicators and therefore cannot reveal demand lost before an order was submitted.

A SKU must have matched product and category information, at least 20 positive-order days and 30 ordered units in 2023, a first positive order by March 1, a final positive order on or after November 1, and no internal zero-order spell longer than 60 days. These filters yield 112 training-eligible SKUs. Requiring positive-order support in 2024 leaves 104, and excluding three products without a usable positive gross-margin proxy gives the final cohort of 101 SKUs in nine categories.

Demand model and information timing. The 2023 observations initialize the demand model and select its preprocessing parameters. All policies start January 1, 2024, with zero physical and recorded inventory and the same demand history. The fixed 2024 paid-order sequence is then treated as latent potential demand and traversed once, in calendar order, for each retention realization. Demand is neither shuffled nor resampled.

The common policy-side demand layer estimates one global day-of-week profile from the full 2023 cohort and updates each SKU's local level online through an exponentially weighted adjustment with a 56-day half-life. Observations from the most recent 365 days are retained. After ordering, a nonstockout sale is an exact demand observation, whereas a stockout supplies a right-censored observation at the realized sales quantity. A rolling Kaplan–Meier estimator combines these observations. Each policy updates this estimator online from its observed sales history. The simulator preserves the dated demand path and reveals neither contemporaneous nor future demand. Parameters calibrated from 2023 remain fixed throughout the 2024 evaluation. The exercise is a retrospective chronological replay designed to preserve operational information timing rather than an out-of-sample forecasting test.

FAO calibration and retention paths. The retailer data contain no historical physical-inventory records, count outcomes, or observed shrinkage. We therefore calibrate a counterfactual retention environment from an archived extract of the Food and Agriculture Organization of the United Nations' *Food Loss and Waste Database*, which contains 30,301 records. We retain records whose activity or supply-chain stage refers to retail, wholesale, distribution, or storage, map commodities to broad families, and take the unweighted mean reported loss fraction within each family. The resulting targets are 2.84% for cereals and bakery, 10.30% for produce, 9.10% for dairy and eggs, and 2.15% for meat and fish. Bakery and pantry products use the cereals-and-bakery target; produce and vegetarian products use the produce target; and meat and fish products use the meat-and-fish target. Prepared meals use the equally weighted mean of the produce, dairy, and meat-and-fish targets, 7.18%. Packaged beverages use a separately specified 1.50% target because the few beverage records in the extract are not representative of the retailer's packaged assortment. These values are calibration targets, not estimates of shrinkage historically experienced by the retailer.

For SKU i, define the exposure-frequency proxy

$$\tau_i = \mathrm{clip}_{[1,30]}\left\lceil \frac{Q_{0.95}(D_{i,2023})}{\overline{D}_{i,2023}} \right\rceil, \qquad p_i = \frac{1}{\tau_i}.$$

The quantity τ_i is a demand-cadence proxy, not a shelf-life estimate; it ranges from 3 to 14 days in the final cohort. For a SKU in category c, the simulator draws

$$\rho_{it} = \begin{cases} 1 - \dfrac{1-\mu_c}{p_i}, & \text{with probability } p_i, \\ 1, & \text{with probability } 1-p_i. \end{cases}$$

Hence $\mathbb{E}[\rho_{it}] = \mu_c$, while a loss event removes the fraction

$$\frac{1-\mu_c}{p_i} = \tau_i(1-\mu_c)$$

of inventory exposed in that period. The specification separates event frequency, which varies with a SKU's demand cadence, from category-level mean retention.

We choose μ_c using only the 365 dated observations from 2023 so that aggregate simulated physical loss divided by aggregate inbound units matches the corresponding category target. Separate simulation paths are used for policy selection, moment fitting, and validation. The fitted daily mean retentions are 0.99940 for meat, 0.99743 for prepared

meals, and 0.99693 for produce. On fresh 2023 paths, the loss-to-inbound ratios are 2.13%, 7.26%, and 10.33%, compared with targets of 2.15%, 7.18%, and 10.30%. These calibrated retention laws define the 2024 simulation environment.

The simulator uses these laws to generate latent physical loss, inventory, stockout, and audit outcomes. These latent quantities are never supplied to the policies. All retention updates use only information revealed through each policy's own operating and audit history.

Policies, economic inputs, and performance measures. For SKU i, P_i is the median valid unit price in its 2023 paid transactions and c_i is the positive `price_net2023` field in the product master, used as a unit-cost proxy. Section 6.3 defines the resulting holding, lost-sales, and audit-cost inputs. They are declared economic assumptions rather than measured retailer accounting costs.

KM–Audit obtains exact retention observations through a physical count or an eligible first-period stockout reconciliation. The retailer implementation uses the common admissible mean-retention interval $[0.95, 0.9999]$ for all SKUs. Its lower endpoint, selected using full cost on a 2023 selection window from $\{0.05, 0.85, 0.90, 0.95\}$, is 0.95. At a replanning epoch KM–Audit optimizes the base-stock level separately for EI and BI over a common dense integer grid and selects the pair with the lower estimated dynamic cost. The retailer ETC variants follow the construction in Section D.1 with exponents 0.10, 0.20, and 0.30; 0.20 is selected using the 2023 chronology.

The service and waste metrics are

$$100\frac{\sum_{i,t}\text{unmet demand}_{it}}{\sum_{i,t}\text{potential demand}_{it}} \quad \text{and} \quad 100\frac{\sum_{i,t}\text{physical shrinkage}_{it}}{\sum_{i,t}\text{inbound units}_{it}},$$

respectively. Numerators and denominators are aggregated before each ratio is taken so that low-volume SKUs do not receive disproportionate weight.

Each SKU is evaluated on 40 independent retention paths with its 2024 demand chronology held fixed, and the same path and random seed are used for every policy. Pointwise 95% intervals use 500 paired bootstrap draws that resample retention realizations within SKU with common indices across policies. They measure simulation uncertainty conditional on the displayed SKUs and fixed demand histories; they are not population-level confidence intervals for the retailer's entire assortment.

D.5. Retailer inventory-record and audit-price comparison

The audit-price experiment uses the 49 displayed SKUs and the common demand and retention paths from the preceding design. Every audit-price cell is replayed from the beginning. Pointwise 95% intervals use 1,000 paired, fixed-SKU bootstrap draws that resample retention paths within each SKU, with common indices across policies and audit-price cells. EI only and BI only restrict the tracking convention but retain the same audit, reconciliation, and base-stock optimization mechanisms. ETC recomputes its cost-balanced count budget at each price.